Asset Pricing, Time Series, and the Limits of Prediction

A scientific compendium of one research session — from general-equilibrium theory to tested, dependency-free code

— A four-level model of a market and its participants
— The mathematics of analysing financial time series
— A SWOT of AI architectures for directional forecasting
— An implemented model with an honest significance gate
— A positive theory of where edge comes from — and why it survives

Compiled by Claude Code · 13 June 2026

Plain-Language Summary

Plain words, no equations — the whole argument for any reader, before the mathematics begins.

The one idea. A market price is not a prediction; it is what someone will pay to hold a risk — and that "someone" cares more about a dollar in bad times than a dollar in good times. So a price already bakes in *fear*, not just odds. That single gap — between the *price* of a future outcome and its *probability* — is what makes investing genuinely hard, and what makes most "I can predict the market" claims quietly false.

What the theory says, in words. You can always describe prices *as if* there were a hidden "fear-weighting" sitting behind them — economists call it the pricing kernel. But you can never dig that weighting back out of prices alone. So from prices you cannot separate what the market *believes* will happen from how much it *fears* it happening. This is not a technicality; it is the reason no formula reliably turns prices into a forecast.

What the data says, in words. Predicting tomorrow's up-or-down is close to a coin flip — about 51–53% right at best, honestly measured. Worse, most published "winning" strategies are mirages: they secretly use information from the future, they are cherry-picked from thousands of attempts, or they ignore trading costs. There is a discipline — testing on data the model has never seen, penalising how many things you tried, charging realistic costs — that separates the real from the imaginary. Almost nothing survives it. So far, this sounds bleak.

Why it is not bleak. A coin-flippy 53% edge sounds useless — until you apply it to *thousands of independent bets*. A tiny edge, repeated with enough breadth and sized with discipline, becomes a great track record. That is the "fundamental law" of active management: skill-per-bet multiplied by the square root of the number of independent bets equals your overall edge. It is exactly how the most successful quantitative fund in history works — Renaissance's Medallion returned roughly 39% a year, after fees, for thirty years, without a single losing year, on an edge so small per trade that it is invisible. And Warren Buffett beat the market for sixty years by finding a few durable advantages early and betting them patiently with cheap borrowed money.

So the honest bottom line is not "you cannot win." It is this: *most people, most methods, most of the time, lose — and a few, in specific niches, with a real and well-defended advantage, win persistently.* The market is beatable. It is just not beatable by the easy thing, and the entire purpose of this book is to be ruthless about telling the two apart — so that scarce time and money flow to the thin sliver that is real, not to the comfortable illusion.

What you can actually do. Look for a real *reason* you should win: better information, faster execution, a structural counterparty who is forced to trade against you, or a behaviour you do not share. Spread that reason across many genuinely independent bets — breadth is the lever. Respect that any edge is limited in size and fades with time. Size it carefully: even a real edge, over-bet, will ruin you. And prove it survives honest, out-of-sample, cost-aware testing before you trust a cent to it. Do all of that and you are doing what the winners do. Do less, and you are donating to them.

The chapters that follow make each of these claims precise — and show their limits.

Preface

Abstract. This compendium assembles, into a single structured volume, the work of one interactive research session. It moves from pure theory to running, tested code and is held together by a single thread: the gap between what prices *are* and what they *say*. Chapter 2 builds a general-equilibrium asset-pricing and microstructure model whose one guardrail is that the pricing kernel *exists but is not identified from prices*. Chapter 3 develops the mathematics of analysing financial time series, ending in the evaluation discipline (purged cross-validation, the deflated Sharpe ratio) that the theory demands. Chapter 4 is an evidence-based SWOT of AI and machine-learning architectures for *directional* stock forecasting, concluding that honest out-of-sample directional accuracy is ~50–55% across every model family. Chapter 5 turns the defensible recipe into a dependency-free, unit-tested implementation. Chapter 6 supplies the constructive counterpart — the Fundamental Law of Active Management, which shows the very edge the empirical chapters call tiny becomes a world-class information ratio once multiplied by breadth, plus why such edges survive. Chapter 7 states the synthesis. The unifying claim, proved from two directions, is that *markets price risk, not odds; models estimate odds, not edge; and the unidentified pricing kernel is exactly what makes both asset pricing hard and machine-learning backtests lie.* The purpose is constructive, not pessimistic: to find the model, regime, and horizon where an edge is real — and the conclusion is explicit that markets *are* beatable, rarely and specifically, as Medallion and Buffett prove. The severity throughout is the filter that protects scarce time from what cannot work; it is not a verdict that nothing does.

How to read it. Each chapter is self-contained and carries its own equations (tagged by logical type), figures, and references. Equations are rendered with KaTeX; figures are original. A methodology appendix documents how the volume was built, including the tooling limits encountered and a candid account of a bug — an in-sample leak — caught inside the project's own demonstration.

Provenance & disclaimer. Compiled by Claude Code from an interactive session on 13 June 2026. It is a research and education artefact, not financial advice. Load-bearing citations were verified against the published record where possible; items that could not be independently re-verified are flagged in place.

Reader’s Guide

This volume is dense by construction — it moves from expected-utility theory to stochastic calculus to gradient-boosted trees — so read it for your purpose, not cover to cover. Its aim is **constructive**: to identify which model fits which sector, regime, and horizon, and to supply a filter severe enough that scarce time is not spent on what cannot work. It is critical because the domain punishes credulity, not because it is pessimistic — its conclusion is that markets *are* beatable, rarely and specifically (Chapter 6 supplies the positive theory; Chapter 7 the synthesis), and the severity is in service of finding where.

- **The thesis in two pages:** the Plain-Language Summary, then the opening of Chapter 6.
- **What a price means (theory):** Chapter 2 — assumes comfort with expected utility and Itô calculus; its boundary and self-consistency sections need no calculus.
- **The empirical and methodological core:** Chapters 3 and 4 — assume basic statistics; the evaluation gate (§10 of Chapter 3) and the SWOT (Chapter 4) are self-contained.
- **The working code:** Chapter 5 — assumes Python; the results table stands alone.
- **A reading aid throughout:** every equation is tagged by logical type — *primitive*, *optimality*, *equilibrium/ market-clearing*, *definition*, or *estimating specification* — so the structure of an argument is legible even without parsing every symbol. Assumptions carry their empirical status inline (*holds / contested / known false*).

A note on the tensions the book reports rather than hides (Chapter 2's self-consistency section): they are of three kinds, and only one would be a defect. *Logical inconsistencies within one model* — none; the consistency check verifies clearing, binding budgets, and a positive kernel. *Seams between deliberately incompatible idealisations* used in different blocks (e.g. the negative prices allowed by the tractable Gaussian microstructure) — the real limit, intrinsic to modelling, since no single tractable model is at once a complete-markets equilibrium, a noisy rational-expectations economy, and a limits-to-arbitrage economy. *Genuinely open empirical problems* (the elasticity clash) — correctly left open, because the field has not settled them and faking a resolution would be the true cop-out.

Contents

CHAPTER 1

Introduction — Four Artefacts, One Argument

The session's thread and the unifying thesis.

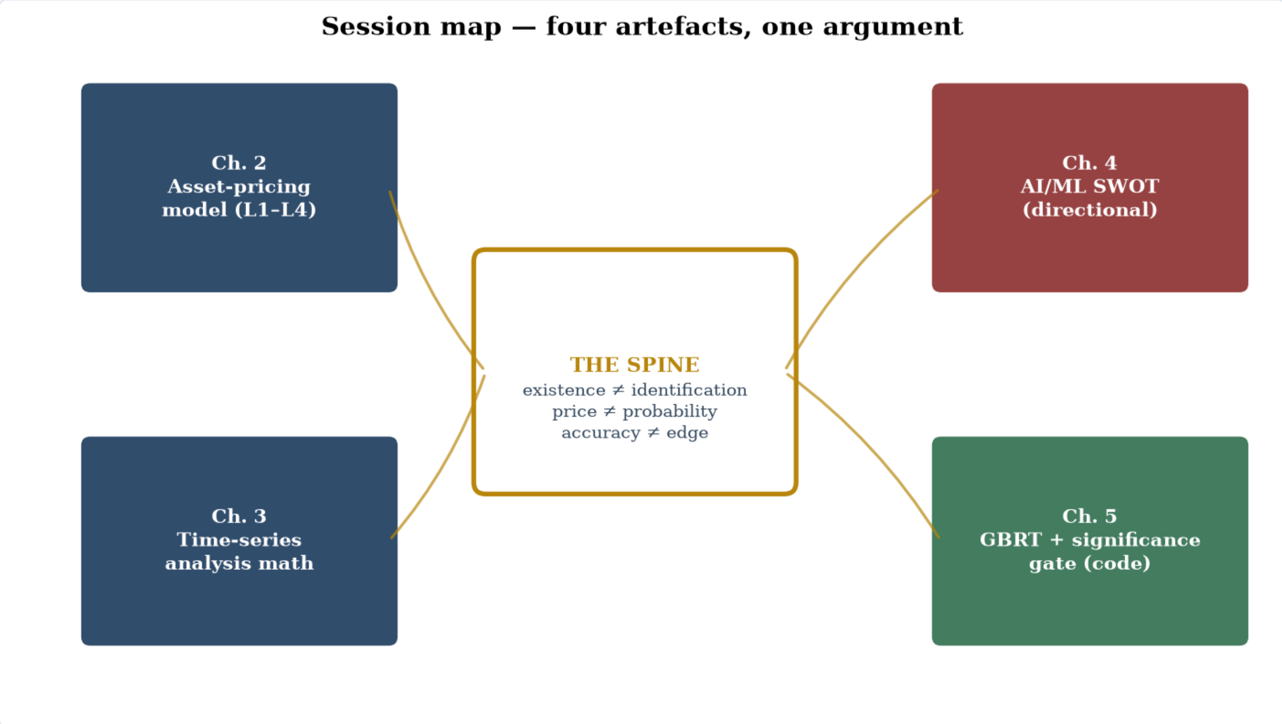

Figure 1.1. The four technical artefacts of the session and the single argument that unifies them.

This repository's `docs/` now holds three formal artefacts that look unrelated but are one argument seen from three angles:

- **`MARKET_EQUILIBRIUM_MODEL.md`** — a general-equilibrium asset-pricing + microstructure model (Levels 1–4), built on one framework (the FTAP: no-arbitrage ⇔ a positive stochastic discount factor) and one guardrail (*existence is not identification*).
- **`TIMESERIES_MATH.md`** — the mathematics of analysing financial time series, from stationarity to the purged-CV / deflated-Sharpe evaluation gate, grounded in what this codebase computes.
- **`AI_TIMESERIES_FORECASTING_SWOT.md`** — a SWOT of AI/ML architectures for *directional* stock forecasting, concluding that honest out-of-sample directional accuracy is ~50–55% across every model family.

The one-sentence thesis that unifies them:

> **Markets price risk, not odds; models estimate odds, not edge; and the gap between them — the unidentified pricing kernel — is exactly what makes both asset pricing hard and machine-learning backtests lie.**

The seven cross-cutting insights

1. The two halves are one theorem from two sides. The asset-pricing model's deepest result — *the kernel exists but is not identified from prices* (recovery fails because the SDF's permanent component is unidentifiable, §IV.L6) — is the **same wall** the ML models hit as the ~50–55% directional ceiling. To turn a price into a forecast you need the kernel; the kernel is not recoverable; so a model "predicting returns" cannot cleanly separate the market's *belief* from the *risk premium*. The non-identification of the physical measure **is** the coin-flip wall, expressed in probability rather than price.

2. "Price ≠ probability" and "accuracy ≠ edge" are the same caution. A 51%-accurate model posting a large pre-cost Sharpe (e.g. the benchmark CatBoost in Rahimikia et al. 2025) is the empirical shadow of the theoretical point that the risk-neutral density is a *valuation*, not a *forecast*. Hit-rate is nearly orthogonal to P&L; what pays is sizing/ranking over the risk-weighted distribution, not getting the sign right more often.

3. The binding constraint is identification and data — never model capacity. Across the SWOT, the *between-family* accuracy spread (~50–55%) is smaller than the *leaked-vs-clean* evaluation spread within any one family. In the asset-pricing model, the unknowable part is a specific object (the permanent kernel component). Both say: more model is the wrong lever; better data, honest evaluation, and naming what is structurally unknowable are the right ones. (`RESEARCH.md` reaching the identical verdict empirically — "the binding constraint is the data, not the model" — is this insight discovered the hard way.)

4. Bias–variance, made economic. On a near-random-walk, low-SNR target, flexible models spend capacity fitting noise — which is *why* transformers and deep RL lose to trees and regularized baselines here (`TIMESERIES_MATH.md` §9b, eq. 9.7). The asset-pricing doc's "GBM is a measure-zero corner" is the continuous-time twin: the world is jumps + stochastic volatility + heavy tails, so a model assuming smooth structure overfits precisely where there is nothing real to fit.

5. The endogenous-kernel rule and the no-look-ahead rule are the same discipline. Both forbid smuggling in the answer. Don't exogenize the discount rate — you'd assume the risk premium you are trying to explain. Don't let future data touch training — you'd assume the prediction you are trying to claim. One sin, two domains; purged + embargoed cross-validation is the econometric image of the endogenous-kernel requirement.

6. Where edge plausibly lives is consistent across both analyses. *Relative* (cross-sectional ranking) over *absolute* (directional timing); risk premia harvested **knowingly** rather than mistaken for alpha; structural / microstructure features over generic price-pattern learning. And the honest ceiling is low *and decaying* — reflexivity / signal decay (McLean–Pontiff) means the act of finding and trading a signal erodes it.

7. The verification gate is not bureaucracy — the math guarantees the naive number will lie. Backtest overfitting yields *negative* out-of-sample returns, not merely zero (Bailey–Borwein–López de Prado–Zhu). So the gate — purged + embargoed CV, the **up-rate** (not 50%) baseline, transaction costs, the deflated Sharpe / PBO / t>3 corrections, point-in-time + delisting-inclusive data, and reporting P&L *and* AUC rather than raw accuracy — is the only thing standing between you and a confidently-wrong result.

From theory to code

Insights 6 and 7 are not just advice — they are implemented in this repository as a runnable, tested recipe:

- **`vpts.ml.gbrt`** — `GradientBoostedTrees`, a dependency-free (numpy) gradient-boosted regression-tree learner, the survey's best-evidence family for noisy tabular financial data (insight 3/6), plus `gbrt_cross_sectional_eval`, which scores it under the existing **purged + embargoed CPCV** (`vpts.validation`) and reports directional accuracy against the **unconditional up-rate**, never 50% (insight 7).
- **`vpts.ml.significance`** — `deflated_sharpe_ratio` (Bailey–López de Prado), `probabilistic_sharpe_ratio`, `expected_max_sharpe`, and `directional_skill` (accuracy minus the majority-direction baseline): the significance gate that corrects a Sharpe for the number of trials and makes a null result correctly fail.

- `examples/gbrt_significance_demo.py` — the full recipe end-to-end on synthetic data: GBRT → CPCV → directional-vs-up-rate → deflated-Sharpe gate. Run it with and without a planted signal (`--null`) to watch the harness find signal when present and collapse to "no edge" when not — *including* the deflated Sharpe, which is computed on the out-of-sample IC stream so the null does not falsely pass.

The demo is also a live illustration of insight 5: an earlier version computed the significance gate on an *in-sample* refit and the null spuriously "passed" — the leakage the documents warn about, leaking into the demo itself. Running the gate on the out-of-sample stream is what makes the discipline real rather than decorative.

Coda — the purpose is constructive, and markets *are* beatable

The severity of these documents is a means, not an end. The point is **constructive**: to find the right model and the sector, regime, horizon, and usage where it works, and to fit the tool to the purpose. In a domain that punishes credulity, an unsparing filter is what protects scarce time — a hallucinated edge trusted, or a dead method pursued, costs more than the filter that would have caught it.

That criticality must not be mistaken for the false dogma that **"the market is unbeatable"** or that **"no one can consistently beat it."** Those statements are true only in an idealized, frictionless, stationary market that does not exist, and — as Keynes warned — "in the long run," in which "we are all dead." In the real, finite, frictional market they are false, and the counterexamples are decisive:

- **Medallion** (Renaissance) compounded at ~63% gross / ~39% net from 1988–2018 with *not one* losing year, and with **negative beta and negative factor loadings** — so it provably *cannot* be a risk premium. Cornell (2019, *JPM*) calls it "the ultimate counterexample" to efficiency: genuine, unexplained alpha.
- **Buffett** beat the market for sixty years (Sharpe ~0.79). Frazzini–Kabiller–Pedersen (2018, *FAJ*) show the alpha is *explainable* — ~1.7× leverage on cheap, safe, high-quality stocks (betting-against-beta + quality-minus-junk) — but he found and sized those durable edges *decades before they were named*, financed by patient insurance float.

Both **exemplify** this framework rather than refute it. The claim was never "edge is impossible"; it was that edge is **rare, specific, real-when-it-exists, and perishable** — living in particular niches and regimes, needing to be genuine rather than a mislabeled premium or an artifact, and needing to be defended against the decay that observation and capital bring. Medallion is true anomaly (capacity-capped near $10B, outside capital returned, fiercely defended); Buffett is premium-harvesting done supremely well. Each is what success *under this description* looks like.

And the joint-hypothesis problem **cuts both ways**: because no test of efficiency is independent of the model assumed, *market efficiency is itself an untestable maintained hypothesis*, not a proven null. Cornell's negative-factor-loading result is the sharper form — there is no risk model under which Medallion is fair compensation, so efficiency is not merely untested there but *refuted*. The base rate (most fail) is not the ceiling (a few persistently win). The severity of this volume is in service of getting into that thin right tail — never of denying it exists.

CHAPTER 2

A General-Equilibrium Model of a Market and Its Participants

Levels 1–4: from a static exchange economy to options, the risk-neutral density, and the non-identification of the physical measure.

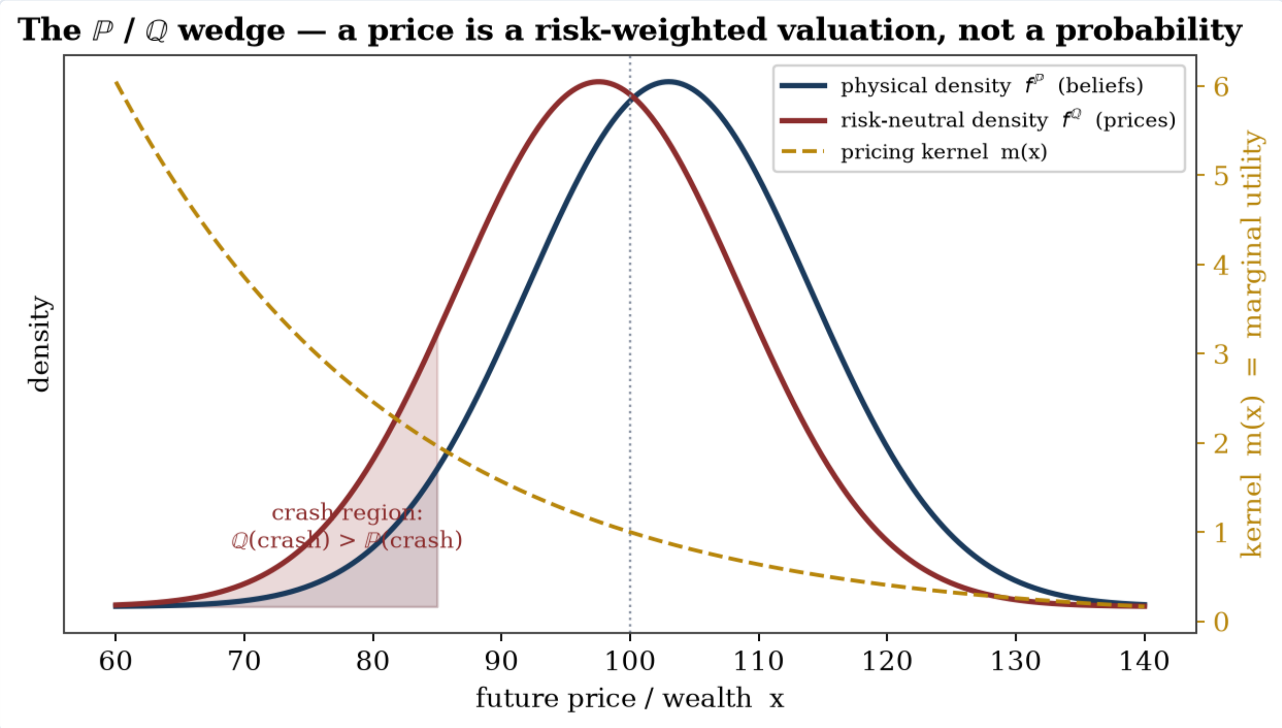

Figure 2.1. The risk-neutral density is the physical density tilted toward bad states by marginal utility — a valuation, not a forecast.

Status of this document. A formal construction in the tradition of general-equilibrium asset pricing and market microstructure. Every equation is tagged with its logical type:

- **[P]** primitive / assumption (with empirical status: *holds / contested / known false*)
- **[FOC]** optimality / first-order condition
- **[EQ]** market-clearing / equilibrium condition
- **[ID]** definition / identity
- **[EC]** econometric specification (an estimating equation, *not* an equilibrium object)

The construction runs the full difficulty ladder: **Part I** (Level 1: one period, finite states, two agent types, competitive equilibrium, closed form), **Part II** (Level 2: dynamic, recursive preferences, noisy rational-expectations equilibrium, market makers and microstructure), **Part III** (Level 3: continuous time, stochastic volatility and jumps, heterogeneous constrained intermediaries, inelastic demand, explicit limits to arbitrage), and **Part IV** (Level 3+: options, the risk-neutral density, and the physical law — what derivative prices do and do not reveal about the likelihood of future states). Each part is self-contained and runs the layers L0–L6. Two closing sections state what the formalization cannot capture and verify its internal consistency, including the tensions that remain.

The pricing kernel is endogenous throughout: it is derived from agents' first-order conditions and market clearing, never posited. Where a block uses a normalization (e.g., a unit gross rate inside a trading window), that is flagged and reconciled in the consistency check.

Methodological core — one framework, one guardrail

Before the symbols, the orientation. The entire construction rests on a single framework and is policed by a single guardrail, and they are two faces of one fact.

Framework — the Fundamental Theorem of Asset Pricing. No-arbitrage $\iff$ the existence of a strictly positive stochastic discount factor m. This is the minimal and most generative structure in the document: it assumes almost nothing, yet every closed form below — Arrow prices (I.9), the Euler equation $\mathbb{E}[mR] = 1$, risk-neutral valuation (IV.4), Breeden–Litzenberger (IV.5), the cross-sectional restrictions — is a corollary of "there exists a positive m." When a result is reduced to what it actually rests on, it terminates here. Keep one framework; keep this one.

Guardrail — existence is not identification. The FTAP *gives* that m exists and is positive; it is *silent* on two things, and that silence is the discipline of the whole field. (i) Under incomplete markets m is **not unique** (a set of pricing measures — (IV.11)). (ii) Even when unique, m does **not** arrive pre-decomposed into *beliefs* $\times$ *risk aversion* — the $\mathbb{P}/\mathbb{Q}$ wedge of (IV.13). Hence the cardinal error this document is built to prevent: **reading a risk-neutral price-density $f^{\mathbb{Q}}$ as if it were a physical-probability belief $f^{\mathbb{P}}$.** This error produces well-formed, perfectly wrong numbers while the model still looks internally consistent — which is why it dominates clearing errors or unit errors as the thing to watch. Two mechanisms enforce the guardrail throughout: the **joint-hypothesis problem** (no test of the kernel is independent of the model — closing-section item 5), which makes the non-identification logical rather than a data limitation; and the **endogenous-kernel rule** (the kernel is the object prices do *not* hand you, so it must be derived from preferences + clearing, never assumed — exogenizing the discount rate smuggles in the risk premium one claims to explain).

Operational compression. A price is the cost of a state weighted by how much you would care about being in it — never the bare odds of it; and no quantity of price data separates the two for you. Use market-implied densities for what is identified (the full $\mathbb{Q}$-surface, bounds on expected returns, risk *premia* paired with a physical forecast); refuse to read them as forecasts.

Notation

Symbols are global across Parts I–III unless the table says otherwise. Where one letter serves two roles, the table says so explicitly and the roles never collide within a part.

Symbol	Meaning
Spaces, time, probability	
$(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$	State space, σ-algebra, filtration $\mathbb{F} = (\mathcal{F}_t)$, physical measure
ω, S	Generic state; number of states (Part I, finite)
$\pi(\omega)$	Common prior probability of state ω (Part I)
t	Time: $t \in \{0, 1\}$ (Part I); $t \in \{0, 1, 2, \ldots\}$ (Part II); $t \in [0, \infty)$ (Part III)
$\mathbb{E}[\cdot]$, $\mathbb{E}_t[\cdot]$	Expectation under $\mathbb{P}$; conditional on $\mathcal{F}_t$
Assets, prices, payoffs	
$j \in \{0, 1, \ldots, J\}$	Assets; $j = 0$ is the one-period riskless bond
$d_j(\omega)$, $\mathbf{D}$	Time-1 payoff of asset j; $S \times J$ payoff matrix (Part I)
D_t	Aggregate dividend / consumption-claim flow (Parts II–III)
p_j, P_t	Asset prices; $q(\omega)$ = Arrow–Debreu state price

Symbol	Meaning
R_j, R_f, r_t	Gross return on j; gross riskless rate; instantaneous riskless rate. **Endogenous everywhere.**
$\bar{\theta}_j$, θ^k	Net supply of asset j; portfolio of agent k
f	Liquidation value of the *microstructure* asset (Parts II–III); μ_f, Σ_0 its prior mean and variance, Σ_1 posterior variance
Agents, preferences	
$k \in \mathcal{K}$	Agent types; each is a tuple $(\mathcal{I}_k, U_k, \mathcal{C}_k, \mathcal{A}_k)$: information, objective, constraint set, action space
c_t^k, C_t	Individual and aggregate consumption; e^k endowments; W_t^k wealth of agent k
$u_k(\cdot)$	Felicity function
γ_k	Absolute risk aversion (CARA contexts: Parts I–II) or relative risk aversion (CRRA/EZ contexts: Parts II–III); the part states which
Γ	Aggregate (harmonic) risk aversion: $\Gamma \equiv \left(\sum_k 1/\gamma_k\right)^{-1}$
β, ϱ	Subjective discount factor (discrete time); subjective discount *rate* (continuous time)
ψ, θ_{EZ}	Elasticity of intertemporal substitution; $\theta_{EZ} \equiv (1-\gamma)/(1-1/\psi)$
V_t, μ_t^{ce}	Continuation value (recursive utility); certainty equivalent $\mu_t^{ce} \equiv (\mathbb{E}_t[V_{t+1}^{1-\gamma}])^{1/(1-\gamma)}$
m, m_{t+1}, m_t	Stochastic discount factor (state-, one-period-, and level-form)
Information and microstructure	
s	Private signal about f
τ_x	Precision of random variable x: $\tau_x \equiv 1/\mathrm{Var}(x)$ (so τ_f, τ_ε, τ_u, $\tau_{\hat{s}}$). *Note: τ is never used for risk tolerance; tolerance is written $1/\gamma_k$.*
ε	Signal noise, $s = f + \varepsilon$
u	Noise-trader net demand, $u \sim N(0, \sigma_u^2)$
φ	Mass of informed investors (Grossman–Stiglitz block); c_I = cost of becoming informed
$\hat{s}$, ϖ	Price-revealed composite signal $\hat{s} = s + \varpi u$; its noise loading ϖ
x, $X(\cdot)$, b_I	Insider order, insider strategy, insider trading intensity (Kyle block)
y	Aggregate order flow $y = x + u$
λ	Kyle's lambda (price impact, dollars per share per share)
a, b, δ_t, ι, $\mathcal{H}_t$	Ask, bid, public posterior $\delta_t = \mathbb{P}(f = f^H \mid \mathcal{H}_t)$, probability a trader is informed, trade history (Glosten–Milgrom block)
Continuous time (Part III)	
B_t^D, B_t^v, B_t^u	Standard Brownian motions (dividend, variance, noise order flow); correlation $d\langle B^D, B^v\rangle_t = \rho\, dt$
$N(dt, dz)$, $\tilde{N}$, $\ell(dz)$	Poisson random measure, its compensated version, jump intensity measure; z = jump size in log dividends
g, v_t, κ, $\bar{v}$, σ_v	Dividend drift; spot variance; variance mean-reversion speed, level, vol-of-vol
η_t^D, η_t^v, $Y(z)$	Market prices of diffusive risk; SDF jump kernel
$\Phi(v)$, $H(n; v)$, $A_\Phi(n)$, $B_\Phi(n)$	Price–dividend ratio; value of the n-maturity dividend strip; its Riccati coefficients
Institutions and frictions (Part III)	

Symbol	Meaning
W^H, W^I, W^A	Wealth of households, intermediaries, arbitrageurs. *(Superscripted W is always wealth; B is always Brownian.)*
ϑ	Intermediary outside-equity multiple (He–Krishnamurthy constraint)
h_j, ξ_t	Margin/haircut on asset j; shadow price of the margin constraint
χ	Proportional transaction cost rate
A_i, $w_i(j)$, $\varkappa_j$, $b_{p,i}$	Assets under management of institution i; its portfolio weight on j; characteristics of j; its price-elasticity coefficient (Koijen–Yogo block)
ζ, $\mathcal{M}$	Aggregate price elasticity of demand; flow-to-price multiplier $\mathcal{M} = 1/\zeta$

Two glyph warnings: φ (informed mass) vs. Φ (price–dividend function) are distinct; Σ_0, Σ_1 (Kyle variances) vs. the summation symbol $\sum$ are distinct.

Part I — Level 1: Static exchange economy, two agent types, closed form

I.L0 — Environment

Time is discrete with two dates, $t \in \{0, 1\}$. Uncertainty: finite $\Omega = \{\omega_1, \dots, \omega_S\}$, $\mathcal{F} = 2^\Omega$, $\mathcal{F}_0 = \{\emptyset, \Omega\}$, $\mathcal{F}_1 = \mathcal{F}$, common prior $\pi(\omega) > 0$.

$$\text{(I.1)} \qquad \big(\Omega, 2^\Omega, (\mathcal{F}_0, \mathcal{F}_1), \mathbb{P}\big), \quad \mathbb{P}(\{\omega\}) = \pi(\omega) > 0.$$

[P] Common prior, objective and known. *Status: contested* (rational-expectations common prior; see the Boundary section).

There is one perishable consumption good per date (the numéraire at each date). Assets: $j = 0$ is riskless with $d_0(\omega) = 1$ for all ω, in zero net supply $\bar\theta_0 = 0$; assets $j = 1, \dots, J$ are risky with payoffs $d_j(\omega) \geq 0$ and net supplies $\bar\theta_j > 0$. Payoff matrix $\mathbf{D} = [d_j(\omega)] \in \mathbb{R}^{S \times J}$.

$$\text{(I.2)} \qquad \operatorname{rank}(\mathbf{D}) = S \quad \text{(complete markets)}.$$

[P] *Status: known false empirically* (markets are incomplete; used here to obtain Arrow prices and closed forms).

Agents $k \in \{A, B\}$ have time-0 endowment $e_0^k > 0$ of the good and initial asset holdings $\bar\theta^k$ with $\sum_k \bar\theta^k = \bar\theta$. Time-1 endowments are zero — all date-1 consumption is financed through assets:

$$\text{(I.3)} \qquad C_0 \equiv e_0^A + e_0^B, \qquad C_1(\omega) \equiv \sum_{j=1}^{J} \bar\theta_j d_j(\omega).$$

[ID] Aggregate resources. (Setting $e_1^k = 0$ is **[P]**, innocuous under (I.2) since any spanned endowment can be repackaged as asset holdings.)

Given (I.2), no arbitrage is equivalent to the existence of strictly positive Arrow prices $q(\omega)$ (state-ω consumption claims), and every asset price is their linear functional:

$$\text{(I.4)} \qquad p_j = \sum_\omega q(\omega)\, d_j(\omega), \qquad p_0 = \sum_\omega q(\omega) \equiv \frac{1}{R_f}.$$

[ID] Law of one price under spanning; R_f is *defined* here but its level is **endogenous** — it comes out of (I.9) below, not assumed. (No-arbitrage ⇔ positive state prices in finite state spaces: Ross 1976-style argument; the general theorem is in Harrison–Kreps 1979, *JET*.)

I.L1 — Information

Both agents observe everything: $\mathcal{I}_A = \mathcal{I}_B = \mathbb{F}$. Signals are degenerate; the prior is common (I.1); updating is trivial. Asymmetric information is deferred to Part II by design — Level 1 isolates risk sharing from information.

[P] Symmetric information. *Status: known false* (insider trading and information asymmetries are documented; Part II repairs this).

I.L2 — Participants as decision problems

Both types are utility-maximizing investors; market makers, noise traders, strategic informed traders, arbitrageurs, and passive demand enter in Parts II–III. For $k \in \{A, B\}$, the tuple is

$$\mathcal{I}_k = \mathbb{F}, \qquad U_k = -\tfrac{1}{\gamma_k} e^{-\gamma_k c_0^k} - \beta \sum_\omega \pi(\omega)\, \tfrac{1}{\gamma_k} e^{-\gamma_k c_1^k(\omega)}, \qquad \mathcal{A}_k = \{(c_0^k, c_1^k(\cdot)) \in \mathbb{R}^{1+S}\},$$

$$\text{(I.5)} \qquad \mathcal{C}_k : \quad c_0^k + \sum_\omega q(\omega)\, c_1^k(\omega) \;\le\; \mathcal{W}^k \equiv e_0^k + \sum_j \bar\theta_j^k\, p_j .$$

[P] CARA felicity $u_k(c) = -\gamma_k^{-1} e^{-\gamma_k c}$, $\gamma_A \,/\, \gamma_B =$allowed; Arrow–Debreu budget (valid given (I.2)). *Status of CARA: known false* (zero wealth effects in risky demand; admits unboundedly negative consumption — this very pathology buys the closed form and removes corner solutions).

Each agent solves $\max_{(c_0, c_1) \in \mathcal{C}_k} U_k$. The Lagrangian first-order conditions, after eliminating the multiplier by dividing the date-1 condition by the date-0 condition, are

$$\text{(I.6)} \qquad q(\omega) \;=\; \beta\, \pi(\omega)\, \frac{u_k'(c_1^k(\omega))}{u_k'(c_0^k)} \;=\; \beta\, \pi(\omega)\, e^{-\gamma_k \left(c_1^k(\omega) - c_0^k\right)}, \qquad k \in \{A, B\},\ \forall \omega.$$

[FOC] Necessary and sufficient by strict concavity of U_k and convexity of $\mathcal{C}_k$; interior by CARA (utility defined on all of $\mathbb{R}$). Budgets bind by strict monotonicity.

I.L3 — Price formation

Walrasian clearing, the appropriate mechanism for symmetric-information price-taking agents (no order flow exists to learn from — microstructure mechanisms are vacuous here and appear in Part II):

$$\text{(I.7)} \qquad \sum_k c_0^k = C_0, \qquad \sum_k c_1^k(\omega) = C_1(\omega) \ \ \forall \omega \qquad \Big(\Longleftrightarrow \sum_k \theta_j^k = \bar\theta_j \ \forall j, \ \text{given (I.2) and binding}$$

[EQ] Goods-market clearing at both dates; asset clearing is equivalent by spanning and Walras's law (verified in the Self-Consistency section).

I.L4 — Equilibrium concept, existence, uniqueness, closed form

Definition (competitive equilibrium). A price system $(q(\omega))_{\omega \in \Omega} \gg 0$ and allocations $(c_0^k, c_1^k(\cdot))_k$ such that (i) each allocation solves agent k's problem (I.5)–(I.6) at those prices, and (ii) markets clear (I.7).

Derivation of the closed form. Solve (I.6) for the consumption spread of each agent:

$$\text{(I.8)} \qquad c_1^k(\omega) - c_0^k = -\frac{1}{\gamma_k}\,\ln\frac{q(\omega)}{\beta\,\pi(\omega)}.$$

[FOC] (Rearrangement of (I.6); the step is a logarithm.)

Sum (I.8) over $k \in \{A, B\}$ and impose (I.7):

$$C_1(\omega) - C_0 = -\left(\tfrac{1}{\gamma_A} + \tfrac{1}{\gamma_B}\right)\ln\frac{q(\omega)}{\beta\pi(\omega)} \quad\Longleftrightarrow\quad \boxed{q(\omega) = \beta\,\pi(\omega)\,e^{-\Gamma\,(C_1(\omega)-C_0)}}, \qquad \Gamma \equiv \left(\tfrac{1}{\gamma_A} + \tfrac{1}{\gamma_B}\right)^{-1}. \tag{I.9}$$

[EQ] Equilibrium state prices in closed form. The market prices risk with the *harmonic mean* of risk aversions — risk tolerances add. This is Wilson's syndicate-aggregation result (Wilson 1968, *Econometrica*): the two-agent economy admits a representative CARA agent with tolerance $1/\Gamma = 1/\gamma_A + 1/\gamma_B$, and here aggregation is *derived*, not assumed.

Existence and uniqueness: (I.9) exhibits the equilibrium price vector explicitly; it is strictly positive since $\pi > 0$, and it is the *unique* one consistent with (I.6)–(I.7), since the chain (I.8)→(I.9) is a chain of equivalences. Allocations follow uniquely:

$$\text{(I.10)} \qquad c_1^k(\omega) = c_0^k + \frac{\Gamma}{\gamma_k}\big(C_1(\omega) - C_0\big),$$

[FOC + EQ] (substitute (I.9) back into (I.8)). Each agent absorbs the share $\Gamma/\gamma_k = (1/\gamma_k)/(1/\gamma_A + 1/\gamma_B)$ of aggregate risk — **linear risk sharing in proportion to risk tolerance** (Wilson 1968; Pareto efficiency follows from the First Welfare Theorem under (I.2), Arrow–Debreu 1954, *Econometrica*; Debreu 1959).

The date-0 levels come from the (binding) budgets. With $Q \equiv \sum_\omega q(\omega)$ and $K \equiv \sum_\omega q(\omega)\,(C_1(\omega) - C_0)$:

$$\text{(I.11)} \qquad c_0^k = \frac{\mathcal{W}^k - \frac{\Gamma}{\gamma_k}K}{1 + Q}.$$

[ID] (Insert (I.10) into (I.5) with equality.)

Portfolio implementation: given (I.2), invert spanning. Equation (I.10) says c_1^k is an affine function of $C_1(\omega) = \sum_j \bar\theta_j d_j(\omega)$, so

$$\text{(I.12)} \qquad \theta_j^k = \frac{\Gamma}{\gamma_k}\,\bar\theta_j \;\; (j \geq 1), \qquad \theta_0^k = c_0^k - \frac{\Gamma}{\gamma_k}C_0 \text{ (bond position making (I.10) levels).}$$

[ID] Two-fund separation: every agent holds the *market portfolio* scaled by relative risk tolerance, plus the bond (mutual-fund separation; cf. Cass–Stiglitz 1970, *JET*, and Rubinstein 1974, *JFE*, for aggregation in securities markets).

I.L5 — Frictions

None at this level: trading is frictionless **[P]**, *status: known false*. Every friction is introduced in Part III as an explicit delta to a Part I/II condition, so the frictionless conditions here are the baseline being perturbed. (One preview: a short-sale constraint $\theta_j^k \geq 0$ would turn (I.6) into an inequality — see (III.20).)

I.L6 — Pricing kernel and cross-sectional restriction

Define the SDF as the state-price density:

$$\text{(I.13)} \qquad m(\omega) \equiv \frac{q(\omega)}{\pi(\omega)} \overset{\text{(I.9)}}{=} \beta\,e^{-\Gamma\,(C_1(\omega)-C_0)} \;>\; 0.$$

[ID + EQ] The kernel is **endogenous**: it is the marginal-rate-of-substitution of either agent (by (I.6) they agree — complete-markets kernel uniqueness), expressed via clearing in terms of *aggregate* consumption. Positivity is automatic (exponential).

Euler equations and the endogenous riskless rate, derived by substituting (I.13) into (I.4):

$$\text{(I.14)} \qquad \mathbb{E}[m\,R_j] = 1 \;\; \forall j, \qquad R_f = \frac{1}{\mathbb{E}[m]} = \frac{1}{\beta\,\mathbb{E}\left[e^{-\Gamma(C_1-C_0)}\right]}.$$

[FOC + EQ] ($R_j \equiv d_j/p_j$.) The discount rate is an output: patient ($\beta\uparrow$), abundant-future ($C_1\downarrow$ risk), or more risk-tolerant ($\Gamma\downarrow$) economies generate it differently.

Cross-sectional restriction (covariance decomposition of (I.14), an algebraic identity given (I.14)):

$$\text{(I.15)} \qquad p_j = \frac{\mathbb{E}[d_j]}{R_f} + \operatorname{Cov}(m, d_j), \qquad \mathbb{E}[R_j] - R_f = -R_f \operatorname{Cov}(m, R_j).$$

[ID] Assets covarying negatively with m — i.e., paying off when aggregate consumption is high, by (I.13) — carry positive premia. This is the finite-state consumption CAPM (Lucas 1978, *Econometrica*, static version; Breeden 1979, *JFE*, for the continuous-time analogue). Fundamental theorem connection: existence of the strictly positive m in (I.13) is equivalent to absence of arbitrage here (finite states: Harrison–Kreps 1979, *JET*).

Part II — Level 2: Dynamics, recursive preferences, noisy REE, market makers

Part II has two interacting blocks, and the architecture is stated honestly up front. **Block C** (consumption core) prices aggregate risk dynamically and produces the endogenous kernel m_{t+1}. **Block M** (microstructure) determines how information gets into the price of an individual security *within* a trading window, through three mechanisms: a noisy rational-expectations market (Grossman–Stiglitz), a strategic batch auction (Kyle), and a sequential dealer market (Glosten–Milgrom). Block M uses the normalization $R_f = 1$ inside the window **[P]** (*status: an approximation, accurate for windows of hours or days; reconciled with Block C's kernel in the Self-Consistency section, where the resulting wedge is quantified*).

II.L0 — Environment

Time $t \in \{0, 1, 2, \dots\}$. A single consumption good. $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ with $\mathbb{F}$ generated by the dividend process and the Block-M randomness.

Assets in Block C: a claim to the aggregate dividend $\{D_t\}$ (equity, net supply 1), and a one-period riskless bond in zero net supply (price $1/R_{f,t}$, **endogenous**). Aggregate dividend growth:

$$\text{(II.1)} \qquad \Delta c_{t+1} \equiv \ln(D_{t+1}/D_t) \;\; \text{i.i.d. } N(\mu_c, \sigma_c^2).$$

[P] Lognormal i.i.d. growth. *Status: contested-to-false* (growth has small persistent components and heteroskedasticity; the i.i.d. case is used to obtain closed forms, and the consequence of relaxing it is stated at (II. 10)).

Asset in Block M: one security with terminal liquidation value f, plus the same bond. In the Grossman–Stiglitz (GS) and Kyle blocks, $f \sim N(\mu_f, \Sigma_0)$, $\Sigma_0 = 1/\tau_f$ **[P]** (*normality: known false for prices of limited-liability assets — it puts positive mass on $f < 0$; flagged again in the consistency check*). In the Glosten–Milgrom (GM) block, $f \in \{f^L, f^H\}$, prior $\delta_0 = \mathbb{P}(f = f^H)$ **[P]**.

II.L1 — Information

Block C: symmetric information, $\mathcal{F}_t = \sigma(D_s, s \le t)$ for all agents **[P]** (*contested*).

Block M: the signal structure is private. Exact updating rules, stated once and used throughout:

Lemma N (normal–normal posterior). If $X \sim N(\mu_X, \tau_X^{-1})$ and, given X, the signal $S = X + \epsilon$ with $\epsilon \sim N(0, \tau_\epsilon^{-1})$ independent, then

$$\text{(II.2)} \qquad X \mid S \;\sim\; N\!\left(\frac{\tau_X\,\mu_X + \tau_\epsilon\,S}{\tau_X + \tau_\epsilon},\; \frac{1}{\tau_X + \tau_\epsilon}\right).$$

[ID] Bayes' rule for Gaussian families: posterior precision is the sum of precisions; posterior mean is the precision-weighted average (DeGroot 1970, *Optimal Statistical Decisions*).

Lemma P (projection theorem). For jointly normal (X, Z): $\mathbb{E}[X \mid Z] = \mu_X + \mathrm{Cov}(X,Z)\mathrm{Var}(Z)^{-1}(Z - \mu_Z)$, $\mathrm{Var}(X \mid Z) = \mathrm{Var}(X) - \mathrm{Cov}(X,Z)\mathrm{Var}(Z)^{-1}\mathrm{Cov}(Z,X)$. **[ID]**

Information sets: informed GS investors have $\mathcal{I}_I = \sigma(s, p)$ with $s = f + \varepsilon$, $\varepsilon \sim N(0, \tau_\varepsilon^{-1})$ independent of f; uninformed have $\mathcal{I}_U = \sigma(p)$; the Kyle insider knows f; the Kyle market maker sees only total order flow y; the GM dealer sees the trade sequence $\mathcal{H}_t$. The common prior over all primitives is (II.1) and the distributions above **[P]** (*common prior: contested*).

II.L2 — Participants as decision problems

(a) Long-horizon households (Block C), Epstein–Zin. Identical preferences across households (aggregation is *proved* below, not assumed).

$$\mathcal{I} = \mathbb{F}, \qquad \mathcal{A} = \{(C_t, \alpha_t)\}_{t\ge 0}\ \text{(consumption, portfolio weights)}, \qquad \mathcal{C}:\ W_{t+1} = R_{w,t+1}\,(W_t - C_t),\ V$$

$$\text{(II.3)} \qquad U:\quad V_t = \left[(1-\beta)\,C_t^{1-1/\psi} + \beta\,\left(\mu_t^{ce}\right)^{1-1/\psi}\right]^{\frac{1}{1-1/\psi}}, \qquad \mu_t^{ce} \equiv \left(\mathbb{E}_t[V_{t+1}^{1-\gamma}]\right)^{\frac{1}{1-\gamma}}.$$

[P] Epstein–Zin–Weil recursive utility (Epstein–Zin 1989, *Econometrica*; Weil 1989, *JME*): risk aversion γ and EIS ψ disentangled. *Status: contested* (axiomatically coherent; empirically debated). $R_{w,t+1} = \sum_j \alpha_{j,t} R_{j,t+1}$ is the return on total wealth.

Derivation of the SDF. The recursion (II.3) is homogeneous of degree one in (C_t, μ_t^{ce}), and the constraint set is linear in wealth, so V_t is homogeneous of degree one in W_t. The consumption FOC equates the marginal utility of date-t consumption with the marginal value of wealth (envelope theorem), and the portfolio FOC prices every traded return against the same marginal rates. Carrying out these two steps (Epstein–Zin 1989, Thm. 3.1, for the full argument) yields the one-period kernel in continuation-value form:

$$\text{(II.4)} \qquad m_{t+1} \;=\; \beta\left(\frac{C_{t+1}}{C_t}\right)^{-1/\psi}\left(\frac{V_{t+1}}{\mu_t^{ce}}\right)^{1/\psi-\gamma}, \qquad \mathbb{E}_t[m_{t+1}R_{j,t+1}] = 1\ \ \forall j.$$

[FOC] Equivalent return-based form: $m_{t+1} = \beta^{\theta_{EZ}}(C_{t+1}/C_t)^{-\theta_{EZ}/\psi} R_{w,t+1}^{\theta_{EZ}-1}$ (Epstein–Zin 1989). Sanity check: $\gamma = 1/\psi$ kills the second factor and (II.4) collapses to the CRRA kernel $\beta(C_{t+1}/C_t)^{-\gamma}$, as it must.

Aggregation (required before any representative-agent use). Households have identical homothetic recursive preferences and markets in Block C are complete over aggregate states. A Pareto optimum with identical homothetic utilities gives each household a constant share of aggregate consumption, $c_t^k = a_k C_t$ with a_k constant (Gorman aggregation; Negishi 1960, *Metroeconomica*, planner construction). Substituting $c_t^k = a_k C_t$ into (II.4), the constants a_k cancel in every ratio, so all households — and hence a representative household consuming C_t —

share the *same* kernel. Heterogeneity that does **not** aggregate (risk aversion, information, constraints) is deliberately housed in Block M and Part III. **[derived]**

(b) Competitive informed investors (Block M, GS). $\mathcal{I}_I = \sigma(s,p)$; CARA over terminal wealth, coefficient γ_I; $\mathcal{A}$: demand $x_I \in \mathbb{R}$; $\mathcal{C}$: $W = W_0 + x_I(f-p)$ (with $R_f = 1$ in-window). For CARA-normal wealth, $\mathbb{E}[-e^{-\gamma W}] = -\exp(-\gamma \mathbb{E}W + \frac{\gamma^2}{2}\operatorname{Var}W)$, so the program is mean–variance and

$$\text{(II.5)} \qquad x_I = \frac{\mathbb{E}[f \mid s,p] - p}{\gamma_I \operatorname{Var}[f \mid s,p]} = \frac{\mathbb{E}[f \mid s] - p}{\gamma_I \operatorname{Var}[f \mid s]}, \qquad \mathbb{E}[f \mid s] = \frac{\tau_f \mu_f + \tau_\varepsilon s}{\tau_f + \tau_\varepsilon}, \quad \operatorname{Var}[f \mid s] = \frac{1}{\tau_f + \tau_\varepsilon}.$$

[FOC], using Lemma N. The second equality holds because, given s, the price (a function of (s,u), shown below) adds only information about $u \perp f$.

(c) Competitive uninformed investors (Block M, GS). Identical except $\mathcal{I}_U = \sigma(p)$, risk aversion γ_U: $x_U = \big(\mathbb{E}[f \mid p] - p\big) / \big(\gamma_U \operatorname{Var}[f \mid p]\big)$ **[FOC]**, with the conditional moments computed in II.L3 once the price function is known — this circularity *is* the rational-expectations fixed point.

(d) Strategic informed trader (Block M, Kyle). Risk-neutral; knows f; internalizes price impact; $\mathcal{A}$: market order $x \in \mathbb{R}$; conjectured pricing rule $p(y)$:

$$\text{(II.6)} \qquad X(f) \in \arg\max_x \, \mathbb{E}\big[(f - p(x+u))\, x \,\big|\, f\big].$$

[P + FOC] Risk neutrality of the insider: *assumption, contested but standard* (short horizon, diversified outside wealth).

(e) Market makers. *Kyle batch auction:* at least two risk-neutral dealers observe total $y = x + u$ and Bertrand-compete to fill it; competition drives expected profit conditional on y to zero:

$$\text{(II.7)} \qquad p(y) = \mathbb{E}[f \mid y].$$

[EQ] Zero-profit / semi-strong-efficiency pricing — an equilibrium condition, not an optimization. *Sequential dealer (GM):* quotes set so each side of the book earns zero expected profit against the mixed pool of traders:

$$\text{(II.8)} \qquad a_t = \mathbb{E}[f \mid \mathcal{H}_{t-1},\ \text{buy at } a_t], \qquad b_t = \mathbb{E}[f \mid \mathcal{H}_{t-1},\ \text{sell at } b_t].$$

[EQ] (Glosten–Milgrom 1985, *JFE*.) Regret-free quotes: the expectation conditions on the *information content of the trade itself*.

(f) Noise / liquidity traders. Exogenous demand $u \sim N(0,\sigma_u^2)$ (GS, Kyle); in GM, a fraction $1-\iota$ of arrivals buy or sell with probability $\frac{1}{2}$ each.

$$\text{(II.9)} \qquad u \sim N(0,\sigma_u^2) \text{ independent of } (f,\varepsilon).$$

[P] *Status: known false as a description of optimization* — these agents lose money on average and their objective is unmodeled. The standard defense is that they proxy for hedging or liquidity-driven trades (Black 1986, *JF*, "Noise"); their budget problem is a real loose end, reported in the Self-Consistency section, and without them prices would be fully revealing and information rents impossible (Grossman–Stiglitz 1980, *AER*).

II.L3 — Price formation: Walrasian REE, batch auction, dealer market — and how they relate

(i) Noisy rational-expectations equilibrium (Walrasian clearing)

Net supply of the Block-M asset is $\bar{x} \geq 0$; mass φ of informed, $1-\varphi$ uninformed. Clearing for every realization (s,u):

$$\text{(II.10)} \qquad \varphi\, x_I(s,p) + (1-\varphi)\, x_U(p) + u = \bar{x}.$$

[EQ]

What the price reveals — derived, not conjectured. Substituting (II.5) into (II.10), $(\varphi\tau_\varepsilon/\gamma_I)s + u$ must be a function of p alone (everything else in (II.10) is). Hence observing p is informationally equivalent to observing the composite signal

$$\text{(II.11)} \qquad \hat{s} \equiv s + \varpi u, \qquad \varpi \equiv \frac{\gamma_I}{\varphi\,\tau_\varepsilon}, \qquad \hat{s} \mid f \sim N\big(f,\ \tau_\varepsilon^{-1} + \varpi^2\sigma_u^2\big), \qquad \tau_{\hat{s}} \equiv \big(\tau_\varepsilon^{-1} + \varpi^2\sigma_u^2\big)^{-1}.$$

[EQ + ID] More informed capital ($\varphi \uparrow$), better signals ($\tau_\varepsilon \uparrow$), or bolder informed trading ($\gamma_I \downarrow$) make prices more revealing. By Lemma N,

$$\text{(II.12)} \qquad \mathbb{E}[f \mid p] = \frac{\tau_f\mu_f + \tau_{\hat{s}}\,\hat{s}}{\tau_f + \tau_{\hat{s}}}, \qquad \mathrm{Var}[f \mid p] = \frac{1}{\tau_f + \tau_{\hat{s}}}.$$

[ID] Exact posterior. Solving (II.10) for p using (II.5), (II.12):

$$\text{(II.13)} \qquad p \;=\; \frac{\varphi\,\frac{\tau_f\mu_f + \tau_\varepsilon s}{\gamma_I} \;+\; (1-\varphi)\,\frac{\tau_f\mu_f + \tau_{\hat{s}}\hat{s}}{\gamma_U} \;+\; u \;-\; \bar{x}}{\varphi\,\frac{\tau_f + \tau_\varepsilon}{\gamma_I} + (1-\varphi)\,\frac{\tau_f + \tau_{\hat{s}}}{\gamma_U}} \;=\; A_p + B_p\,\hat{s},$$

[EQ] with A_p, B_p constants ($B_p > 0$), because $\varphi\tau_\varepsilon s/\gamma_I + u = (\varphi\tau_\varepsilon/\gamma_I)\,\hat{s}$ — the price is a strictly increasing function of $\hat{s}$ alone, which *verifies* the informational equivalence used at (II.11). The fixed point closes: beliefs in (II.12) are computed from the true law of (II.13). The $-\bar{x}$ term is the risk-premium discount: with positive supply the price sits below the risk-neutral posterior mean even at neutral signals. (Grossman–Stiglitz 1980, *AER*; Hellwig 1980, *JET*.)

Endogenous information acquisition. Let becoming informed cost c_I in numéraire. For a CARA-γ_I agent with information set $\mathcal{G} \supseteq \sigma(p)$, substituting the optimal demand into the objective gives conditional expected utility $-e^{-\gamma_I W_0}\exp\big(-(\mathbb{E}[f|\mathcal{G}] - p)^2/(2\mathrm{Var}[f|\mathcal{G}])\big)$. Taking unconditional expectations with the Gaussian quadratic-form formula, and using the total-variance identity $\mathrm{Var}(f-p) = \mathbb{E}[\mathrm{Var}(f-p \mid \mathcal{G})] + \mathrm{Var}(\mathbb{E}[f-p \mid \mathcal{G}])$ — the same for both types since both condition on p — all terms cancel in the ratio except:

$$\text{(II.14)} \qquad \frac{\mathbb{E}U_{\text{informed}}}{\mathbb{E}U_{\text{uninformed}}} = e^{\gamma_I c_I}\sqrt{\frac{\mathrm{Var}[f \mid s]}{\mathrm{Var}[f \mid p]}} \qquad \Longrightarrow \qquad e^{2\gamma_I c_I} \;=\; \frac{\mathrm{Var}[f \mid p]}{\mathrm{Var}[f \mid s]} = \frac{\tau_f + \tau_\varepsilon}{\tau_f + \tau_{\hat{s}}(\varphi^*)} \quad \text{at an}$$

[EQ] Indifference condition determining φ^* (Grossman–Stiglitz 1980). **The Grossman–Stiglitz impossibility follows**: if the price were fully revealing ($\tau_{\hat{s}} = \tau_\varepsilon$, RHS $= 1$) and $c_I > 0$, the LHS exceeds 1 — contradiction. Informationally efficient prices and costly information are jointly inconsistent; equilibrium informativeness is interior. **[derived]**

(ii) Strategic batch auction (Kyle)

Conjecture linear rules $p(y) = \mu_f + \lambda y$ and $X(f) = b_I(f - \mu_f)$; both are *verified*. The insider problem (II.6) given the linear rule: $\max_x x(f - \mu_f - \lambda x)$, strictly concave iff $\lambda > 0$, giving

$$\text{(II.15)} \qquad X(f) = \frac{f - \mu_f}{2\lambda} \quad \Rightarrow \quad b_I = \frac{1}{2\lambda}.$$

[FOC] The dealer condition (II.7) with Lemma P applied to (f, y), $y = b_I(f - \mu_f) + u$:

$$\text{(II.16)} \qquad \lambda = \frac{\mathrm{Cov}(f, y)}{\mathrm{Var}(y)} = \frac{b_I\Sigma_0}{b_I^2\Sigma_0 + \sigma_u^2}.$$

[EQ] Solving (II.15)–(II.16) simultaneously:

$$\text{(II.17)} \qquad \boxed{\lambda = \frac{1}{2}\sqrt{\frac{\Sigma_0}{\sigma_u^2}}, \qquad b_I = \sigma_u\,\Sigma_0^{-1/2}, \qquad \Sigma_1 \equiv \mathrm{Var}(f \mid y) = \frac{\Sigma_0}{2}, \qquad \mathbb{E}[\pi_{\text{insider}}] = \frac{\sigma_u\sqrt{\Sigma_0}}{2}}$$

[EQ] (Kyle 1985, *Econometrica*.) Price impact rises with the value of private information (Σ_0) and falls with noise depth (σ_u); exactly half the private information enters the price; the insider's expected gain equals, dollar-for-dollar, the noise traders' expected loss, $\mathbb{E}[(f-p)u] = -\lambda\sigma_u^2 = -\sigma_u\sqrt{\Sigma_0}/2$, with dealers breaking even — the zero-sum audit is run in the Self-Consistency section. The SOC requires $\lambda > 0$: satisfied. Uniqueness holds within the linear class; uniqueness over all measurable equilibria is delicate and not claimed here.

(iii) Sequential dealer market (Glosten–Milgrom)

$f \in \{f^L, f^H\}$, public belief δ_t. Each arrival is informed with probability ι (buys iff $f = f^H$, sells iff $f = f^L$), noise otherwise (buy/sell with prob. $\frac{1}{2}$). Exact Bayes updates:

$$\text{(II.18)} \qquad \delta_t^+ \equiv \mathbb{P}(f^H \mid \mathcal{H}_{t-1}, \text{buy}) = \frac{\delta_t(1+\iota)}{1+\iota(2\delta_t-1)}, \qquad \delta_t^- \equiv \mathbb{P}(f^H \mid \mathcal{H}_{t-1}, \text{sell}) = \frac{\delta_t(1-\iota)}{1+\iota(1-2\delta_t)}.$$

[ID] (Bayes' rule with $\mathbb{P}(\text{buy} \mid f^H) = \iota + \frac{1-\iota}{2}$, $\mathbb{P}(\text{buy} \mid f^L) = \frac{1-\iota}{2}$.) Quotes from (II.8):

$$\text{(II.19)} \qquad a_t = f^L + \delta_t^+\,(f^H - f^L), \qquad b_t = f^L + \delta_t^-\,(f^H - f^L), \qquad a_t - b_t\Big|_{\delta_t = 1/2} = \iota\,(f^H - f^L).$$

[EQ] The spread is *pure adverse selection* — it is positive with zero order-handling cost iff $\iota > 0$, and proportional to both the share of informed trading and the stakes. Because each transaction price is a conditional expectation under the public filtration, the law of iterated expectations makes transaction prices a **martingale** with respect to public information **[derived]** (Glosten–Milgrom 1985, *JFE*).

(iv) Relating the mechanisms

Proposition II.1 (dealer pricing is the risk-neutral limit of Walrasian REE). In (II.13), let the uninformed sector become risk-neutral, $\gamma_U \to 0$. The uninformed terms dominate the numerator and denominator, and

$$\text{(II.20)} \qquad p \;\longrightarrow\; \frac{\tau_f\mu_f + \tau_{\hat{s}}\,\hat{s}}{\tau_f + \tau_{\hat{s}}} \;=\; \mathbb{E}[f \mid p].$$

[derived] The Walrasian price converges to the conditional expectation of value given the price's information — formally the same object as the Kyle dealer rule (II.7), with the composite signal $\hat{s}$ playing the role of order flow y. The two mechanisms differ in (a) *who* absorbs noise (risk-averse investors at a price concession vs. risk-neutral dealers at zero expected profit) and (b) *strategic* behavior (the Kyle insider shades b_I to manage impact; GS investors are price takers). Kyle 1989 (*REStud*) builds the bridge model — strategic traders submitting demand schedules — that nests both. The supply term $-\bar{x}$ in (II.13), absent in (II.20), is the risk premium that risk-neutral dealer pricing cannot carry; this matters for the cross-block consistency check.

II.L4 — Equilibrium concepts

Definition (noisy REE, Block M-GS). A price function $P : (s,u) \mapsto p$ and demands (x_I, x_U) such that (i) x_I and x_U are optimal given beliefs computed by Bayes' rule from the *true* joint law of $(f, s, u, P(s,u))$; (ii) (II.10) holds for a.e. (s,u); (iii) endogenous φ^* satisfies (II.14). Existence in the linear class: explicit construction (II.13) — for any $\varphi \in (0,1]$ a linear REE exists and is unique *within the linear class* (Hellwig 1980; Grossman–Stiglitz 1980). Global uniqueness is not guaranteed in general; Breon-Drish (2015, *REStud* 82(3), 868–921) gives a constructive existence proof and uniqueness within continuous-price-function equilibria for a class nesting GS/Hellwig without joint normality.

Definition (Kyle equilibrium). A pair $(X, p(\cdot))$ such that X solves (II.6) given $p(\cdot)$, and $p(\cdot)$ satisfies (II.7) given X — a Bayesian–Nash equilibrium between the insider and a competitive dealer sector. (II.17) is its unique linear representative.

Definition (GM equilibrium). Quote processes satisfying (II.8) with beliefs (II.18) — sequential equilibrium of the trading game.

Block C equilibrium is Radner: price processes and plans such that households optimize (II.4) and both markets clear, $C_t = D_t$ (goods) and bond in zero net supply **[EQ]** (Radner 1972, *Econometrica*; Lucas 1978 exchange-economy structure).

II.L5 — Frictions at this level

Two are already endogenous to Level 2, and both are *informational*: (i) the cost of information c_I, which by (II.14) makes equilibrium prices necessarily *imperfectly* revealing; (ii) the adverse-selection spread (II.19) and price impact (II.17), which are equilibrium compensation for trading against better-informed counterparties — not physical costs. Physical/contractual frictions (transaction costs, margins, short-sale bans, balance-sheet limits) are introduced as deltas in Part III L5.

II.L6 — The endogenous kernel and what it prices

Under (II.1) and the proven aggregation, conjecture (and verify) that wealth is proportional to consumption, $V_t = \bar{\phi}\, C_t$ for a constant $\bar{\phi}$ (homogeneity of (II.3) plus i.i.d. growth). Then $\mu_t^{ce} = \bar{\phi}\, C_t \big(\mathbb{E}[(C_{t+1}/C_t)^{1-\gamma}]\big)^{1/(1-\gamma)} = \bar{\phi} C_t \cdot$ const, and (II.4) collapses to

$$\text{(II.21)} \qquad m_{t+1} = \tilde{\beta}\left(\frac{C_{t+1}}{C_t}\right)^{-\gamma}, \qquad \tilde{\beta} \equiv \beta\left(\mathbb{E}\big[(C_{t+1}/C_t)^{1-\gamma}\big]\right)^{\frac{1/\psi-\gamma}{1-\gamma}} \ (\text{a constant}).$$

[derived from (II.4) + (II.1)] Under i.i.d. growth, Epstein–Zin is observationally equivalent to CRRA for *asset returns* — the EIS moves only the constant $\tilde{\beta}$ (Kocherlakota 1990, *JF*). Recursive preferences earn their keep only with predictable or long-run consumption risk (Bansal–Yaron 2004, *JF*) — stated as the relevant extension, not derived here.

Endogenous riskless rate and equity premium (lognormal algebra on (II.21) and $\mathbb{E}_t[m_{t+1}R_{t+1}] = 1$):

$$\text{(II.22)} \qquad \ln R_f = -\ln\tilde{\beta} + \gamma\mu_c - \tfrac{1}{2}\gamma^2\sigma_c^2, \qquad \ln \mathbb{E}_t[R_{e,t+1}] - \ln R_f = \gamma\,\sigma_c^2,$$

[EQ] where R_e is the return on the consumption claim (whose log return is const $+\,\Delta c_{t+1}$ under i.i.d. growth, so $-\mathrm{Cov}(\ln m, \ln R_e) = \gamma\sigma_c^2$). *Empirical status, stated plainly:* with $\sigma_c \approx 2\%$ per year, (II.22) delivers a premium of 0.0004γ; matching a ~6% historical premium needs $\gamma \approx 150$ — the **equity premium puzzle** (Mehra–Prescott 1985, *JME*), equivalently a violation of the Hansen–Jagannathan (1991, *JPE*) volatility bound at plausible γ. The model is internally consistent and empirically rejected at this point; Part III's intermediary and disaster channels are the modern repairs.

Cross-sectional restriction, every traded claim including Block M's asset:

$$\text{(II.23)} \qquad \mathbb{E}_t[R_{j,t+1}] - R_{f,t} = -R_{f,t}\,\mathrm{Cov}_t(m_{t+1}, R_{j,t+1}) \qquad \forall j.$$

[ID given (II.4)] Block M prices at $\mathbb{E}[f \mid \cdot]$, which obeys (II.23) only if $\mathrm{Cov}_t(m_{t+1}, f) \approx 0$ — the diversifiability assumption behind dealer risk neutrality. The wedge when it fails is quantified in the Self-Consistency section.

Part III — Level 3: Continuous time, stochastic volatility, jumps, constrained intermediaries, inelastic demand, limits to arbitrage

III.L0 — Environment

$t \in [0,\infty)$ on $(\Omega,\mathcal{F},\mathbb{F},\mathbb{P})$, $\mathbb{F}$ generated by independent Brownian motions B^D, B^v, B^u (with $d\langle B^D, B^v\rangle_t = \rho\, dt$) and a Poisson random measure $N(dt,dz)$ with compensator $\ell(dz)\,dt$, $\int(e^z-1)^2\ell(dz) < \infty$.

Aggregate dividend (also aggregate consumption in equilibrium):

$$\text{(III.1)}\qquad \frac{dD_t}{D_{t^-}} = g\,dt + \sqrt{v_t}\,dB_t^D + \int_{\mathbb{R}} (e^z - 1)\,N(dt,dz),$$

$$\text{(III.2)}\qquad dv_t = \kappa(\bar v - v_t)\,dt + \sigma_v\sqrt{v_t}\,dB_t^v, \qquad 2\kappa\bar v \ge \sigma_v^2 \ (\text{Feller: } v_t > 0).$$

[P] Jump-diffusion with square-root stochastic variance (Heston 1993, *RFS*; Merton 1976, *JFE*, for the jump component; the affine class is Duffie–Pan–Singleton 2000, *Econometrica*). *Status: constant-volatility models are known false* (volatility clustering: Engle 1982, *Econometrica*) — that is why v_t is a state variable; *known* $\ell(dz)$ is itself *contested-to-false* (see Boundary: peso problems).

Assets: the equity claim to D (net supply 1); an instantaneous riskless bond in zero net supply, rate r_t **endogenous**; a cross-section of claims j with exposures $(\sigma_{jD}, \sigma_{jv}, J_j(z))$ to the three risks; and a *redundant pair* of claims with identical cash flows but different margins (for the limits-to-arbitrage block). Trading is continuous **[P]** (*frictionless continuous trading: known false; the L5 deltas are the corrections*).

III.L1 — Information

Common information $\mathbb{F}$ for the macro block **[P]** (*contested*); the microstructure asymmetry of Part II carries over inside trading windows via the continuous-time Kyle model (Back 1992, *RFS*), used in III.L3(ii).

III.L2 — Participants

(a) Households H (mass 1): stochastic differential utility (the continuous-time Epstein–Zin), value V^H:

$$\text{(III.3)}\qquad V_t^H = \mathbb{E}_t\int_t^\infty \phi(C_s, V_s^H)\,ds, \qquad \phi(C,V) = \frac{\varrho}{1-1/\psi}\,\frac{C^{1-1/\psi} - \big((1-\gamma)V\big)^{\frac{1-1/\psi}{1-\gamma}}}{\big((1-\gamma)V\big)^{\frac{1-1/\psi}{1-\gamma}-1}},$$

[P] (Duffie–Epstein 1992, *Econometrica*, stochastic differential utility; reduces to time-additive CRRA with rate ϱ when $\psi = 1/\gamma$.) Constraint: self-financing wealth with consumption outflow; actions: (C_t, α_t). The HJB equation and FOCs:

$$\text{(III.4)}\qquad 0 = \max_{C,\alpha}\Big\{\phi(C, V^H) + \mathcal{L}^{W,v}V^H\Big\}, \qquad [\text{FOC}]\text{: } \phi_C = V_W^H, \quad \alpha:\ \text{(III.13) below.}$$

[FOC] ($\mathcal{L}^{W,v}$ = generator of wealth and state dynamics.) The utility-gradient representation of the kernel for SDU is $m_t = \exp\big(\int_0^t \phi_V(C_s,V_s)\,ds\big)\,\phi_C(C_t,V_t)$ (Duffie–Skiadas 1994, *J. Math. Econ.* 23, 107–131).

(b) Intermediaries I (specialists): CRRA-γ_I over their consumption stream; uniquely able to hold the full risky menu; households can take equity stakes in intermediaries only up to a multiple of inside capital:

$$\text{(III.5)}\qquad \text{outside equity}_t \ \le\ \vartheta\, W_t^I.$$

[P] Skin-in-the-game / agency constraint (He–Krishnamurthy 2013, *AER*). *Status: a reduced form for moral hazard; the constraint's tightness is contested, its existence is well-evidenced (intermediary balance sheets price assets: Adrian–Etula–Muir 2014, JF; He–Kelly–Manela 2017, JFE).* While the intermediary itself trades without binding portfolio constraints, its Euler equation prices the risky menu:

$$\text{(III.6)} \qquad \mathbb{E}_t[dR_j] - r_t\,dt = \gamma_I\,\mathrm{Cov}_t\!\Big(dR_j,\ \frac{dW_t^I}{W_t^I}\Big) \qquad \text{(for assets the intermediary is marginal in).}$$

[FOC] When (III.5) binds, household capital cannot flow in, the intermediary's leverage α_t^I rises as W_t^I falls, dW^I/W^I loads more on returns, and premia in (III.6) rise — the **amplification region**: $\partial(\text{premium})/\partial W^I < 0$. Global dynamics of such economies (occupation of the constrained region, endogenous volatility, the "volatility paradox" — lower exogenous risk inviting higher leverage and larger endogenous risk) are characterized in Brunnermeier–Sannikov (2014, *AER*).

(c) Arbitrageurs A: specialize in the redundant pair; mean–variance flow objective; margin constraint:

$$\text{(III.7)} \qquad \max_\theta\ \mathbb{E}_t[dW^A] - \frac{\gamma_A}{2}\mathrm{Var}_t[dW^A] \quad \text{s.t.} \quad \sum_j h_j\,|\theta_j|\,P_j \ \le\ W_t^A.$$

[P + objective] Margins $h_j \in (0,1]$ per dollar of position (Brunnermeier–Pedersen 2009, *RFS*). The Lagrangian FOC with multiplier $\xi_t \ge 0$, for a long position in j:

$$\text{(III.8)} \qquad \mathbb{E}_t[dR_j] - r_t\,dt = \gamma_A\,\mathrm{Cov}_t(dR_j, dW^A/W^A) + \xi_t\,h_j\,dt.$$

[FOC] The **margin CAPM**: expected returns carry a funding premium proportional to the position's margin use (Gârleanu–Pedersen 2011, *RFS*). Funding tightness $\xi_t > 0$ is an equilibrium object (scarcity of arbitrage capital).

(d) Passive / price-inelastic institutions (mass of institutions i, AUM A_i): portfolio weights given by a characteristics-based demand system,

$$\text{(III.9)} \qquad \frac{w_i(j)}{w_i(0)} = \exp\!\big(b_{p,i}\ \ln P_j + b'_{x,i}\,\varkappa_j + \epsilon_{i,j}\big), \qquad b_{p,i} < 1,$$

[EC] — an **econometric specification**, deliberately tagged as such: within this model it is *not* derived from an optimization problem (Koijen–Yogo 2019, *JPE*, derive it from a restricted mean–variance problem with characteristics-spanned beliefs and then estimate it via instruments; here it is grafted on as institutional demand). The condition $b_{p,i} < 1$ makes excess demand downward-sloping in price, which underwrites uniqueness of the demand-system fixed point (Koijen–Yogo 2019). The aggregate price elasticity ζ implied by (III.9) is *finite and empirically small*; the flow-to-price multiplier is

$$\text{(III.10)} \qquad \mathcal{M} = \frac{1}{\zeta}, \qquad \text{empirically } \mathcal{M} \approx 5 \text{ at the aggregate level,}$$

[EC] (Gabaix–Koijen, "In Search of the Origins of Financial Fluctuations: The Inelastic Markets Hypothesis," NBER WP 28967, 2021 — still a working paper; the \$\approx\\$5$-per-$\\$1\$-flow multiplier is their granular-instrumental-variables estimate, and the GIV methodology itself is published as Gabaix–Koijen 2024, *JPE*). The frictionless blocks of this model imply ζ orders of magnitude larger; this unresolved tension is reported, not smoothed, in the Self-Consistency section.

(e) Noise order flow: cumulative exogenous flow $dq_t^u = \sigma_q\,dB_t^u$ **[P]** (*known false as optimization; same defense and same budget caveat as (II.9)*).

III.L3 — Price formation: two tiers

(i) Walrasian tier (low frequency). For each asset, designated marginal pricers' demands plus institutional demand (III.9) plus noise flow clear against supply:

$$(\text{III.11})\qquad \theta^H_t+\theta^I_t+\theta^A_t+\sum_i\frac{A_i\,w_i(\cdot)}{P_\cdot}+q^u_t=\bar\theta\qquad\forall t.$$

[EQ] Goods market: $C^H_t+C^I_t+C^A_t=D_t$ **[EQ]**.

(ii) Microstructure tier (within the window). Transaction prices are set by dealers filtering order flow (continuous-time Kyle: Back 1992, *RFS*): the quote midpoint is $p^e_t=\mathbb{E}[f\mid\mathcal{F}^{MM}_t]$ with price impact λ_t, where f is the *Walrasian value of the asset* — the value (III.11) and the kernel (III.14) assign once current private information becomes public. Observed price decomposes as

$$(\text{III.12})\qquad p^{obs}_t=p^e_t+\varsigma_t,\qquad p^e_t\text{ a martingale on }\mathcal{F}^{MM},\qquad \varsigma_t\text{ stationary (inventory, tick, transient in}$$

[EC] the efficient-price/noise decomposition used to *measure* microstructure quality (Hasbrouck 1993, *RFS* — econometric identity, not an equilibrium condition). The bridge to the Walrasian tier is the Part II result (II.20) taken to the limit: as private information is revealed ($\Sigma_t\to0$ at the end of the trading window in Kyle/Back), $p^e_t\to f=P^W$ — dealer prices converge to the Walrasian price, and microstructure deviations are transient. **The two tiers are consistent by construction at the window boundary; within the window, the microstructure price can deviate from Walrasian value by exactly the unrevealed-information and inventory terms.**

III.L4 — Equilibrium concept

Definition (Radner equilibrium with constraints). Processes $\{P_t,r_t\}$ and plans $\{C^k_t,\theta^k_t\}_{k\in\{H,I,A\}}$ such that (i) each agent's plan solves its HJB problem (III.4)/(III.6)-program/(III.7) given prices and constraints (III.5), (III.7); (ii) institutional demand follows (III.9); (iii) all markets clear (III.11) for all t; (iv) all agents share the (correct) law of motion of the aggregate state (D_t,v_t,W^I_t,W^A_t).

Existence: for the frictionless complete-markets exchange-economy core, equilibrium existence in continuous time is classical (Duffie–Huang 1985, *Econometrica* — dynamic implementation of Arrow–Debreu; Karatzas–Lehoczky–Shreve 1990, *Math. OR* 15, 80–128). With the constraint (III.5) and heterogeneous agents, no general existence/uniqueness theorem applies; equilibria are constructed as solutions to the PDE system in the state $(v_t,\omega^I_t\equiv W^I_t/(W^I_t+W^H_t))$ and verified numerically (He–Krishnamurthy 2013; Brunnermeier–Sannikov 2014). **Multiplicity is real, not hypothetical**: margin-spiral economies admit multiple equilibria for the same fundamentals (Brunnermeier–Pedersen 2009), and the selection used here — the equilibrium branch continuous in fundamentals at $\xi=0$ — is a *convention*, flagged again in the Boundary section.

III.L5 — Frictions as explicit deltas

Each friction is written as the modification it makes to a frictionless condition derived earlier. Frictionless benchmark (discrete-step notation for clarity of the perturbation argument; $X_{t+1}\equiv P_{t+1}+D_{t+1}$): $P_t=\mathbb{E}_t[m_{t+1}X_{t+1}]$ — equation (II.23)'s level form.

Δ1. Proportional transaction costs χ. Buying costs $(1+\chi)P_t$, selling yields $(1-\chi)P_t$. The buy-now/sell-next-period perturbation must not raise utility, and symmetrically for shorting:

$$(\text{III.19})\qquad \mathbb{E}_t\big[m_{t+1}\left(D_{t+1}+(1-\chi)P_{t+1}\right)\big]\ \le\ (1+\chi)\,P_t,\qquad \mathbb{E}_t\big[m_{t+1}\left(D_{t+1}+(1+\chi)P_{t+1}\right)\big]\ \ge\ (1$$

[FOC, inequality form] The Euler *equation* becomes a **band**; inside it the agent does not trade (the no-trade region of Constantinides 1986, *JPE*; Davis–Norman 1990, *Math. OR*, characterize the continuous-time cone). The

FTAP extends to bid–ask spreads: no arbitrage is equivalent to the existence of an equivalent measure under which some process *lying between the bid and ask price processes* is a martingale (Jouini–Kallal 1995, *JET* 66, 178–197) — the band form of (III.19) is the agent-level shadow of that theorem.

Δ2. Margin / leverage limits. Already derived as (III.8): the delta to the frictionless Euler equation is the additive term $\xi_t h_j\, dt$. Two assets with **identical cash flows** but margins $h_1 < h_2$ must satisfy, by differencing (III.8),

$$\text{(III.20)} \qquad \mathbb{E}_t[dR_2 - dR_1] = \xi_t\,(h_2 - h_1)\,dt \;>\; 0 \;\text{ when } \xi_t > 0,$$

[derived] — a **violation of the law of one price sustained in equilibrium**: the basis is the present value of expected funding-tightness differentials (Gârleanu–Pedersen 2011; the post-2008 persistence of covered interest parity deviations is the canonical empirical instance: Du–Tepper–Verdelhan 2018, *JF*).

Δ3. Short-sale constraint $\theta_j^k \ge 0$. KKT turns the Euler equation into

$$\text{(III.21)} \qquad \mathbb{E}_t[m^k_{t+1} R_{j,t+1}] \le 1, \quad \text{with equality iff } \theta_j^k > 0.$$

[FOC, complementary slackness] With heterogeneous beliefs, pessimists are sidelined and the price reflects optimists only (Miller 1977, *JF*); dynamically, the option to resell to future optimists pushes the price *above even the most optimistic* static valuation — the speculative premium (Harrison–Kreps 1978, *QJE*).

Δ4. Funding liquidity and spirals. Make margins endogenous: $h_j = h(\hat\sigma_{j,t})$ increasing in measured volatility **[P]**. Then a price drop → higher measured volatility → higher h → tighter (III.7) → forced liquidation → further price drop: the demand curve can bend backward and **two stable equilibria** (liquid/illiquid) coexist for the same fundamentals (Brunnermeier–Pedersen 2009, *RFS*). The model's selection convention is stated in III.L4.

Δ5. Intermediary balance-sheet constraint. Already (III.5)–(III.6): the delta is that the *household* Euler equation fails to price risky assets with equality when (III.5) binds (households cannot reach the exposure), so

$$\text{(III.22)} \qquad \mathbb{E}_t[m^H\, dR_j] \le r_t\,\frac{dt}{1} \text{ (inequality for households);} \qquad \text{(III.6) holds with equality for } I.$$

[FOC, segmented] Pricing migrates from the household kernel to the intermediary kernel exactly in the states where intermediary capital is scarce — the identifying insight of intermediary asset pricing (He–Krishnamurthy 2013; empirical kernel: He–Kelly–Manela 2017 **[EC]**). Limits to arbitrage in the Shleifer–Vishny (1997, *JF*) sense sit on top of Δ2/Δ5: arbitrage capital is *performance-sensitive* — losses trigger outflows precisely when spreads are widest, so W^A shrinks when $\xi_t h_j$ compensation is highest, and convergence trades can diverge before they converge.

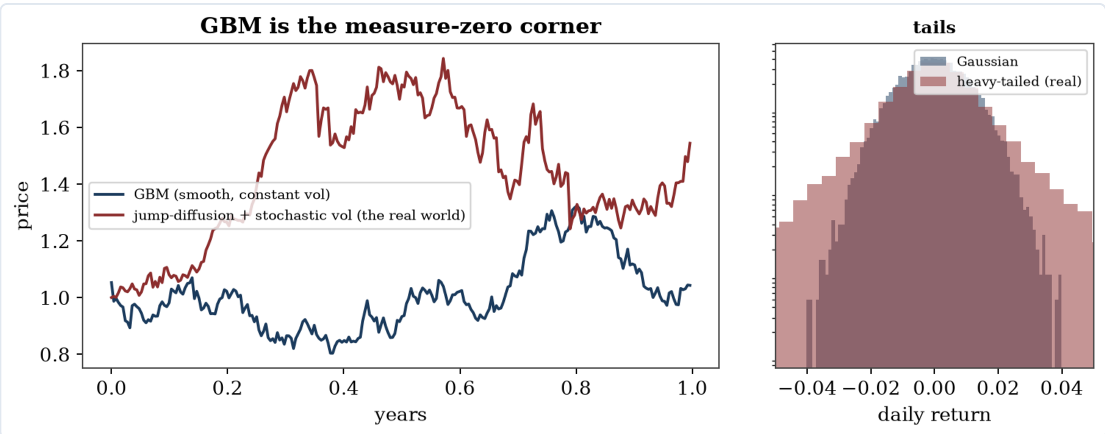

Figure 2.2. Geometric Brownian motion is the measure-zero corner: the real world is jumps, stochastic volatility, and heavy tails.

III.L6 — Aggregate dynamics and the pricing kernel

For the closed-form spine, take the CRRA special case of (III.3) ($\psi = 1/\gamma$, rate ϱ) in the **unconstrained region** (where households are marginal and aggregation as in Part II applies); the constrained-region kernel swaps in W^I per (III.6). Equilibrium goods clearing forces $C_t = D_t$, so the kernel is **endogenous**:

$$\text{(III.13)} \qquad m_t = e^{-\varrho t}\, C_t^{-\gamma} = e^{-\varrho t}\, D_t^{-\gamma}.$$

[FOC + EQ] Apply Itô's formula with jumps to (III.13) using (III.1):

$$\text{(III.14)} \qquad \frac{dm_t}{m_{t^-}} = \Big[-\varrho - \gamma g + \frac{\gamma(\gamma+1)}{2} v_t\Big]\, dt \;-\; \underbrace{\gamma\sqrt{v_t}}_{\eta_t^D}\, dB_t^D \;+\; \int_{\mathbb{R}} \big(\underbrace{e^{-\gamma z}}_{Y(z)} - 1\big)\, N(dt, dz).$$

[derived] The market prices of risk are read off: diffusive consumption risk $\eta_t^D = \gamma\sqrt{v_t}$; **variance risk** $\eta_t^v = 0$ **under CRRA** — a sharp, honest implication: time-additive power utility attaches no premium to volatility shocks per se; a non-trivial variance risk premium requires $\gamma \neq 1/\psi$ (recursive utility: Drechsler–Yaron 2011, *RFS*) or habits. Jump risk is priced by the kernel's jump kernel $Y(z) = e^{-\gamma z}$: the risk-neutral jump measure is $\ell^{\mathbb{Q}}(dz) = e^{-\gamma z}\ell(dz)$ — bad jumps ($z < 0$) are *over-weighted*, the rare-disaster premium channel (Rietz 1988, *JME*; Barro 2006, *QJE*).

Endogenous riskless rate — no drift in m beyond $-r_t$ (definition of the shadow rate, equivalently the bond's Euler equation):

$$\text{(III.15)} \qquad r_t = \varrho + \gamma g - \frac{\gamma(\gamma+1)}{2}\, v_t - \int_{\mathbb{R}} \big(e^{-\gamma z} - 1\big)\, \ell(dz).$$

[EQ] Precautionary terms: high spot variance and negative-jump risk *lower* the equilibrium rate.

Euler equation as a martingale restriction. For any claim with price P_t and cash-flow rate D_t:

$$\text{(III.16)} \qquad m_t P_t + \int_0^t m_s D_s\, ds \;\text{ is a } \mathbb{P}\text{-martingale.}$$

[FOC] (The continuous-time $\mathbb{E}[mR] = 1$.) For the equity claim, the ansatz $P_t = D_t\, \Phi(v_t)$ with $\Phi(v) = \int_0^\infty H(n; v)\, dn$, where $H(n; v) = \mathbb{E}_t[(m_{t+n} D_{t+n})/(m_t D_t)]$ is the n-maturity dividend strip, gives via Feynman–Kac the exponential-affine solution $H = e^{A_\Phi(n) + B_\Phi(n) v}$ with Riccati ODEs

$$\text{(III.17)} \qquad \begin{aligned} A_\Phi' &= -\varrho + (1-\gamma) g + \int \big(e^{(1-\gamma)z} - 1\big)\ell(dz) + \kappa\bar{v}\, B_\Phi, \\ B_\Phi' &= -\frac{\gamma(1-\gamma)}{2} + \big(\rho\sigma_v(1-\gamma) - \kappa\big) B_\Phi + \frac{\sigma_v^2}{2} B_\Phi^2, \end{aligned} \qquad A_\Phi(0) = B_\Phi(0) = 0,$$

[derived] (affine machinery: Duffie–Pan–Singleton 2000, *Econometrica*), with the transversality requirement that $A_\Phi(n) + B_\Phi(n) v \to -\infty$ fast enough for $\int_0^\infty H\, dn < \infty$ (a joint restriction on ϱ, g, γ, ℓ). *Verification special case:* $\gamma = 1$ (log utility) gives $B_\Phi' = 0 \Rightarrow B_\Phi \equiv 0$, $A_\Phi' = -\varrho$, so $\Phi = 1/\varrho$: the price–dividend ratio is constant — the known log-utility result, a check that (III.17) is not mis-derived.

Cross-sectional restriction. For any asset with $dR_j = \mu_j\, dt + \sigma_{jD}\, dB^D + \sigma_{jv}\, dB^v + \int J_j(z)\, N(dt, dz)$, the martingale property of $m \times$ (gain) yields

$$\text{(III.18)} \qquad \mu_j + \int J_j(z)\ell(dz) - r_t = \underbrace{\gamma\, v_t \frac{\sigma_{jD}}{\sqrt{v_t}}}_{=\,\eta_t^D\,\sigma_{jD}} + \underbrace{0 \cdot \sigma_{jv}}_{\eta_t^v\,\sigma_{jv}} + \int J_j(z)\, \big(1 - e^{-\gamma z}\big)\, \ell(dz).$$

[derived] Equity itself ($\sigma_{jD} = \sqrt{v_t}$, $J_j = e^z - 1$, plus the $\Phi'(v)/\Phi$ loading on B^v priced at zero under CRRA) earns $\gamma v_t + \int (e^z - 1)(1 - e^{-\gamma z})\ell(dz)$: a time-varying premium moving with v_t — **endogenous, state-dependent expected returns from constant preferences**, plus a level premium for jump risk.

Reduction to geometric Brownian motion. Set $\sigma_v = 0$ with $v_0 = \bar{v}$ (variance constant) and $\ell \equiv 0$ (no jumps): then B_Φ is irrelevant, Φ is constant, $P_t \propto D_t$, and $dP/P = (g)\,dt + \sqrt{\bar{v}}\,dB^D$ — exact GBM, the Black–Scholes environment, with r constant by (III.15). **Every assumption needed for GBM is individually known false** (volatility clusters; jumps happen); GBM is the measure-zero corner of this model, which is the precise sense in which Black–Scholes is a limiting case rather than a foundation.

Fundamental theorem of asset pricing. The strictly positive kernel (III.13)/(III.14) defines an equivalent (local) martingale measure $\mathbb{Q}$ via $d\mathbb{Q}/d\mathbb{P}|_{\mathcal{F}_t} = m_t e^{\int_0^t r_s ds}$; conversely, absence of arbitrage in the NFLVR sense is equivalent to the existence of such a measure (Harrison–Kreps 1979, *JET*; Harrison–Pliska 1981, *SPA*; general semimartingale version: Delbaen–Schachermayer 1994, *Mathematische Annalen*). With the L5 frictions, equality-FTAP relaxes to no-arbitrage *bands* (Δ1) and kernel *segmentation* (Δ5): there exist positive kernels, but not a unique one shared by all agents — each agent's shadow kernel prices what that agent can trade frictionlessly.

Part IV — Level 3+: Options, the risk-neutral density, and the physical law

This part is the payoff for everything built above. Options are the instrument through which the abstract objects of Parts I–III — Arrow prices, the SDF, the measures $\mathbb{P}$ and $\mathbb{Q}$ — become *observable*. The organizing question is the one the title of the section poses: **what do derivative prices reveal about the likelihood of future states, and about expected future returns?** The disciplined answer requires separating two densities that are constantly conflated in practice:

- $f^{\mathbb{P}}$ — the **physical** (statistical, real-world) density: the actual likelihood of future outcomes. This is what "expectations/likelihood" colloquially means.
- $f^{\mathbb{Q}}$ — the **risk-neutral** (pricing) density: a risk-adjusted, marginal-utility-weighted pseudo-density that prices claims.

The bridge between them is the pricing kernel of Parts I–III. Part IV derives exactly what options pin down ($f^{\mathbb{Q}}$, completely and model-free), what they bound ($\mathbb{P}$-moments), and what they cannot identify without auxiliary, empirically-false restrictions ($f^{\mathbb{P}}$ itself). It runs L0–L6; the closing sections are extended to cover it.

Notation additions (Part IV). $\mathbb{Q}$ = risk-neutral measure; $M_{t,T} \equiv m_T/m_t$ = SDF over $[t,T]$; $B(t,T) \equiv e^{-\int_t^T r_s\,ds}$ = bond price; $F_{t,T} \equiv S_t/B(t,T)$ = forward; $C(K), P(K)$ = call/put prices at strike K; $f_t^{\mathbb{P}}, f_t^{\mathbb{Q}}$ = physical/risk-neutral densities of S_T conditional on $\mathcal{F}_t$; $m_t(x) \equiv \mathbb{E}[M_{t,T} \mid S_T = x]$ = kernel **projected** onto the underlying; $\sigma^{BS}(K,T)$ = Black–Scholes implied volatility; $w(K,T) \equiv \sigma^{BS}(K,T)^2(T-t)$ = total implied variance; VRP = variance risk premium; SVIX^2 = Martin's option-implied return-variance index; $\varphi(x), \eta, \mathsf{M}^{\text{mart}}$ = eigenfunction, eigenvalue, and martingale (permanent) component of the SDF factorization. *Glyph note: φ here is the recovery eigenfunction, distinct from the informed-mass φ of Part II — they never appear together.*

IV.L0 — Environment

Inherit the Part III environment exactly: underlying S_t (the equity claim, $S_t \equiv P_t$ from III.L6) with $\mathbb{P}$-dynamics (III.1)–(III.2) — jump-diffusion with stochastic variance v_t. Add a menu of **European options**: claims paying $g(S_T)$ at a fixed maturity T, for arbitrary measurable g; the leading cases are the call $g(S) = (S-K)^+$ and put $g(S) = (K-S)^+$ at strikes K ranging over a continuum (idealization **[P]**, *status: known false* — strikes are discrete and bounded; this is what makes empirical $f^{\mathbb{Q}}$ estimation an interpolation/extrapolation problem rather than a read-off).

$$\text{(IV.0)} \qquad O_t(g,T) \;=\; \text{price at } t \text{ of the claim paying } g(S_T) \text{ at } T, \qquad \bar{\theta}_{\text{opt}} = 0 \text{ (options in zero net su}$$

[P] Zero net supply — every long is someone's short; this matters for the demand-pressure friction (IV.L5).

IV.L1 — Information: options are not redundant

The decisive structural fact. In the complete-markets corner (Part I, or the GBM corner of III.L6), options are **redundant**: spanned by dynamic trading in (S, bond), they add no payoffs and carry no independent information. Under stochastic volatility and jumps (III.1)–(III.2) the market is **incomplete** — the Brownian B^v and the jump measure $N(dt, dz)$ are extra risk sources that the stock and bond alone do not span — so options are *genuinely new assets* that complete (or partially complete) the market by spanning variance and jump risk.

$$\text{(IV.1)} \qquad \dim(\text{traded risks with } S, \text{bond}) < \dim(\text{state } (S, v, N)) \implies \text{options span the gap.}$$

[P / structural] *Status: holds* (the empirical non-redundancy of options is the entire basis of the variance- and jump-risk-premium literatures). This is **why option prices carry information about the kernel's pricing of volatility and jumps** that the underlying alone cannot reveal — the thread running through all of IV.L6.

IV.L2 — Participants: the option in the marginal investor's problem

No new agents. The option is priced by the same marginal investor whose Euler equation produced the kernel (III. 13)/(III.16). For any agent k marginal in the option, the first-order condition of the III.L2 program applied to the claim $g(S_T)$ gives

$$\text{(IV.2)} \qquad m_t\, O_t(g,T) \;=\; \mathbb{E}_t\big[m_T\, g(S_T)\big] \quad\Longleftrightarrow\quad O_t(g,T) \;=\; \mathbb{E}_t\big[M_{t,T}\, g(S_T)\big], \qquad M_{t,T} \equiv \frac{m_T}{m_t}.$$

[FOC] Derived from the same Euler/martingale condition (III.16); the kernel is the *endogenous* one of (III.13). When perfect hedging fails (incompleteness), the option market-maker is a constrained intermediary in the sense of III.L2(b)/(c) — the demand-based wedge this creates is IV.L5.

IV.L3 — Price formation: from the kernel to the risk-neutral density

(i) Risk-neutral valuation

Define the bond and the change of measure. With r_s the *endogenous* short rate (III.15),

$$\text{(IV.3)} \qquad B(t,T) \equiv \mathbb{E}_t[M_{t,T}] = e^{-\int_t^T r_s\, ds}, \qquad \left.\frac{d\mathbb{Q}}{d\mathbb{P}}\right|_{\mathcal{F}_T} \equiv \frac{M_{t,T}}{B(t,T)} \;>\; 0.$$

[ID + EQ] Positivity is the FTAP content (III.L6): $m > 0 \Rightarrow$ a legitimate Radon–Nikodym derivative with $\mathbb{E}_t^{\mathbb{P}}[d\mathbb{Q}/d\mathbb{P}] = 1$. Substituting into (IV.2),

$$\text{(IV.4)} \qquad O_t(g,T) \;=\; B(t,T)\, \mathbb{E}_t^{\mathbb{Q}}\big[g(S_T)\big].$$

[EQ] Risk-neutral valuation: the price is the discounted $\mathbb{Q}$-expectation of the payoff. Consistency with the underlying (checked in IV.L4) requires $e^{-\int r} S$ to be a $\mathbb{Q}$-martingale, i.e. $\mathbb{E}_t^{\mathbb{Q}}[S_T] = F_{t,T}$ — **the $\mathbb{Q}$-mean of the future price is the forward, not the expected future price.** Already here the central caution appears: even the *center* of the option-implied density is the risk-free-growth forward, displaced from $\mathbb{E}_t^{\mathbb{P}}[S_T]$ by the risk premium.

(ii) Breeden–Litzenberger: options reveal the entire $\mathbb{Q}$-density

Take $g(S) = (S-K)^+$ in (IV.4) and differentiate in the strike K (Leibniz; $\partial_K (S-K)^+ = -\mathbf{1}\{S > K\}$):

$$\text{(IV.5)} \qquad \frac{\partial C(K)}{\partial K} = -B(t,T)\,\mathbb{Q}_t(S_T > K), \qquad \boxed{\frac{\partial^2 C(K)}{\partial K^2} = B(t,T)\, f_t^{\mathbb{Q}}(K)}.$$

[ID] (Breeden–Litzenberger 1978, *J. Business* 51, 621–651.) **A complete strip of call prices across strikes at maturity T reveals the entire risk-neutral density of S_T — model-free, no Black–Scholes assumed.** The butterfly spread $\partial_K^2 C\,dK$ is precisely the price today of an Arrow–Debreu claim on $\{S_T \in [K, K+dK]\}$: this is the Part I state price $q(\omega)$ of (I.4) **made observable** by the options market. Options are the empirical window onto the state-price vector the whole document is built on.

(iii) Implied volatility as a re-encoding of $f^{\mathbb{Q}}$

Define implied volatility $\sigma^{BS}(K,T)$ as the number solving $C^{\text{mkt}}(K,T) = C^{BS}\big(S_t, K, T, r, \sigma^{BS}\big)$.

$$\text{(IV.6)} \qquad C^{\text{mkt}}(K,T) = C^{BS}\big(S_t, K, T, r, \sigma^{BS}(K,T)\big).$$

[ID / definition] It is a strictly monotone reparametrization of $C(K,T)$, hence a bijective re-encoding of $f^{\mathbb{Q}}$ (Dupire/Derman–Kani local-vol theory makes the map explicit). Reading the surface: - **Flat surface** $\iff$ $f^{\mathbb{Q}}$ lognormal $\iff$ the GBM/Black–Scholes corner of III.L6. Every non-flatness is a measured departure of $f^{\mathbb{Q}}$ from lognormality. - **Skew/smirk** (puts richer than calls, σ^{BS} decreasing in K) $\iff$ negative $\mathbb{Q}$-skewness $\iff$ crash-risk pricing (the dominant feature of index options post-1987). - **Smile curvature** $\iff$ excess $\mathbb{Q}$-kurtosis (fat tails). - **Term structure of ATM σ^{BS}** $\iff$ term structure of expected $\mathbb{Q}$-variance.

(iv) Black–Scholes as the kernel-free corner

In the GBM-complete corner the option is replicable by a dynamic (Δ_t, bond) portfolio; Itô plus no-arbitrage gives the PDE

$$\text{(IV.7)} \qquad \frac{\partial O}{\partial t} + rS\frac{\partial O}{\partial S} + \tfrac{1}{2}\sigma^2 S^2 \frac{\partial^2 O}{\partial S^2} - rO = 0,$$

[derived] in which the **physical drift μ does not appear** — the kernel drops out of the *option price* because the payoff is spanned. Reconciliation with the equilibrium valuation (IV.2): μ enters both the $\mathbb{P}$-dynamics and the kernel, and in a complete market these offsetting appearances cancel in the spanned price, so (IV.7) and (IV.2) agree. The moment markets are incomplete (stochastic v, jumps), replication fails, the kernel no longer cancels, and the *shape* of $f^{\mathbb{Q}}$ starts carrying genuine information about risk prices — which is the content of IV.L6.

(v) Model-free implied variance (the log contract / VIX)

The spanning identity (Carr–Madan 1998, in *Volatility*, ed. Jarrow; Britten-Jones–Neuberger 2000, *JF* 55, 839–866): any twice-differentiable payoff replicates statically as bonds + forward + a continuum of options,

$$\text{(IV.8)} \qquad H(S_T) = H(F) + H'(F)(S_T - F) + \int_0^F H''(K)(K - S_T)^+ dK + \int_F^\infty H''(K)(S_T - K)^+ dK$$

[ID] Apply to the **log contract** $H(S) = -2\ln(S/F)$, so $H''(K) = 2/K^2$. Under $\mathbb{Q}$ with *continuous* S, $d\ln S = (r - \frac{1}{2}v)\,dt + \sqrt{v}\,dB^{\mathbb{Q}}$, whence $-2\ln(S_T/F) = \int_t^T v_s\,ds - 2\int \sqrt{v}\,dB^{\mathbb{Q}}$ and the $\mathbb{Q}$-expectation of the stochastic integral vanishes:

$$\text{(IV.9)} \qquad \mathbb{E}_t^{\mathbb{Q}}\Big[\int_t^T v_s\,ds\Big] = \frac{2}{B(t,T)}\Big[\int_0^F \frac{P(K)}{K^2}dK + \int_F^\infty \frac{C(K)}{K^2}dK\Big].$$

[derived / EC] This is the (squared) VIX construction (Demeterfi–Derman–Kamal–Zou 1999; CBOE). With jumps the log contract additionally captures $\mathbb{E}^{\mathbb{Q}}[2(z-(e^z-1))]$ cumulant terms, so the gap between model-free implied variance and true $\mathbb{Q}$-variance is itself a *jump/skew* measure — the underpinning of the SKEW index and of (vii).

(vi) Risk-neutral higher moments

Bakshi–Kapadia–Madan (2003, *RFS* 16, 101–143) give model-free $\mathbb{Q}$-skewness and $\mathbb{Q}$-kurtosis as prices of specific OTM-option portfolios (the "volatility, cubic, quartic" contracts) via the same spanning logic (IV.8). The strongly negative index $\mathbb{Q}$-skew is the moment-space statement of the smirk and the crash premium. **[EC]**

(vii) Static no-arbitrage shape restrictions on the surface

These are the equilibrium-consistency constraints — the surface analogue of "markets clear":

$$\text{(IV.10)} \qquad \underbrace{\partial_K^2 C \ge 0}_{f^{\mathbb{Q}}\ge 0,\ \text{no butterfly arb}} \quad , \quad \underbrace{-B \le \partial_K C \le 0}_{\text{monotonicity}}, \quad \underbrace{\partial_T\, w(K,T) \ge 0}_{\text{no calendar arb}}.$$

[EQ] Convexity in strike is *equivalent* to a nonnegative density (IV.5); total implied variance w must be nondecreasing in maturity. Arbitrage-free parametrizations (Gatheral's SVI) are built to satisfy (IV.10). A surface violating them admits a static arbitrage — the model-free, kernel-independent floor of consistency.

IV.L4 — Equilibrium concept: one kernel, one $\mathbb{Q}$, a price *interval* under incompleteness

Definition (option-consistent equilibrium). Prices $\{O_t(g,T)\}$, the underlying S_t, and the bond are option-consistent if a single positive kernel m (equivalently one measure $\mathbb{Q}$) prices all of them via (IV.2)/(IV.4) simultaneously — i.e. the $f^{\mathbb{Q}}$ extracted from any maturity-T option sub-strip (IV.5) is the same density that reprices the underlying's forward and the bond.

Under **completeness** (the GBM corner) $\mathbb{Q}$ is unique and (IV.4) pins every option to a single arbitrage price. Under **incompleteness** (stochastic v, jumps) the set of equivalent martingale measures $\mathcal{Q}$ is a *continuum*; (IV.4) holds for each $\mathbb{Q} \in \mathcal{Q}$, and an unspanned option's no-arbitrage price lies in an interval

$$\text{(IV.11)} \qquad O_t \in \left[\inf_{\mathbb{Q}\in\mathcal{Q}} B\,\mathbb{E}^{\mathbb{Q}}[g(S_T)]\ ,\ \sup_{\mathbb{Q}\in\mathcal{Q}} B\,\mathbb{E}^{\mathbb{Q}}[g(S_T)] \right] = [\text{sub-replication price},\ \text{super-replication}$$

[EQ] the dual of the super-/sub-hedging problem (El Karoui–Quenez; the bounds are the cost of the cheapest dominating / dearest dominated static-plus-dynamic hedge). The market *selects* one $\mathbb{Q} \in \mathcal{Q}$ — through the marginal investor's kernel (IV.2) and, when intermediaries cannot hedge, through demand (IV.L5). Selection within $\mathcal{Q}$ is exactly the incompleteness analogue of the equilibrium-selection problem flagged for Parts II–III.

IV.L5 — Frictions: incompleteness and demand pressure as deltas

Δ6. Incompleteness wedge. Already (IV.11): the frictionless single-price (IV.4) is replaced by an interval; the delta is that the option price is no longer a pure function of (S, bond) but depends on the *selected* $\mathbb{Q}$, i.e. on preferences and demand, not arbitrage alone.

Δ7. Demand-based option pricing. When market-makers are risk-averse and *cannot* perfectly hedge their option inventory (the generic incomplete case), net end-user demand moves prices. The equilibrium deviation of an option's price from a reference-$\mathbb{Q}$ value is, to first order, proportional to the **variance of the unhedgeable part** of the option's payoff times the dealer's net position:

$$\text{(IV.12)} \qquad O_t^{\text{obs}} - O_t^{\mathbb{Q}\text{-ref}} \approx \gamma_{MM} \sum_{T',K'} \mathrm{Cov}_t\big(\text{unhedgeable}(g),\, \text{unhedgeable}(g_{K'T'})\big)\, d_{K'T'},$$

[FOC, intermediary] where $d_{K'T'}$ is net end-user demand and γ_{MM} the dealer's effective risk aversion (Gârleanu–Pedersen–Poteshman 2009, *RFS* 22, 4259–4299). This is the **options-market instance of the inelastic-demand / limits-to-arbitrage mechanism** of Part III (Δ2–Δ5): it explains the index skew's level (chronic end-user demand for protective puts) as a *demand* phenomenon on top of the kernel, and predicts that expensive options are the ones dealers are most short and least able to hedge. *Status: holds* — a documented departure from any single-$\mathbb{Q}$ no-arbitrage account.

Δ8. Transaction costs in options. The Δ1 bid–ask band (III.19) applies a fortiori to options, whose spreads are wide; deep-OTM-tail densities (IV.5) are the least identified precisely where spreads are widest and strikes sparsest — the empirical tail of $f^{\mathbb{Q}}$ is extrapolated, not measured. **[P, known false to ignore]**

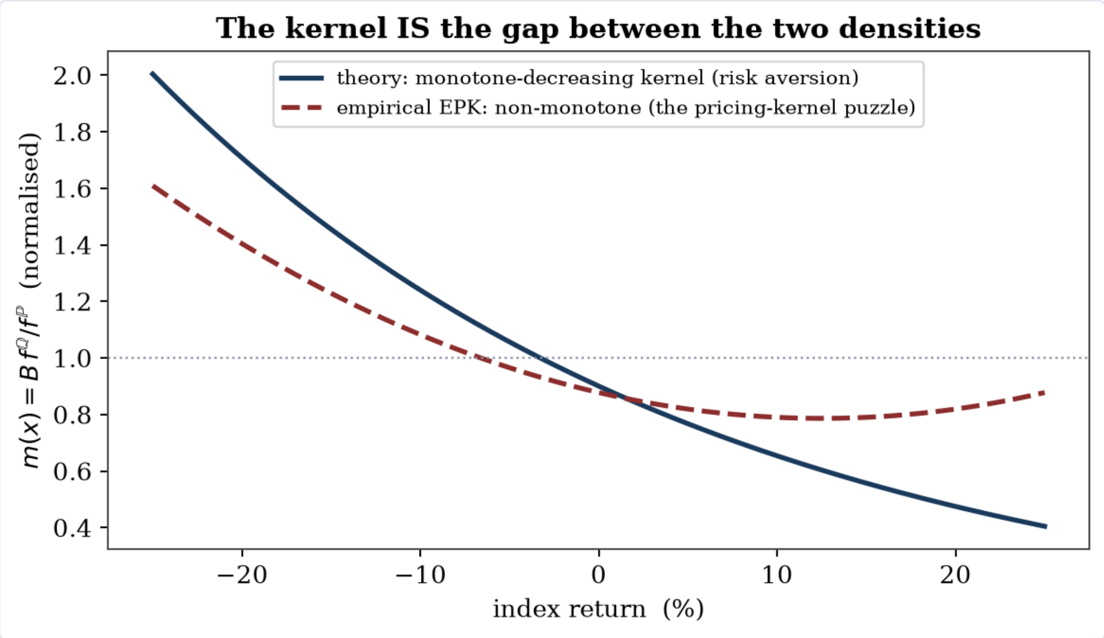

Figure 2.3. The empirical pricing kernel is the ratio of the two densities; its observed non-monotonicity is the pricing-kernel puzzle.

IV.L6 — The kernel, recovery, and the implication for $\mathbb{P}$

This is the heart of the user's question. Everything above delivers $f^{\mathbb{Q}}$. What can be said about $f^{\mathbb{P}}$ — the actual *likelihood* of future states — and about expected future returns?

(i) The empirical pricing kernel: the gap *is* the kernel

Project the SDF onto the terminal underlying, $m_t(x) \equiv \mathbb{E}[M_{t,T} \mid S_T = x]$. Then by iterated expectations on the Radon–Nikodym derivative (IV.3),

$$\text{(IV.13)} \qquad f_t^{\mathbb{Q}}(x) = \frac{m_t(x)}{B(t,T)}\, f_t^{\mathbb{P}}(x) \iff \boxed{\frac{m_t(x)}{B(t,T)} = \frac{f_t^{\mathbb{Q}}(x)}{f_t^{\mathbb{P}}(x)}}.$$

[ID] The ratio of the two densities **is** the (projected, normalized) pricing kernel — the *empirical pricing kernel* (EPK). With standard risk aversion (m decreasing in aggregate wealth, S_T increasing in wealth), $m_t(x)$ is **decreasing in** x, so $f^{\mathbb{Q}}$ is tilted toward low-x (bad) states relative to $f^{\mathbb{P}}$:

$$\text{(IV.14)} \qquad \mathbb{Q}_t(\text{crash}) > \mathbb{P}_t(\text{crash}), \qquad \mathbb{E}_t^{\mathbb{Q}}[S_T] = F_{t,T} < \mathbb{E}_t^{\mathbb{P}}[S_T].$$

[derived] This is the precise sense in which option-implied densities overstate physical crash likelihood: the excess is risk premium — bad states are overweighted in price because marginal utility is high there. Quoting the risk-neutral density as "the market's probability of a crash" double-counts: it is probability times marginal utility. This is the macro-density analogue of the Part II result that the dealer/risk-neutral price (II.20) drops the risk-discount term $-\bar{x}$ present in the Walrasian price (II.13).

(ii) The pricing-kernel puzzle (contested)

Empirically, the EPK estimated as $B\, f^{\mathbb{Q}}/f^{\mathbb{P}}$ — $f^{\mathbb{Q}}$ from options (IV.5), $f^{\mathbb{P}}$ from realized index returns — is often **non-monotone** (U-shaped/humped) over the index range (Jackwerth 2000, *RFS* 13, 433–451; Aït-Sahalia–Lo 2000, *J. Econometrics* 94, 9–51; Rosenberg–Engle 2002, *JFE*). A locally *increasing* kernel means marginal utility rising with wealth — a violation of basic risk aversion. *Status: contested, unresolved.* Candidate resolutions, each with a cost: (a) **state-dependent utility** — the true kernel is monotone in wealth *and* volatility, and projecting a variance-pricing kernel onto the index *alone* manufactures spurious non-monotonicity (consistent with the III.L6 point that variance must be a separately priced state for any variance premium to exist); (b) heterogeneous beliefs / sentiment and probability weighting; (c) the index is a poor proxy for the wealth/consumption argument of the true kernel; (d) the $f^{\mathbb{P}}$ and $f^{\mathbb{Q}}$ are estimated under different conditioning information. The puzzle is a live caution that (IV.13) read naively can misidentify the kernel.

(iii) Variance and tail risk premia — and forecasting future returns

The robust, identified objects are *differences* $\mathbb{P} - \mathbb{Q}$, available once a physical forecast is paired with the option-implied $\mathbb{Q}$-quantity:

$$\text{(IV.15)} \qquad \mathrm{VRP}_t \;\equiv\; \mathbb{E}_t^{\mathbb{P}}\Big[\int_t^T v\,ds\Big] - \mathbb{E}_t^{\mathbb{Q}}\Big[\int_t^T v\,ds\Big] \;<\; 0\ (\text{typically}),$$

[EC] with the $\mathbb{Q}$-term read from options (IV.9). The variance risk premium **predicts future equity returns**: high $-$VRP forecasts high returns, dominating P/E, default-spread and cay at the quarterly horizon (Bollerslev–Tauchen–Zhou 2009, *RFS* 22, 4463–4492). The tail/jump component carries a disproportionate share: deep-OTM puts identify a large jump-tail risk premium and a time-varying "Investor Fears" index (Bollerslev–Todorov 2011, *JF* 66, 2165–2211), the empirical incarnation of the disaster term $\int (e^z - 1)(1 - e^{-\gamma z})\ell(dz)$ in (III.18). These are concrete "implications for future asset-pricing expectations": option prices, differenced against physical forecasts, deliver *predictors* of the future equity premium.

(iv) A model-free *bound* on the expected return (the cleanest $\mathbb{Q}\to\mathbb{P}$ bridge)

Martin (2017, *QJE* 132, 367–433) shows that under the **negative correlation condition** (NCC: $\mathrm{Cov}_t^{\mathbb{P}}(M_{t,T}R_m,\, R_m) \le 0$, satisfied in essentially every standard model), the equity premium is bounded *below* by an entirely option-implied quantity:

$$\text{(IV.16)} \qquad \mathbb{E}_t^{\mathbb{P}}[R_{m,t\to T}] - R_{f,t\to T} \;\ge\; \mathrm{SVIX}_t^2, \qquad \mathrm{SVIX}_t^2 \;=\; R_{f,t}\cdot \mathrm{Var}_t^{\mathbb{Q}}(R_{m,t\to T}),$$

[derived + EC] where $\mathrm{Var}_t^{\mathbb{Q}}(R_m)$ is computed model-free from the option strip via the spanning logic (IV.8)–(IV.9). Empirically the bound averages $\approx 5\%$/yr, is approximately tight, and spiked above 20% in late 2008. The cross-sectional analogue prices individual stocks with *no free parameters* from index and single-name option variances (Martin–Wagner 2019, *JF* 74, 1887–1929). This is the strongest robust statement available: **option prices place a tight, tradeable lower bound on the physical expected return, requiring only a sign condition — not knowledge of the full kernel.**

(v) Full recovery of $\mathbb{P}$: the Ross theorem and why it fails

Can one go all the way — recover the *entire* physical density $f^{\mathbb{P}}$ from option prices alone? Ross (2015, *JF* 70, 615–648) says yes, under two strong restrictions: a **finite-state, time-homogeneous Markov** driving state, and a **transition-independent** kernel $m_{ij} = \delta\,\varphi_j/\varphi_i$ (marginal utility depends only on the state, with a constant discount δ). Let P be the observable matrix of one-step state prices (extracted from options across maturities and strikes via (IV.5)). Since the physical transition matrix F has unit row sums and $P_{ij} = \delta\,F_{ij}\,\varphi_j/\varphi_i$,

$$\text{(IV.17)} \qquad \sum_j P_{ij}\,\frac{1}{\varphi_j} = \delta\,\frac{1}{\varphi_i} \quad \Longleftrightarrow \quad P\,w = \delta\,w, \quad w_i \equiv 1/\varphi_i > 0.$$

[derived] By Perron–Frobenius, P (positive) has a *unique* positive eigenvector w with eigenvalue equal to its spectral radius δ; this recovers φ (marginal utility), δ (time preference), and hence $F =$ the physical transitions and $f^{\mathbb{P}}$ — from prices alone. A beautiful result.

It does not survive contact with the data — and the reason is exactly the kernel structure of Part III. By the Hansen–Scheinkman (2009, *Econometrica* 77, 177–234) factorization, every SDF decomposes as

$$\text{(IV.18)} \qquad M_{t,t+T} \;=\; \underbrace{e^{-\eta T}}_{\text{discount}}\;\underbrace{\frac{\varphi(X_t)}{\varphi(X_{t+T})}}_{\text{transitory}}\;\underbrace{\mathsf{M}^{\text{mart}}_{t,t+T}}_{\text{permanent (martingale)}}\;,$$

[ID] and Ross's transition-independence is precisely the assumption $\mathsf{M}^{\text{mart}} \equiv 1$ — **no permanent component.** Borovička–Hansen–Scheinkman (2016, *JF* 71, 2493–2544) prove that when M^{mart} is nondegenerate, the eigen-procedure (IV.17) recovers not $\mathbb{P}$ but the **long-term-risk-neutral measure** associated with the eigenfunction — a *different* measure that has absorbed all permanent risk adjustment. And the permanent component is not a minor term: Alvarez–Jermann (2005, *Econometrica* 73, 1977–2016) show, from the low average return on long-term bonds relative to the Hansen–Jagannathan bound, that M^{mart}'s volatility is *at least as large as the entire SDF's* — it dominates. Long-run-risk and most structural models (II.L6; Bansal–Yaron 2004) have exactly this feature.

$$\text{(IV.19)} \qquad \text{Recovery identifies } \mathbb{P} \iff \mathsf{M}^{\text{mart}} \equiv 1, \quad \text{but the data require } \operatorname{Vol}(\mathsf{M}^{\text{mart}}) \gtrsim \operatorname{Vol}(M).$$

[EQ / impossibility] Conclusion: option prices do not identify the physical likelihood of future states. The map $\mathbb{Q} \to \mathbb{P}$ requires the kernel, the kernel has a large permanent component, and that component is unidentified from a cross-section of prices — recovery silently sets it to one and returns the wrong measure. This is the option-pricing incarnation of the joint-hypothesis problem (closing section) and a hard limit, not a calibration nuisance.

(vi) What is left: disciplined, assumption-flagged transforms

Between "the full density (impossible)" and "a one-sided bound (robust)" sit parametric methods that *assume* a kernel and own the assumption. Fix a one-parameter family — power/exponential utility with relative risk aversion γ — and invert (IV.13):

$$\text{(IV.20)} \qquad f_t^{\mathbb{P}}(x) \;\propto\; x^{\gamma}\, f_t^{\mathbb{Q}}(x), \qquad \gamma \text{ chosen so the implied } f^{\mathbb{P}} \text{ is calibrated to realized returns.}$$

[EC] (Bliss–Panigirtzoglou 2004, *JF* 59, 407–446; Aït-Sahalia–Lo 2000.) Honest about what it buys: a physical density *conditional on* a postulated monotone kernel — useful, but it imports the very risk-aversion structure that recovery tried to avoid assuming, and it inherits the pricing-kernel puzzle (ii) when the assumed monotone form is wrong.

(vii) Synthesis — options and "future expectations/likelihood"

The disciplined ledger, which is the answer to the question:

1. **Fully identified, model-free:** the entire risk-neutral density $f^{\mathbb{Q}}$ (IV.5), forward-looking and conditional on today's information — its mean is the forward, its left tail is fattened by risk premia. It is a *price*, not a belief.
2. **Robustly bounded:** the physical expected return is bounded below by an option-implied index under a sign condition (IV.16), and individual expected returns are pinned by option variances (Martin–Wagner).
3. **Identified as a premium when paired with a physical forecast:** variance and jump-tail risk premia (IV.15), which *predict* future returns (Bollerslev–Tauchen–Zhou; Bollerslev–Todorov).
4. **Not identified from prices alone:** the physical density $f^{\mathbb{P}}$ itself — the actual likelihood of future states. Recovery (IV.17) appears to deliver it but is misspecified whenever the SDF has a permanent component (IV.18)–(IV.19), which the data demand.

So option prices say a great deal about the *prices of future risk* and impose tight, tradeable discipline on expected returns — but they do **not** hand over the physical probability distribution of future prices. The market's risk-neutral density is its *valuation* of future states, not its *forecast* of them, and the wedge between the two is the entire subject of asset pricing.

What this construction cannot capture, and why — concretely.

1. **Non-stationarity and regime change.** Every parameter ($g, \kappa, \bar{v}, \sigma_v, \ell, \iota, \sigma_u, \vartheta$, the demand coefficients $b_{p,i}$) is a constant of a single stationary law $\mathbb{P}$. Decimalization, electronic and high-frequency market making, post-2008 dealer regulation, and QE each *moved* the "constants" of Parts II–III (λ, spreads, ζ, ϑ). Regime-switching extensions (Hamilton 1989, *Econometrica*) remain stationary meta-models — a known switching law is just a bigger $\mathbb{P}$. Genuine structural change — policy regimes altering decision rules (Lucas 1976, *Carnegie-Rochester*) — is outside the model class, by construction.

2. **Reflexivity and adaptation-to-measurement.** The GS fixed point (II.14) already shows informativeness is self-limiting; the stronger empirical fact is that the *map itself decays under observation*: across 97 published return predictors, portfolio returns are 26% lower out-of-sample and 58% lower post-publication (McLean–Pontiff 2016, *JF* 71, 5–32). Formally, the model would need an equilibrium over the space of *models agents hold*, with the act of estimating an equation like (III.9) shifting the coefficients of (III.9). Nothing in L0–L6 has that structure (informal antecedents: Soros's reflexivity; Lo 2004, *J. Portfolio Management*, adaptive markets). Any econometric use of this document's [EC] equations is conditional on the strategy not yet having eaten its own signal.

3. **Knightian / model uncertainty.** All agents know $\mathbb{P}$ exactly — including the jump measure $\ell(dz)$ governing events that may never have occurred in sample. Under ambiguity, the Euler equation becomes a worst-case condition over a prior set (maxmin: Gilboa–Schmeidler 1989, *JME*; robust control: Hansen–Sargent 2008, *Robustness*), and ambiguity premia are observationally entangled with the risk premia of (III.18); this model cannot separate them, and *no* model can without auxiliary identifying assumptions.

4. **Heavy tails and rare disasters outside the assumed law.** The model has jumps, but with *known, fixed* ℓ. Empirical return tails are approximately power-law with exponent near 3 (Gabaix–Gopikrishnan–Plerou–Stanley 2003, *Nature*); finite samples cannot pin the tail of ℓ (the peso problem), so the disaster-premium terms in (III.15)/(III.18) are identified mostly by prior, not data (Rietz 1988; Barro 2006 uses cross-country century-scale data precisely because within-market samples cannot). A disaster whose *possibility* is not in ℓ's support is not "risk" in this model at all.

5. **The joint-hypothesis problem.** Every empirical test of (I.14)/(II.23)/(III.18) is a joint test of the kernel specification *and* market efficiency/informational assumptions (Fama 1970, *JF*); Hansen–Jagannathan (1991) bounds discipline candidate kernels but do not escape the joint hypothesis. Microstructure tests inherit the same problem through their auxiliary structure (e.g., (III.12) identifies "noise" only relative to the assumed efficient-price dynamics). No test of this model is independent of this model.
6. **Equilibrium multiplicity and selection.** The document *selects* throughout: linear equilibria in (II.13)/(II.17) (nonlinear ones not ruled out); the fundamentals-continuous branch in Δ4's spiral region; uniqueness of the demand-system fixed point only under $b_{p,i} < 1$. Sunspot equilibria exist in related economies even with complete markets broken only slightly (Cass–Shell 1983, *JPE*), and run-type multiplicity is endemic to liquidity provision (Diamond–Dybvig 1983, *JPE*). The model has no positive theory of selection; comparative statics in the multiple-equilibrium regions are conditional on the conventions stated.
7. **Architecture limits (self-inflicted, disclosed).** Block M prices a single security in partial equilibrium under an in-window normalization; production, endogenous cash flows, taxes, and agency problems inside the institutions of (III.9) are absent; noise traders' budgets are open (see check 4 below).
8. **Non-identification of the physical measure from prices (Part IV).** The sharpest boundary the options layer exposes: $f^{\mathbb{Q}}$ is fully observable (IV.5) but $f^{\mathbb{P}}$ is not recoverable from a cross-section of prices, because the map between them is the kernel and the kernel's permanent (martingale) component (IV.18) is unidentified — Ross recovery (IV.17) returns it only by assuming that component away, counterfactually (IV.19). This is not a measurement gap that better data closes; it is the joint-hypothesis problem (item 5) in density form. Any claim to read "the market's probability of a crash" off option prices is reading a risk-adjusted price as a belief. What *is* identified — the full $\mathbb{Q}$-density, lower bounds on expected returns (IV.16), and risk premia paired with physical forecasts (IV.15) — is bounded precisely by where the kernel stops being inferable.

Self-consistency check

Run against the construction; tensions are reported, not smoothed.

1. **Markets clear / Walras's law (Part I).** Goods clearing was *imposed* in deriving (I.9); asset clearing must follow. Check: summing (I.11) over k, using $\sum_k \Gamma/\gamma_k = 1$ and $\sum_k \mathcal{W}^k = C_0 + \sum_j \bar{\theta}_j p_j$, gives $\sum_k c_0^k = C_0 \iff \sum_j \bar{\theta}_j p_j = \sum_\omega q(\omega) C_1(\omega)$ — which holds term-by-term by (I.3)–(I.4). ✔ Portfolios (I.12) sum to $\bar{\theta}$ since $\sum_k \Gamma/\gamma_k = 1$. ✔
2. **Budget constraints bind.** Strict monotonicity of all utilities ⇒ multipliers strictly positive ⇒ (I.5) and Block-C budgets hold with equality. ✔ Inequality constraints (III.5), (III.7), Δ3 satisfy complementary slackness by construction of the FOCs. ✔
3. **Kyle zero-sum audit.** With (II.17): $f - p = \frac{1}{2}(f - \mu_f) - \lambda u$, so $\mathbb{E}[\pi_{\text{insider}}] = \mathbb{E}[(f-p)X] = \sigma_u\sqrt{\Sigma_0}/2$; $\mathbb{E}[\pi_{\text{noise}}] = \mathbb{E}[(f-p)u] = -\lambda\sigma_u^2 = -\sigma_u\sqrt{\Sigma_0}/2$; dealers: zero by (II.7). Sum: 0. ✔ The trading game conserves wealth.
4. **Noise-trader budgets — open tension (T1).** In GS, noise traders' expected loss is $B_p \varpi \sigma_u^2 > 0$ per round (they buy into their own price impact); in GM they pay the spread every round. Their wealth process and participation rationale are outside the model: repeated play requires an unmodeled resource inflow. This is a genuine non-conservation at the boundary of the model, standard in the literature and still a real loose end (Black 1986).

5. **SDF positivity.** (I.13): exponential ⇒ >0. ✔ (II.4)/(II.21): powers of positive consumption and value ⇒ >0 given $C_t>0$, which (II.1) guarantees. ✔ (III.13): >0 given $D_t>0$, guaranteed by (III.1) (geometric dynamics) and Feller for v. ✔ **Tension (T2):** the CARA-normal Block M allows $f<0$ and $p<0$ with positive probability — limited liability is violated inside the microstructure block even while the macro blocks enforce positivity. Known false assumption, retained for tractability, quarantined to Block M.

6. **Units and dimensions.** $q(\omega)$: date-0 goods per unit of state-ω goods; $m=q/\pi$: same units (probabilities dimensionless); $\mathbb{E}[mR]$: dimensionless = 1. ✔ Kyle: $[\lambda] = \$/\text{share}^2$ and $\lambda = \tfrac12\sqrt{\Sigma_0}/\sigma_u = (\$/\text{share})/\text{share}$ ✔; $[b_I] = \text{share}^2/\$$ and $b_I=\sigma_u/\sqrt{\Sigma_0}$ ✔; precisions τ carry inverse squared price units, so $\varpi=\gamma_I/(\varphi\tau_\varepsilon)$ has units (price²·tolerance/shares)… consistent with ϖu being price-of-signal units. ✔ Continuous time: $r,\varrho,\kappa,\ell(\mathbb{R}),v$ all per unit time; $\sqrt{v}\,dB$ is dimensionless per √time·√time. ✔

7. **FOCs consistent with equilibrium prices.** Part I: prices (I.9) were *constructed* from the FOCs plus clearing; substituting back, (I.6) holds for both agents simultaneously because (I.10) makes their MRSs equal state by state. ✔ Part II-GS: the conjectured information content of the price was *derived* from clearing at (II.11) and re-verified at (II.13) ($B_p>0$, invertible). ✔ Kyle: SOC $\lambda>0$ holds at (II.17). ✔ GM: quotes are fixed points of (II.8) by construction of (II.18). ✔

8. **Cross-block kernel consistency — quantified wedge (T3).** Block C's Euler (II.23) applied to Block M's asset requires $p=\mathbb{E}_t[m_{t+1}f]\,/\,\mathbb{E}_t[m_{t+1}]\cdot(1/R_f)\cdot R_f=\mathbb{E}_t[f]/R_f+\mathrm{Cov}_t(m_{t+1},f)$, while Block M's dealers set $p=\mathbb{E}[f\mid\text{flow}]$ with $R_f\equiv 1$ in-window. The blocks agree iff (a) the window is short ($R_f\approx 1$ over hours: fine) and (b) $\mathrm{Cov}_t(m_{t+1},f)\approx 0$ (idiosyncratic asset). For an asset with systematic exposure, the microstructure block misprices by exactly $-R_f\,\mathrm{Cov}_t(m_{t+1},f)$ — visible in (II.13) as the $-\bar{x}$ risk-discount term that survives in the Walrasian price but vanishes from the risk-neutral dealer price (II.20). Integration of the two is *partial by design*; the wedge is the price of tractability and is now on the record.

9. **Who is marginal where — segmentation bookkeeping (T4).** With binding constraints, agents' shadow kernels differ. The model's assignment: households price the bond always and risky claims in the unconstrained region; intermediaries price the risky menu when (III.5) binds (III.6); arbitrageurs price the redundant pair (III.8). Consistency requires that no asset is priced *with equality* by two agents whose kernels disagree on its payoff span — enforced here by the participation structure (households cannot hold the risky menu directly in the constrained region; only arbitrageurs trade the basis pair). This is an assumption about market access doing real work, and it is what makes (III.20)'s law-of-one-price violation an equilibrium rather than a contradiction. ✔ as bookkeeping; flagged as economics.

10. **Elasticity clash (T5) — a tension between lenses, not a logical contradiction.** The optimizing CARA blocks imply per-capita demand slopes $\partial x/\partial p=-(\tau_f+\tau_\varepsilon)/\gamma_I$ that aggregate to enormous price elasticities, while the estimated demand system (III.9)–(III.10) imposes a small finite ζ (multiplier $\mathcal{M}\approx 5$). These are **not contradictory propositions within one model** — they are two different *lenses* (a frictionless optimizing micro-foundation versus an econometric demand system estimated on real holdings) that imply different elasticities for the same market. That they disagree is not a defect of internal logic — the consistency checks above (clearing, binding budgets, positive SDF, units) all pass; it is the model faithfully *surfacing* one of the field's central open problems, the **inelastic-markets hypothesis** (Gabaix–Koijen): measured aggregate demand is one-to-two orders of magnitude more inelastic than frictionless theory predicts, and *why* is unresolved — candidate channels include mandate rigidity, benchmarking, infrequent rebalancing, leverage/risk constraints, and the very limits to arbitrage of §III.L5. The model's reconciliation — most wealth in inelastic mandates, a thin elastic arbitrage fringe — *is* the inelastic-markets hypothesis, stated as a hypothesis, not asserted as a result. "Reported, not smoothed" is here the **correct** scientific posture, not a cop-out: to *resolve* the clash would mean taking a side in a live, unsettled dispute, which

would be overclaiming. This is the model's largest live tension, and it is an open empirical frontier — not a bookkeeping error, and (notably) it is the *constructive* mechanism of edge in the companion `EDGE_METHODOLOGY.md` §5: the inelastic counterparty is who the active trader trades against.

11. **Endogeneity of the discount rate.** R_f in (I.14), $R_{f,t}$ in (II.22), r_t in (III.15): each is derived from preferences + clearing, never assumed. The only exogenous rate anywhere is the in-window normalization $R_f = 1$ of Block M, disclosed at the top of Part II and bounded in check 8. ✔
12. **Special-case verifications.** $\gamma = 1/\psi$ collapses (II.4) to CRRA ✔; i.i.d. growth collapses EZ pricing to (II.21) ✔ (Kocherlakota 1990); $\gamma = 1$ gives constant price–dividend ratio in (III.17) ✔; $\sigma_v = 0, \ell = 0$ gives exact GBM ✔. The model degrades gracefully to its classical corners.
13. **Options layer uses one kernel (Part IV).** The option Euler equation (IV.2) is the *same* (III.16) restricted to a terminal-payoff claim — no new kernel is introduced, so the option, the underlying, and the bond are priced by one m. Density positivity: $f^{\mathbb{Q}} = B^{-1}\partial_K^2 C \geq 0$ holds iff the surface is butterfly-arbitrage-free (IV.10), the exact convexity condition. ✔ Mean consistency: $\mathbb{E}_t^{\mathbb{Q}}[S_T] = F_{t,T}$ requires $e^{-\int r}S$ to be a $\mathbb{Q}$-martingale, which is the underlying's *own* Euler equation under (IV.3) — verified, not assumed. ✔ **Tension (T6):** the EPK identity (IV.13) is internally exact, but its *empirical* version is estimated by combining a forward-looking $f^{\mathbb{Q}}$ (today's options) with a backward-estimated $f^{\mathbb{P}}$ (historical returns) under different conditioning sets; the pricing-kernel puzzle (IV.L6.ii) may be partly an artifact of that mismatch rather than a preference anomaly. Reported, not resolved. **Limited-liability note:** unlike the CARA-normal Block M (T2), the Part IV underlying inherits $S_t > 0$ from (III.1), so $f^{\mathbb{Q}}$ is supported on $(0, \infty)$ and the BL density is a genuine probability density. ✔

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Confidence key: ★ = standard result, attribution confident (entries marked "Verified" were checked against the published record on 2026-06-10); ◖ = attribution confident, details (year/journal/exact scope) not independently re-verified; ○ = reconstructed from memory — none remain after the verification pass.

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CHAPTER 3

The Mathematics of Analysing Financial Time Series

From stationarity and the Wold decomposition to the purged-CV and deflated-Sharpe evaluation gate.

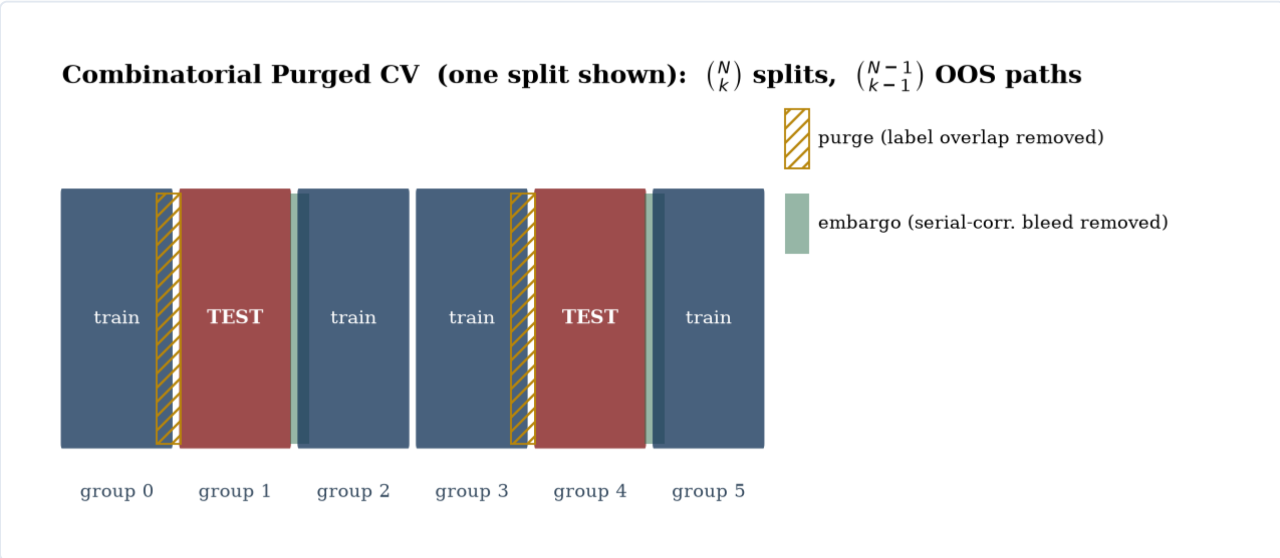

Figure 3.1. Combinatorial purged cross-validation: contiguous groups, every k-subset as the test set, with purging and embargo to stop label leakage.

Status of this document. A rigorous, self-contained reference for the mathematics used to analyse time series — built around *this* repository's actual pipeline (volume-profile feature construction, labelling, and the purged-CV + permutation validation harness in `vpts/`), and extended to the full canon so the harness sits in its proper statistical context. It is the methodological companion to `MARKET_EQUILIBRIUM_MODEL.md`: that document asks *what a price means*; this one asks *what can be learned from a sequence of them*. The two share one guardrail, stated at the end of §0 and cashed out in §11.

Every numbered equation is tagged with its logical role:

- **[DEF]** definition / identity (true by construction)
- **[MODEL]** a generative model or assumption (with empirical status: *holds / contested / known false*)
- **[EST]** an estimator or fitting procedure (a function of the sample)
- **[STAT]** a test statistic / inferential object (with its null)
- **[VALID]** a validation, resampling, or multiple-testing-control procedure
- **[REPO]** implemented in this codebase — with a pointer to the file, so the math maps 1:1 to the code

The cardinal discipline, inherited from the companion document: **never present an estimator as if it were the truth it estimates, and never present an in-sample fit as out-of-sample evidence.** A backtest is an estimate with a standard error and a multiple-testing burden; the whole back half of this document is about taking that seriously.

§0 — Conventions, notation, and the standing caveat

Notation

Symbol	Meaning
$(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$	Probability space with a filtration (information available up to t)

Symbol	Meaning
$\{X_t\}_{t\in\mathbb{Z}}$	A (real-valued) discrete-time stochastic process / time series
$P_t,\ H_t,\ L_t,\ C_t,\ V_t$	Bar OHLCV: close (also P_t), high, low, close, volume at bar t
r_t	Return at t: simple $R_t = P_t/P_{t-1} - 1$ or log $r_t = \ln(P_t/P_{t-1})$ — the doc states which
$\mu,\ \sigma^2,\ \gamma(k),\ \rho(k)$	Mean, variance, autocovariance at lag k, autocorrelation at lag k
L	Lag (backshift) operator: $LX_t = X_{t-1}$, $L^k X_t = X_{t-k}$
$\varepsilon_t,\ \{\varepsilon_t\} \sim WN(0,\sigma^2)$	Innovation / white-noise shock
$\phi(\cdot),\ \theta(\cdot)$	AR and MA lag polynomials
$f(\lambda)$	Spectral density at frequency λ
$\sigma_t^2,\ v_t$	Conditional variance (GARCH) / spot variance
$\hat{\theta},\ \theta_0$	An estimator and the true parameter it targets
$1\{\cdot\}$	Indicator function
$\Phi,\ \phi_{\mathcal{N}}$	Standard-normal CDF, PDF
IC	Information coefficient (cross-sectional rank correlation of signal vs forward return)
$\Pi(\cdot)$	Volume-at-price distribution (the volume profile)
POC, VAH, VAL	Point of control (mode of Π), value-area high/low
ATR_n	Average true range over n bars
CLV_t	Close location value $\in [-1, 1]$
$N,\ k,\ \varphi[N,k]$	CPCV: number of groups, test-groups per split, number of backtest paths
$\mathrm{SR},\ \mathrm{DSR},\ \mathrm{PBO}$	Sharpe ratio, deflated Sharpe ratio, probability of backtest overfitting

Two glyph notes: C_t is the close, distinct from $C(\cdot)$ a count or combination; ρ is autocorrelation here, distinct from the cross-asset correlation of the companion document.

The standing caveat (read once, applies everywhere)

Almost every estimator below assumes some form of **stationarity** — that the law generating the data does not move. Financial series violate this: means, variances, and dependence structures drift and break (regime change, microstructure evolution, policy shifts). So every number this document teaches you to compute is *conditional on a stationarity assumption that is known false in general*. The discipline is not to pretend otherwise but to (i) state the assumption, (ii) test it where possible (§5), and (iii) validate out-of-sample under resampling that respects dependence (§10).

The shared guardrail. The companion document's lesson — *a price is a risk-weighted valuation, not a probability* — has a time-series twin: **a statistically predictive signal is not the same as a tradeable edge, and an in-sample edge is not the same as an out-of-sample one.** The gap is made of three things this document quantifies: dependence (which inflates naïve significance), multiplicity (trying many things), and non-stationarity (the future is drawn from a different law). This repository's own research log (`RESEARCH.md`) is a case study: an apparent +14.5% backtest that collapses to −0.68% per path under purged CV, and a real out-of-sample correlation that turns out to be survivorship, not signal.

§1 — Foundations

Time series as a stochastic process. A time series is one realisation of a process $\{X_t\}$ on $(\Omega, \mathcal{F}, \mathbb{P})$; we see a single path and must infer the law. This is the original sin of the field: cross-sectional statistics average over independent draws, but here we have *one* draw observed through time, and inference is only possible if time-averaging can substitute for ensemble-averaging — i.e. under **ergodicity**.

$$(1.1) \qquad \frac{1}{T}\sum_{t=1}^{T} g(X_t) \xrightarrow{\text{a.s.}} \mathbb{E}[g(X_t)] \quad \text{as } T \to \infty.$$

[MODEL] Ergodic theorem (Birkhoff). *Status: assumed, untestable from one path.* Without it, sample means estimate nothing.

Stationarity. Strict: the joint law of $(X_{t_1}, \ldots, X_{t_n})$ is invariant to time shifts. The workable weakening is **covariance (weak) stationarity**:

$$(1.2) \qquad \mathbb{E}[X_t] = \mu \; \forall t, \qquad \mathrm{Cov}(X_t, X_{t+k}) = \gamma(k) \text{ depends on } k \text{ only.}$$

[MODEL] *Status: contested for returns (roughly holds short-horizon), known false for prices (unit-root non-stationary — §2).*

Autocovariance / autocorrelation function (ACF).

$$(1.3) \qquad \gamma(k) = \mathrm{Cov}(X_t, X_{t+k}), \qquad \rho(k) = \frac{\gamma(k)}{\gamma(0)}, \qquad \hat{\gamma}(k) = \frac{1}{T}\sum_{t=1}^{T-k}(X_t - \bar{X})(X_{t+k} - \bar{X}).$$

[DEF]/[EST] The sample ACF $\hat{\rho}(k)$ is the workhorse diagnostic. Under white noise, $\hat{\rho}(k) \approx \mathcal{N}(0, 1/T)$, giving the $\pm 1.96/\sqrt{T}$ bands. The **partial** ACF (PACF) $\alpha(k)$ is the correlation of X_t, X_{t+k} with intermediate lags projected out — it identifies AR order.

White noise and the Wold decomposition. $\{\varepsilon_t\} \sim WN(0, \sigma^2)$ iff mean 0, variance σ^2, $\gamma(k) = 0$ for $k \neq 0$. The structural foundation of linear time-series analysis:

$$(1.4) \qquad X_t = \sum_{j=0}^{\infty} \psi_j \, \varepsilon_{t-j} + \eta_t, \qquad \psi_0 = 1, \; \sum \psi_j^2 < \infty,$$

[DEF] Wold (1938): *every* covariance-stationary process is an infinite MA of its own innovations plus a deterministic part η_t. This is why ARMA models (§2) are not arbitrary — they are parsimonious rational approximations to the Wold representation.

Returns algebra (the first modelling choice).

$$(1.5) \qquad r_t = \ln\frac{P_t}{P_{t-1}} = \ln(1 + R_t), \qquad \sum_t r_t = \ln\frac{P_T}{P_0} \text{ (log returns add)}, \qquad \prod_t (1 + R_t) = \frac{P_T}{P_0} \text{ (sin}$$

[DEF] Log returns are time-additive (convenient for horizon aggregation and Gaussian modelling); simple returns are asset-additive across a portfolio. Confusing them is a units error. *Note: the RESEARCH.md backtest illusion is partly (1.5) — compounding a positive drift over a bull market manufactures a large terminal number from no per-bar edge.*

The stylized facts (what any model of returns must respect). Across markets and frequencies, return series exhibit a robust, model-independent set of empirical regularities (Cont 2001, *Quantitative Finance* 1, 223–236):

$$(1.6) \qquad \rho_r(k) \approx 0 \; (k \geq 1) \quad \text{but} \quad \rho_{|r|}(k), \; \rho_{r^2}(k) > 0 \text{ slowly decaying;} \qquad \mathbb{P}(|r| > x) \sim x^{-\alpha}, \; \alpha \in (2$$

[MODEL] *Status: holds.* Returns are nearly **uncorrelated** (so the mean is unforecastable — §2) yet **not independent**: absolute/squared returns are strongly autocorrelated (**volatility clustering** — §3). Distributions are **heavy-tailed** (power-law tail index $\alpha \approx 3$, finite variance but infinite/large higher moments), **negatively skewed** with a **leverage effect** (down-moves raise future volatility more), and display **aggregational Gaussianity** (the tail thins as the horizon lengthens). Any model that produces i.i.d. Gaussian returns is, by (1.6), known false — and every later section is in some sense a response to one of these facts.

The random-walk / martingale hypothesis. The weakest efficient-markets statement is that prices are a martingale, $\mathbb{E}[P_{t+1} \mid \mathcal{F}_t] = P_t$ — i.e. returns are unforecastable in mean. Stronger forms (i.i.d. increments, RW1; independent, RW2; uncorrelated, RW3) are nested and separately testable. The canonical test is the **variance ratio** (Lo–MacKinlay 1988, *RFS* 1, 41–66): under a random walk, variance scales linearly with horizon, so

$$(1.7) \qquad \mathrm{VR}(q) = \frac{\mathrm{Var}(r_t^{(q)})}{q\ \mathrm{Var}(r_t)} = 1 + 2\sum_{k=1}^{q-1}\left(1 - \tfrac{k}{q}\right)\rho(k) \xrightarrow{H_0} 1,$$

[STAT] where $r_t^{(q)}$ is the q-period return. $\mathrm{VR}(q) > 1$ indicates positive autocorrelation/trending; < 1 mean reversion. Lo–MacKinlay reject RW for weekly US index returns — note the tension with §10's finding that this rejection is rarely *tradeable* after costs (the joint-hypothesis problem, §11). The companion document's $\mathbb{Q}$-martingale (IV.4) is the *risk-neutral* analogue; here the question is whether the **physical** price is a martingale, which it very nearly — but not exactly — is.

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§2 — Linear models: ARMA, spectra, unit roots, cointegration, VAR

ARMA(p,q). With lag polynomials $\phi(L) = 1 - \phi_1 L - \cdots - \phi_p L^p$ and $\theta(L) = 1 + \theta_1 L + \cdots + \theta_q L^q$:

$$(2.1) \qquad \phi(L)\, X_t = \theta(L)\, \varepsilon_t.$$

[MODEL] Stationary iff the roots of $\phi(z) = 0$ lie outside the unit circle; invertible iff $\theta(z)$'s do. *Status: a parsimonious linear approximation; returns are weakly autocorrelated at best, so ARMA captures little of the mean but frames everything else.*

Estimation and identification. Yule–Walker equations (method of moments on the ACF) for pure AR; conditional/exact Gaussian MLE for ARMA. Order chosen by the **Box–Jenkins** loop (identify via ACF/PACF → estimate → diagnose residual whiteness via Ljung–Box) and information criteria:

$$(2.2) \qquad \mathrm{AIC} = -2\ln\hat{L} + 2k, \qquad \mathrm{BIC} = -2\ln\hat{L} + k\ln T, \qquad Q_{LB} = T(T+2)\sum_{k=1}^{m}\frac{\hat{\rho}(k)^2}{T-k}.$$

[EST]/[STAT] AIC targets predictive loss (does not penalise enough to be consistent); BIC is consistent for the true order if it is finite; $Q_{LB} \sim \chi^2_{m-p-q}$ under correct specification (Ljung–Box 1978, *Biometrika*).

Spectral analysis. The frequency-domain dual of the ACF:

$$(2.3) \qquad f(\lambda) = \frac{1}{2\pi}\sum_{k=-\infty}^{\infty}\gamma(k)e^{-ik\lambda}, \qquad \gamma(k) = \int_{-\pi}^{\pi} f(\lambda)e^{ik\lambda}\,d\lambda, \qquad I_T(\lambda) = \frac{1}{2\pi T}\left|\sum_{t=1}^{T} X_t e^{-it\lambda}\right|^2.$$

[DEF]/[EST] f is the spectral density (Fourier transform of the ACF — Wiener–Khinchin); the **periodogram** I_T is its raw, inconsistent estimator (variance does not vanish; must be smoothed). Power-law $f(\lambda) \sim \lambda^{-2d}$ near 0 signals long memory (§3).

Unit roots and integration. A series is $I(1)$ if its first difference is stationary. The **augmented Dickey–Fuller** regression tests H_0 : unit root ($\beta = 0$):

$$(2.4)\qquad \Delta X_t = \alpha + \delta t + \beta X_{t-1} + \sum_{i=1}^{p} \zeta_i \, \Delta X_{t-i} + \varepsilon_t, \qquad \mathrm{ADF} = \frac{\hat{\beta}}{\mathrm{se}(\hat{\beta})} \text{ (non-standard, Dickey–Ful}$$

[STAT] Dickey–Fuller (1979, *JASA*); Phillips–Perron (1988, *Biometrika*) is the HAC-robust variant. **KPSS** (Kwiatkowski–Phillips–Schmidt–Shin 1992, *J. Econometrics*) flips the null to *stationarity* — running both is the honest practice, since each has low power against the other's alternative. *Prices are typically $I(1)$, returns $I(0)$ — which is why one models returns.*

Cointegration and error correction. Two $I(1)$ series can share a stationary linear combination ($I(0)$): they are tied by a long-run equilibrium even as each wanders.

$$(2.5)\qquad y_t - \beta x_t = u_t \sim I(0) \;\Rightarrow\; \Delta y_t = \alpha\,(y_{t-1} - \beta x_{t-1}) + \cdots + \varepsilon_t, \quad \alpha < 0.$$

[MODEL]/[EST] Engle–Granger (1987, *Econometrica*) two-step; Johansen (1991, *Econometrica*) the system MLE / VECM with rank tests for the number of cointegrating vectors. This is the statistical engine of pairs/relative-value trading — and note the link to the companion document's Δ2: a cointegrating residual is a *convergence trade*, and §III's limits-to-arbitrage say it can diverge before it converges.

Vector autoregression, Granger causality, impulse responses.

$$(2.6)\qquad \mathbf{X}_t = \mathbf{c} + \sum_{i=1}^{p} \mathbf{A}_i \mathbf{X}_{t-i} + \boldsymbol{\varepsilon}_t, \qquad \mathbf{X}_t = \boldsymbol{\mu} + \sum_{j\ge 0} \boldsymbol{\Psi}_j \boldsymbol{\varepsilon}_{t-j} \text{ (MA}\to\text{IRF).}$$

[MODEL] Sims (1980, *Econometrica*). x *Granger-causes* y if past x improves the forecast of y given past y (Granger 1969, *Econometrica*) — a statement about predictability, **not** structural causation. Impulse-response and forecast-error-variance decomposition require an identification scheme (Cholesky ordering, etc.) to map reduced-form $\boldsymbol{\varepsilon}$ to structural shocks; the choice is an assumption, not data.

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§3 — Second moments: volatility, jumps, long memory

The mean is nearly unpredictable; the *variance* is highly predictable (volatility clusters). This is where most of the structure in financial series actually lives.

ARCH / GARCH. Engle (1982, *Econometrica*) and Bollerslev (1986, *J. Econometrics*):

$$(3.1)\qquad r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t,\; z_t \sim (0,1), \qquad \sigma_t^2 = \omega + \sum_{i=1}^{q} \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^{p} \beta_j \sigma_{t-j}^2.$$

[MODEL] GARCH(1,1) ($\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$) is the workhorse; covariance-stationary iff $\alpha + \beta < 1$, with unconditional variance $\omega/(1-\alpha-\beta)$ and persistence $\alpha+\beta$ (often ≈ 0.99 daily). *Status: captures clustering — holds; assumes a symmetric response — known false.* Extensions encode the **leverage effect** (down-moves raise vol more): EGARCH (Nelson 1991, *Econometrica*, log-variance with sign asymmetry) and GJR-GARCH (Glosten–Jagannathan–Runkle 1993, *JF*, a threshold term $\gamma\varepsilon_{t-1}^2 1\{\varepsilon_{t-1} < 0\}$). Estimated by **QMLE** (Gaussian quasi-likelihood is consistent even if z_t is non-Gaussian — §9). This is the discrete-time image of the companion document's stochastic-variance state v_t (III.2).

Realised variance and jumps. With intraday returns $r_{t,i}$, $i = 1..M$:

$$(3.2)\qquad \mathrm{RV}_t = \sum_{i=1}^{M} r_{t,i}^2 \xrightarrow{p} \int_{t-1}^{t} v_s\, ds + \sum_{\text{jumps}} J^2, \qquad \mathrm{BV}_t = \frac{\pi}{2} \sum_{i=2}^{M} |r_{t,i-1}||r_{t,i}| \xrightarrow{p} \int_{t-1}^{t} v_s\, ds.$$

[EST] Realised variance converges to total quadratic variation; **bipower variation** (Barndorff-Nielsen–Shephard 2004, *J. Financial Econometrics*) is jump-robust, so RV − BV estimates the **jump** contribution — the empirical handle on the companion document's jump measure $\ell(dz)$ (III.1). *Daily OHLC bars (this repo's data) cannot form RV; the repo proxies volatility with ATR — see §6.*

Long memory. When the ACF decays hyperbolically ($\rho(k) \sim k^{2d-1}$, $0 < d < \frac{1}{2}$) rather than geometrically, shocks persist far longer than ARMA allows.

$$(3.3) \qquad (1-L)^d X_t = \varepsilon_t \text{ (ARFIMA)}, \qquad \mathbb{E}\left[\frac{R(n)}{S(n)}\right] \sim n^H, \qquad H = d + \tfrac{1}{2}.$$

[MODEL]/[EST] Fractional integration (Granger–Joyeux; Hosking 1981, *Biometrika*); the **Hurst exponent** H estimated by rescaled-range (R/S) analysis (Hurst 1951; Mandelbrot–Wallis) or detrended fluctuation analysis (DFA, Peng et al. 1994). $H = \frac{1}{2}$ is a random walk, $H > \frac{1}{2}$ trending/persistent, $H < \frac{1}{2}$ mean-reverting. *Status: contested — R/S is biased in small samples and confounded by short-memory and non-stationarity; treat H estimates as suggestive, not decisive.*

Forecasting volatility: the HAR model. The dominant practical volatility forecaster sidesteps fractional integration by regressing realised variance on its own averages over a *cascade* of horizons (daily, weekly, monthly), approximating long memory with three lags (Corsi 2009, *J. Financial Econometrics* 7, 174–196):

$$(3.4) \qquad \mathrm{RV}_{t+1} = \beta_0 + \beta_d \,\mathrm{RV}_t^{(d)} + \beta_w \,\mathrm{RV}_t^{(w)} + \beta_m \,\mathrm{RV}_t^{(m)} + \varepsilon_{t+1},$$

[MODEL]/[EST] where $\mathrm{RV}^{(w)}, \mathrm{RV}^{(m)}$ are 5- and 22-day averages. The **HAR-RV** is a plain OLS regression yet forecasts as well as far more complex long-memory models — a recurring lesson (echoed in this repo's experiments 4 and 9) that *parsimony beats sophistication* out-of-sample. *On daily OHLC bars one substitutes a range estimator (Parkinson/Garman–Klass) or ATR (§6) for RV.*

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§4 — State space, filtering, and regimes

State-space form unifies a huge class of models (ARMA, time-varying parameters, stochastic vol, unobserved trends):

$$(4.1) \qquad \underbrace{\mathbf{x}_t = \mathbf{F}\mathbf{x}_{t-1} + \mathbf{w}_t\,,}_{\text{state (transition)}} \qquad \underbrace{y_t = \mathbf{H}\mathbf{x}_t + v_t\,,}_{\text{observation}} \qquad \mathbf{w}_t \sim \mathcal{N}(0, \mathbf{Q}),\ v_t \sim \mathcal{N}(0, R).$$

[MODEL] The hidden state $\mathbf{x}_t$ is observed only through noisy y_t.

Kalman filter. The optimal (MMSE, and under Gaussianity the exact Bayesian) recursive estimator of $\mathbf{x}_t$ given $y_{1:t}$ — it is the dynamic version of the projection theorem (Lemma P in the companion document). Predict then update:

$$(4.2) \qquad \begin{aligned} &\hat{\mathbf{x}}_{t|t-1} = \mathbf{F}\hat{\mathbf{x}}_{t-1}, \quad \mathbf{P}_{t|t-1} = \mathbf{F}\mathbf{P}_{t-1}\mathbf{F}' + \mathbf{Q}, \\ &\mathbf{K}_t = \mathbf{P}_{t|t-1}\mathbf{H}'(\mathbf{H}\mathbf{P}_{t|t-1}\mathbf{H}' + R)^{-1}, \\ &\hat{\mathbf{x}}_t = \hat{\mathbf{x}}_{t|t-1} + \mathbf{K}_t(y_t - \mathbf{H}\hat{\mathbf{x}}_{t|t-1}), \quad \mathbf{P}_t = (\mathbf{I} - \mathbf{K}_t\mathbf{H})\mathbf{P}_{t|t-1}. \end{aligned}$$

[EST] Kalman (1960). The gain $\mathbf{K}_t$ is the precision-weighting of new information against the prior — exactly the normal–normal update (companion Lemma N) made recursive. The innovations $y_t - \mathbf{H}\hat{\mathbf{x}}_{t|t-1}$ are a white-noise sequence whose Gaussian likelihood gives the prediction-error decomposition for MLE; the RTS smoother runs it backward for $\mathbf{x}_t \mid y_{1:T}$. Nonlinear/non-Gaussian extensions: extended/unscented Kalman, and **particle filters** (sequential Monte Carlo) for general state spaces.

Regime-switching / hidden Markov models. A discrete latent state $S_t \in \{1, \ldots, m\}$ following a Markov chain with transition matrix $\mathbf{P} = [p_{ij}]$ governs the parameters:

$$(4.3) \qquad y_t \mid S_t = j \;\sim\; \mathcal{N}(\mu_j, \sigma_j^2), \qquad \mathbb{P}(S_t = j \mid S_{t-1} = i) = p_{ij}.$$

[MODEL] Hamilton (1989, *Econometrica*) for switching regressions; the latent path is inferred by the forward–backward (Baum–Welch, an EM algorithm) for the likelihood and **Viterbi** for the most-likely state sequence. This is the formal version of "the market is in a different regime" — and a sober reminder (companion §Boundary) that a *known* switching law is still one stationary meta-model. **[REPO]** `vpts/regime/quiet.py` implements a simpler, non-probabilistic regime read: a "quiet-phase" score from volatility/volume compression rather than a fitted HMM — a hand-built detector occupying the same conceptual slot.

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§5 — Non-stationarity and nonlinearity: breaks, trends, filters, regimes-in-the-mean

Because the standing caveat is real, detecting and handling parameter change — and nonlinearity — is not optional.

Structural breaks. Chow (1960) tests a break at a *known* date; **CUSUM** (Brown–Durbin–Evans 1975, *JRSS-B*) monitors cumulative recursive residuals for parameter drift; **Bai–Perron** (1998, *Econometrica*; 2003, *J. Applied Econometrics*) estimates *multiple* breaks at *unknown* dates by minimising total SSR with a dynamic program. **[STAT]** A break detected is the stationarity assumption failing in-sample — the honest response is to model the change, not to extend the window through it.

Trend/cycle decomposition. The Hodrick–Prescott filter (1997, *JMCB*) splits $y_t = \tau_t + c_t$ by penalised least squares:

$$(5.1) \qquad \min_{\{\tau_t\}} \sum_t (y_t - \tau_t)^2 + \lambda \sum_t \big[(\tau_{t+1} - \tau_t) - (\tau_t - \tau_{t-1})\big]^2.$$

[EST] λ trades fit against smoothness (1600 for quarterly data, by convention). *Known issues: spurious end-point cycles and the look-ahead of a two-sided filter — using HP-filtered features in a backtest is a classic leakage bug (§10).* Band-pass alternatives: Baxter–King; multiresolution alternatives: the **wavelet** transform, which localises variance jointly in time and frequency (better than the global Fourier spectrum for non-stationary series). Change-point detection (CUSUM, Bayesian online change-point, PELT) is the online cousin.

Nonlinear models in the mean. Linear ARMA cannot capture state-dependent dynamics (trending in one regime, mean-reverting in another). The threshold/regime family makes the parameters depend on an observable or its own lag:

$$(5.2) \qquad X_t = \begin{cases} \phi^{(1)}(L)X_t + \varepsilon_t, & q_{t-d} \le c \\ \phi^{(2)}(L)X_t + \varepsilon_t, & q_{t-d} > c \end{cases} \quad \text{(SETAR)}, \qquad X_t = \phi^{(1)} X_{t-1} + \big(\phi^{(2)} - \phi^{(1)}\big) X_{t-1}\, G(q_{t\cdot}$$

[MODEL] Self-exciting threshold AR (Tong 1990, *Non-Linear Time Series*) switches sharply at threshold c; smooth-transition AR (Teräsvirta 1994, *JASA*) interpolates via a logistic/exponential $G \in [0,1]$. These are the *in-the-mean* cousins of the Hamilton regime-switching model (§4, which switches on a *latent* state) and of this repo's regime gating (§6). *Status: flexible but easy to overfit — the bias–variance warning of §9 applies sharply.*

Testing for nonlinearity / i.i.d. The **BDS test** (Brock–Dechert–Scheinkman–LeBaron 1996, *Econometric Reviews* 15) uses the correlation integral to test the null that a series is i.i.d. against unspecified (possibly nonlinear/chaotic) dependence — typically applied to model *residuals* to check whether linear filtering left structure behind. **[STAT]** Rejection says "there is more here," not "it is tradeable."

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§6 — The signal math this repository computes

This section documents `vpts`'s feature construction as time-series mathematics. All of it operates **without look-ahead** (features at bar t use data $\leq t$), which is itself the most important modelling constraint (§10).

Average true range (volatility scale). [REPO] `vpts/regime/indicators.py`, `profile/calculator.py`:

$$(6.1) \qquad \mathrm{TR}_t = \max\big(H_t - L_t,\ |H_t - C_{t-1}|,\ |L_t - C_{t-1}|\big), \qquad \mathrm{ATR}_n = \frac{1}{n}\sum_{i=t-n+1}^{t} \mathrm{TR}_i.$$

[EST] A robust, OHLC-only volatility proxy (Wilder 1978). It scales bin widths, level tolerances, and — crucially — the barrier distances in labelling (§7), so that targets are defined in volatility units, not price units.

The volume profile as a price-conditional volume density. [REPO] `vpts/profile/calculator.py`. Distribute each bar's volume across price bins in proportion to the overlap of $[L_t, H_t]$ with each bin (the volume-conserving "uniform" estimator), giving an empirical *volume-at-price* distribution Π:

$$(6.2) \qquad \Pi(b) = \sum_t V_t \cdot \frac{\big|[L_t, H_t] \cap \mathrm{bin}_b\big|}{H_t - L_t}, \qquad \mathrm{POC} = \arg\max_b \Pi(b).$$

[DEF]/[EST] This is a kernel-smoothed histogram (a KDE with a uniform/box kernel of bar-dependent width) of traded volume over price. The **POC** is its mode; HVN/LVN are its peaks/valleys (found by `scipy.signal.find_peaks` on a Gaussian-smoothed Π — equivalently a mode/antimode hunt). The **value area** is built by the Market-Profile expansion rule — start at the POC and greedily annex the heavier of the two adjacent pairs until 70% of volume is enclosed:

$$(6.3) \qquad \mathrm{VA} = [\mathrm{VAL}, \mathrm{VAH}] \text{ minimal-ish interval s.t.} \sum_{b \in \mathrm{VA}} \Pi(b) \geq 0.70 \sum_b \Pi(b).$$

[DEF] Statistically this is a (greedy, not exact) **70% highest-density region** of Π — the volume analogue of a credible interval around the modal price. ATR-adaptive binning (quarter-ATR bins) makes resolution regime-dependent: finer in quiet markets, coarser in volatile ones.

Close location value & synthetic order-flow delta. [REPO] `vpts/structure/analytics.py`. With no tick data, infer aggressor side from where a bar closes in its range:

$$(6.4) \qquad \mathrm{CLV}_t = \frac{(C_t - L_t) - (H_t - C_t)}{H_t - L_t} \in [-1, 1], \qquad \delta_t = V_t \cdot \mathrm{CLV}_t, \qquad \delta^{\mathrm{net}} = \frac{\sum_t \delta_t}{\sum_t V_t}.$$

[EST] δ_t is a signed-volume **synthetic cumulative volume delta** — an OHLC estimator of the buy/sell imbalance that the companion document's microstructure block (order flow y, Kyle/GM) models with real signed trades. The *POC-delta fraction* (signed volume at the modal price) is the repo's "buy-the-dip / passive accumulation" tell. It is an *estimate* of order flow, with all the bias that an OHLC proxy carries — flagged as such.

Volume-weighted shape moments. [REPO] Treating Π as a distribution with $p_b = \Pi(b)/\sum \Pi$:

$$(6.5) \qquad m = \sum_b p_b x_b, \quad s^2 = \sum_b p_b (x_b - m)^2, \quad \mathrm{skew} = \sum_b p_b \Big(\tfrac{x_b - m}{s}\Big)^3, \quad \mathrm{kurt} = \sum_b p_b \Big(\tfrac{x_b - m}{s}\Big)^4.$$

[DEF] Standardised 3rd/4th moments of the volume-at-price distribution — the same moment machinery the companion document reads off the *risk-neutral* density (BKM, IV.L3.vi), here applied to the realised volume density to classify profile shape (P / b / D / B).

Time-decayed gravity (cost-basis migration). [REPO] An exponentially-weighted profile with half-life h reveals whether recent trade is migrating to a new fair value:

$$(6.6) \qquad \Pi^{\text{decay}}(b) = \sum_t V_t \, 2^{-\text{age}(t)/h} \cdot (\text{overlap}_b), \qquad \text{decayed POC} = \arg\max_b \Pi^{\text{decay}}(b).$$

[EST] The $2^{-\text{age}/h}$ kernel is an EMA; the gap between the decayed and lifetime POC is a slow drift estimator (cf. the EMA as a one-sided low-pass filter — a causal cousin of the HP trend in §5, but look-ahead-free).

Rolling standardisation (the z-score trigger). [REPO] The value-area-compression ratio $\text{VACR} = (\text{VAH} - \text{VAL})/P$ is turned into a stationary signal by a **rolling z-score**:

$$(6.7) \qquad z_t = \frac{\text{VACR}_t - \hat{\mu}_t^{(w)}}{\hat{\sigma}_t^{(w)}}, \qquad \hat{\mu}_t^{(w)}, \hat{\sigma}_t^{(w)} \text{ computed on } [t - w + 1,\ t] \text{ only.}$$

[EST] Rolling standardisation is the practical concession to non-stationarity: it removes a slowly-varying local mean/scale so a level becomes comparable across regimes. The trailing window is what keeps it causal.

The confluence score (linear fusion). [REPO] `vpts/scoring/scorer.py`. The hand-set strategy is a weighted linear combination of component strengths $g_c \in [0, 1]$ with signs $d_c \in \{-1, 0, 1\}$:

$$(6.8) \qquad \text{quality} = 100 \cdot \frac{\sum_c w_c g_c}{\sum_c w_c} \in [0, 100], \qquad \text{bias} = 100 \cdot \frac{\sum_c w_c g_c d_c}{\sum_c w_c} \in [-100, 100].$$

[DEF] A fixed-weight linear scorecard — the "primitive/assumption" layer of the strategy. Experiment 2 in `RESEARCH.md` *learns* these w_c by ridge regression (§8) and finds no out-of-sample improvement over the hand weights: the linear functional form, not the weights, is the binding object.

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§7 — Labelling: defining the target as a first-passage problem

A predictor needs something to predict. The repo uses **triple-barrier labelling** (López de Prado, *AFML* 2018, ch. 3) — a first-passage formulation that is far more honest than fixed-horizon returns because it encodes a path-dependent, risk-defined exit. **[REPO]** `vpts/ml/labeling.py`.

For an event at t taken in side $s \in \{+1, -1\}$, with volatility-scaled barriers (profit $u = \text{pt} \cdot \sigma_t$, stop $\ell = \text{sl} \cdot \sigma_t$, vertical T bars):

$$(7.1) \qquad \tau = \inf\Big\{ j > 0 :\ s\, r_{t,t+j} \ge u \ \text{ or } \ s\, r_{t,t+j} \le -\ell \ \text{ or } \ j = T \Big\}, \qquad y_t = \text{sign}\left(s\, r_{t,t+\tau} \right).$$

[DEF] The label is determined by which barrier the (volatility-normalised) path hits first — a **first-passage / barrier-hitting time** of the price process, exactly the mathematics the companion document uses for option exercise and the Kyle revelation horizon. Scaling barriers by σ_t (here ATR/close) makes the label stationary across volatility regimes — a target with constant difficulty. The breakeven win-rate is set by the reward:risk u/ℓ (2:1 → 33%).

Meta-labelling. A *primary* model picks the side $s = \text{sign}(\text{bias})$; the triple barrier yields a binary **meta-label** $1\{\text{the bet won}\}$; a *secondary* model learns $\mathbb{P}(\text{win} \mid \text{features})$ — i.e. it learns *whether to act*, not *which way*. **[VALID-adjacent]** This cleanly separates **direction** from **selectivity**, and RESEARCH.md exploits exactly that split: the directional call is survivorship-driven and inverts, but the *selectivity* lift is the most survivorship-resilient signal in the whole study (it degrades rather than flips). **MFE/MAE** labelling is the continuous cousin — label by whether maximum favourable excursion beat maximum adverse excursion, a volatility-scaled triple barrier in disguise.

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§8 — Dependence, the cross-section, and the information coefficient

Cross-correlation and lead–lag. $\rho_{xy}(k) = \mathrm{Corr}(x_t, y_{t+k})$; a peak at $k > 0$ says x leads y — again predictability, not causation (§2's Granger caveat).

The information coefficient. The central performance statistic for a cross-sectional signal: the rank correlation between today's signal and the forward return, across names, each rebalance date. **[REPO]** `vpts/ml`, `vpts/scoring`:

$$(8.1) \qquad \mathrm{IC}_d = \rho_{\mathrm{Spearman}}\big(\mathrm{signal}_{i,d},\ r^{\mathrm{fwd}}_{i,d}\big)_{i=1..N_d}, \qquad t_{\mathrm{IC}} = \overline{\mathrm{IC}}\,\frac{\sqrt{D}}{\sigma_{\mathrm{IC}}}, \qquad \mathrm{se}(\mathrm{IC}_d) \approx \frac{1}{\sqrt{N_d - 1}}.$$

[STAT] Spearman (rank) IC is robust to the non-normal, outlier-heavy nature of returns. Two facts the repo leans on hard: (i) per-date IC noise scales like $1/\sqrt{N_d}$, so a thin cross-section ($N \approx 20$) produces a huge $\sigma_{\mathrm{IC}} \approx 0.28$ and spurious near-misses — *width* (more names) is the decisive test, and the 20-name "+0.021, p=0.10" washes to "−0.009, p=0.86" at 88 names (experiments 5→6); (ii) the single-feature t-stat t_{IC} understates the multiple-testing burden of having tried many features (§10).

Factor models and dimension reduction. Cross-sectional regressions $r_i = \boldsymbol{\beta}_i'\mathbf{f} + \epsilon_i$; **PCA** (eigendecomposition of the return covariance) extracts statistical factors; the companion document's $\mathbb{Q}/\mathbb{P}$ point applies — a "factor premium" is compensation for risk, not free money. **Copulas** (Sklar's theorem: any joint law factors into marginals plus a copula C) model tail dependence beyond linear ρ; **DCC-GARCH** (Engle 2002, *JBES*) makes the correlation matrix itself time-varying — the multivariate image of §3.

Nonlinear dependence (information-theoretic). Correlation and rank-IC miss nonlinear/asymmetric dependence (the stylized fact (1.6) that returns are uncorrelated but *not* independent). Information theory measures the full dependence:

$$(8.2) \qquad I(X;Y) = \sum_{x,y} p(x,y)\ln\frac{p(x,y)}{p(x)p(y)} \ \geq 0, \qquad T_{Y\to X} = \sum p(x_{t+1}, x_t^{(\cdot)}, y_t^{(\cdot)})\ln\frac{p(x_{t+1} \mid x_t^{(\cdot)}, y_t^{(\cdot)}}{p(x_{t+1} \mid x_t^{(\cdot)})}$$

[STAT] Mutual information $I(X;Y)$ is zero iff $X \perp Y$ (it catches any dependence, not just linear); **transfer entropy** $T_{Y\to X}$ (Schreiber 2000, *Phys. Rev. Lett.* 85) is its directed, lagged version — a model-free, nonlinear generalisation of Granger causality (§2). *Caveat: both need a lot of data to estimate densities and are easy to over-read in finite, noisy samples — the multiple-testing and overfitting cautions of §§9–10 apply.*

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§9 — Estimation and inference machinery

Maximum likelihood / QMLE. Most models above are fit by maximising the (Gaussian prediction-error) likelihood; under misspecified innovations the Gaussian **quasi**-MLE remains consistent and asymptotically normal with a sandwich covariance (Bollerslev–Wooldridge), which is why GARCH estimates survive fat-tailed z_t. **[EST]**

GMM. When a model implies moment conditions $\mathbb{E}[g(X_t, \theta_0)] = 0$ (e.g. the Euler equation $\mathbb{E}[mR - 1] = 0$ of the companion document), estimate by minimising $\bar{g}(\theta)'\mathbf{W}\bar{g}(\theta)$ (Hansen 1982, *Econometrica*). **[EST]** The optimal $\mathbf{W}$ is the inverse long-run variance — which requires:

HAC standard errors. Serial correlation and heteroskedasticity make naïve standard errors wrong (usually too small). The Newey–West (1987, *Econometrica*) estimator corrects them:

$$(9.1) \qquad \hat{S} = \hat{\Gamma}_0 + \sum_{k=1}^{m}\Big(1 - \frac{k}{m+1}\Big)\big(\hat{\Gamma}_k + \hat{\Gamma}_k'\big).$$

[EST] The Bartlett weights guarantee a positive-definite estimate; m is the bandwidth. *This is the single most common fix for the "my t-stat is inflated by autocorrelation" problem — the time-series analogue of clustering.*

The bootstrap (resampling under dependence). I.i.d. bootstrap is invalid for dependent data — it destroys the autocorrelation. Two repairs:

$$(9.2)\qquad \textbf{block bootstrap:}\ \text{resample contiguous blocks of length } b; \qquad \textbf{stationary bootstrap:}\ \text{ge}$$

[VALID] The stationary bootstrap (Politis–Romano 1994, *JASA*) randomises block length so the resampled series is itself stationary — the standard tool for confidence intervals on Sharpe ratios, IC, and other path statistics. Block length $b \sim T^{1/3}$ (or $1/p$) must grow with the dependence horizon.

Permutation / randomisation tests. [REPO] `vpts/ml` — the decisive significance test in the whole research log. Destroy the feature→label link by shuffling labels (per-row for a single series; **within-date** for a cross-section, to preserve cross-sectional structure), recompute the statistic B times, and read off:

$$(9.3)\qquad p = \frac{1 + \#\{b:\ \hat{\theta}^{(b)}_{\text{shuffled}}\ \geq\ \hat{\theta}_{\text{observed}}\}}{1 + B}.$$

[VALID] A model-free null that asks "could structure-free noise have produced a statistic this large?" The $+1$s make it a valid finite-sample test. *An effect that cannot clear its own shuffled null is reported as no edge* — this is what kills meta-labelling (p 0.005 on survivors → 0.80 with delisted names injected).

§9b — Regularised regression, classification, and learning machines

The fitted models in this repo (and modern empirical finance generally) are penalised regressions and tree ensembles. Their math, and the one decomposition that explains when they fail.

Ridge regression. [REPO] `vpts/ml/factor_model.py` learns the confluence weights by exactly this closed form (train-only standardisation, target centred):

$$(9.4)\qquad \hat{\mathbf{w}} = \arg\min_{\mathbf{w}}\ \|\mathbf{y} - \mathbf{X}\mathbf{w}\|^2 + \alpha\|\mathbf{w}\|_2^2 = (\mathbf{X}'\mathbf{X} + \alpha\mathbf{I})^{-1}\mathbf{X}'\mathbf{y}.$$

[EST] Hoerl–Kennard (1970, *Technometrics* 12). The ℓ_2 penalty α shrinks weights toward 0, trading a little bias for a large variance reduction — essential when features are collinear (confluence factors are) and the signal is faint. *In experiment 2, ridge shrank every learned weight to ≈ 0 and did not beat the hand-set baseline: the data did not support the extra degrees of freedom.*

LASSO and elastic net. Swap the penalty: ℓ_1 ($\alpha\|\mathbf{w}\|_1$, Tibshirani 1996, *JRSS-B* 58, 267–288) drives some weights *exactly* to zero (selection); the elastic net (Zou–Hastie 2005, *JRSS-B* 67) mixes $\ell_1 + \ell_2$ to keep selection while handling correlated groups. **[EST]** These are the standard tools for the high-dimensional factor zoo — and, with the Harvey–Liu–Zhu hurdle (§10), the honest way to fight selection bias.

Logistic regression (the meta-model). [REPO] `vpts/ml/meta_model.py` minimises the L2-regularised log-loss by gradient descent to learn $\mathbb{P}(\text{win} \mid \text{features})$:

$$(9.5)\qquad p_i = \sigma(\mathbf{w}'\mathbf{x}_i + b) = \frac{1}{1 + e^{-(\mathbf{w}'\mathbf{x}_i + b)}}, \qquad \hat{\mathbf{w}} = \arg\min_{\mathbf{w}}\ -\sum_i \big[y_i \ln p_i + (1 - y_i)\ln(1 - p_i)\big] +$$

[EST] The binary cross-entropy is convex; the secondary model *filters* primary signals (meta-labelling, §7), so it is scored by classification metrics (§10), not by direction.

Gradient-boosted trees (XGBoost). [REPO] A forward stagewise additive ensemble $F_M(\mathbf{x}) = \sum_{m=1}^{M} \nu\, f_m(\mathbf{x})$ of regression trees, each fit to the (regularised) functional gradient of the loss (Friedman 2001, *Annals of Statistics* 29; XGBoost: Chen–Guestrin 2016, *KDD*, adds a second-order objective and explicit tree penalties):

$$f_m = \arg\min_f \sum_i \ell\big(y_i,\, F_{m-1}(\mathbf{x}_i) + f(\mathbf{x}_i)\big) + \Omega(f), \qquad \Omega(f) = \gamma T + \tfrac{1}{2}\lambda\|\mathbf{w}\|^2. \tag{9.6}$$

[EST] Hugely flexible — and hugely prone to overfit on low-signal financial data. In RESEARCH.md experiment 9, XGBoost memorised the training set (in-sample AUC **0.943**) yet scored **0.496 out-of-sample — below 0.5, worse than the linear logistic (0.529)**. The gaudy in-sample number is precisely the trap §10 exists to catch.

The bias–variance decomposition (why this happens). For squared-error loss, expected test error at $\mathbf{x}$ factors as

$$\mathbb{E}\big[(y - \hat{f}(\mathbf{x}))^2\big] = \underbrace{\big(\mathbb{E}[\hat{f}(\mathbf{x})] - f(\mathbf{x})\big)^2}_{\text{bias}^2} + \underbrace{\mathrm{Var}\big(\hat{f}(\mathbf{x})\big)}_{\text{variance}} + \underbrace{\sigma_\varepsilon^2}_{\text{irreducible}}. \tag{9.7}$$

[DEF] Hastie–Tibshirani–Friedman (2009, *ESL*). Flexible models (XGBoost) cut bias but inflate variance; in a regime where the irreducible noise σ_ε^2 dominates the signal — exactly daily return prediction (§11) — the variance term swamps any bias gain, so the *more flexible* model generalises *worse*. Regularisation (α, λ, γ, tree depth, ν) is the dial that buys variance reduction with bias; cross-validation (§10) is how it is set honestly. This single equation is why "use a bigger model" is usually the wrong answer here, and why this repo's linear book beat its gradient-boosted one.

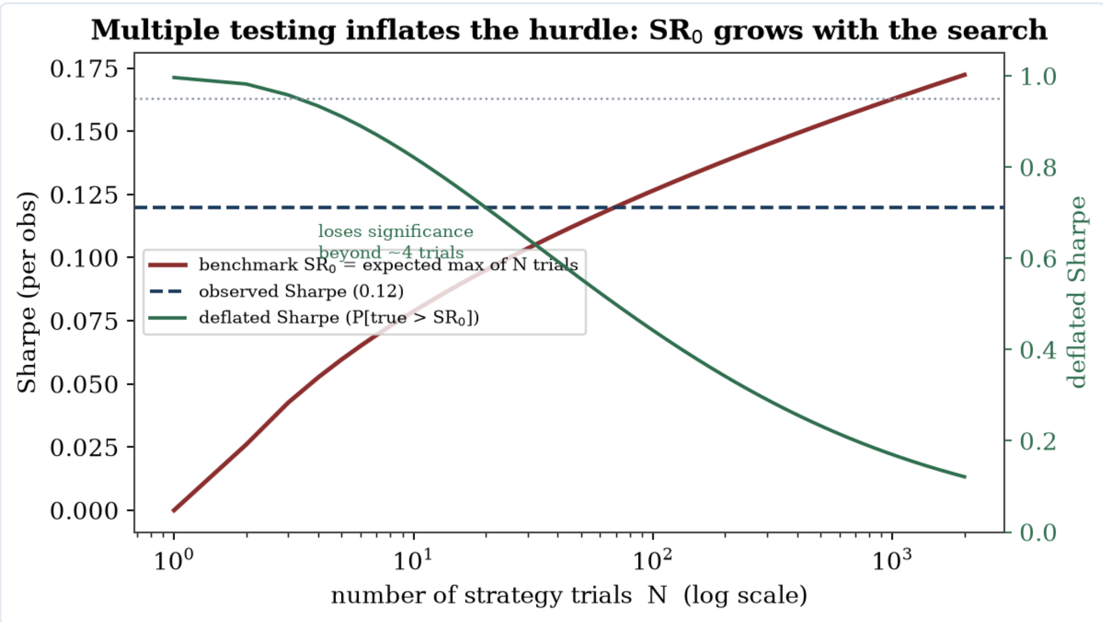

Figure 3.2. Multiple testing inflates the significance hurdle: the benchmark Sharpe grows with the number of trials, so an observed Sharpe loses significance as the search widens.

§10 — Forecast evaluation and the overfitting problem (the core of the harness)

This is where the document earns its keep, and where this repository's discipline lives. The enemy is not bad models; it is **good-looking in-sample numbers that do not survive honest out-of-sample evaluation under dependence and multiplicity.**

Forecast loss and comparison. For forecasts with errors e_t, the **Diebold–Mariano** (1995, *JBES*) test compares two models' loss differentials $d_t = L(e_{1t}) - L(e_{2t})$ with a HAC variance:

$$\mathrm{DM} = \frac{\bar{d}}{\sqrt{\widehat{\mathrm{LRV}}(\bar{d})/T}} \xrightarrow{d} \mathcal{N}(0,1) \text{ under } H_0 : \mathbb{E}[d_t] = 0. \tag{10.1}$$

[STAT] Note the HAC variance (§9) — predictive accuracy comparisons are autocorrelated and must be deflated.

Why ordinary cross-validation is invalid here, and the fix. Standard k-fold CV assumes i.i.d. rows. Financial labels span a forward horizon, so a train row near a test block **leaks** its label into the test set. The fix has three parts (López de Prado, *AFML* 2018):

1. **Purging** — drop train observations whose label window $[t, t + \text{horizon}]$ overlaps any test block. **[VALID] [REPO]** `vpts/validation/cpcv.py`.
2. **Embargo** — additionally drop a few train observations *immediately after* each test block, to break the serial-correlation bleed across the boundary. **[VALID][REPO]**
3. **Combinatorial Purged CV (CPCV)** — instead of one train/test partition, split the timeline into N groups, use *every* combination of k as the test set, and aggregate the recombined out-of-sample segments into a **distribution** of performance, not a point:

$$\#\text{splits} = \binom{N}{k}, \qquad \#\text{backtest paths } \varphi[N,k] = \frac{k}{N}\binom{N}{k} = \binom{N-1}{k-1}. \tag{10.2}$$

[VALID] (E.g. $N = 6, k = 2$: 15 splits, $\varphi = 5$ paths.) The output is a *sampling distribution* of OOS return/Sharpe — median, dispersion, fraction of paths profitable — which is what turns "+14.5% in one backtest" into "−0.68% per path, only 36% profitable" (experiment 1). *Honest scope (from the code's own docstring): with a no-parameter strategy CPCV measures robustness/dispersion, not selection-overfitting protection — that protection matters the moment any parameter is fit on the folds.*

Data-snooping across many strategies. When you try M configurations and report the best, its in-sample Sharpe is biased up by the maximum-of-M effect. Controls:

- **White's Reality Check** (2000, *Econometrica*) and **Hansen's SPA** (2005, *JBES*): bootstrap the null that the *best* strategy does not beat the benchmark, correcting for the full set searched. **[VALID]**
- **Multiple-testing hurdles**: with hundreds of tested factors, the conventional $|t| > 1.96$ is far too lax — Harvey–Liu–Zhu (2016, *RFS* 29, 5–68) argue a new factor needs $|t| \gtrsim 3.0$ once Bonferroni/Holm/BHY corrections for the search are applied. **[VALID]**

Backtest overfitting, quantified. The frontier tools, all directly relevant to a research log like this one:

$$\text{PBO} = \mathbb{P}\big(\text{the in-sample-best config is below-median out-of-sample}\big), \tag{10.3}$$

[VALID] estimated by **combinatorially-symmetric cross-validation** (Bailey–Borwein–López de Prado–Zhu 2017, *J. Computational Finance* — "The Probability of Backtest Overfitting"). And the **Deflated Sharpe Ratio** (Bailey–López de Prado 2014, *J. Portfolio Management* 40(5), 94–107), which discounts a Sharpe for the number of trials M, the skew γ_3 and kurtosis γ_4 of returns, and the sample length:

$$\text{DSR} = \Phi\left(\frac{(\widehat{\text{SR}} - \text{SR}_0)\sqrt{T-1}}{\sqrt{1 - \gamma_3\widehat{\text{SR}} + \frac{\gamma_4 - 1}{4}\widehat{\text{SR}}^2}}\right), \qquad \text{SR}_0 \approx \sigma_{\text{SR}}\Big[(1-\gamma_E)\Phi^{-1}\big(1 - \tfrac{1}{M}\big) + \gamma_E\Phi^{-1}\big(1 - \tag{10.4}$$

[VALID] where SR_0 is the Sharpe you'd expect from the *luckiest* of M random trials (γ_E = Euler–Mascheroni). A strategy is significant only if it beats that inflated benchmark.

The Sharpe ratio is itself a noisy estimate. It has a sampling distribution, and autocorrelation corrupts its annualisation (Lo 2002, *FAJ* 58(4), 36–52):

$$\text{se}(\widehat{\text{SR}}) \approx \sqrt{\frac{1 + \frac{1}{2}\text{SR}^2}{T}}\ (\text{iid}), \qquad \text{SR}_{\text{annual}} = \frac{q\,\text{SR}}{\sqrt{q + 2\sum_{k=1}^{q-1}(q-k)\rho_k}} \quad / \quad \sqrt{q}\,\text{SR unless } \rho_k \tag{10.5}$$

[STAT] Positive autocorrelation (smoothed returns) inflates the naïvely-annualised Sharpe; the $\sqrt{q}$ rule is valid only for serially-uncorrelated returns. *Reporting a Sharpe without its standard error, or* $\sqrt{252}$*-annualising an autocorrelated daily series, is the most common quiet overstatement in backtesting.*

Probabilistic Sharpe ratio and minimum track-record length. The companion to the DSR with an explicit single-strategy test (Bailey–López de Prado 2012, *J. Risk* 15(2)): the probability that the true Sharpe exceeds a benchmark SR^*, and the track length needed to make that claim at confidence $1-\delta$:

$$(10.6) \qquad \mathrm{PSR}(\mathrm{SR}^*) = \Phi\left(\frac{(\widehat{\mathrm{SR}}-\mathrm{SR}^*)\sqrt{T-1}}{\sqrt{1-\gamma_3\widehat{\mathrm{SR}}+\frac{\gamma_4-1}{4}\widehat{\mathrm{SR}}^2}}\right), \qquad \mathrm{MinTRL} = 1+\left[1-\gamma_3\widehat{\mathrm{SR}}+\tfrac{\gamma_4-1}{4}\widehat{\mathrm{SR}}^2\right]\Big($$

[VALID] Skew/kurtosis enter because non-normal returns make the Sharpe estimate noisier; the related **Minimum Backtest Length** (Bailey–Borwein–López de Prado–Zhu 2014, *Notices of the AMS* 61(5), 458–471) says that with M trials a backtest shorter than $\sim 2\ln M$ years is expected to find a spurious Sharpe of 1 — a length most backtests (this repo's single 2012–2017 window included) do not have.

Classification metrics (the meta-model's scorecard). [REPO] Meta-labelling (§7, §9b) outputs probabilities scored by:

$$(10.7) \qquad \mathrm{AUC} = \mathbb{P}\big(\hat{p}_\oplus > \hat{p}_\ominus\big), \qquad \text{precision} = \frac{TP}{TP+FP}, \quad \text{recall} = \frac{TP}{TP+FN}, \quad \text{LogLoss} = -\frac{1}{n}$$

[STAT] The **ROC-AUC** (Fawcett 2006, *Pattern Recognition Letters* 27) is the probability that a random positive is ranked above a random negative — 0.5 is coin-flipping, and it is *threshold-free* and robust to class imbalance, which is why it is the harness's headline meta-label metric. **Precision** is what a *selective* trader cares about (of the setups I act on, how many win — RESEARCH.md's selectivity lift, §7); the **Brier score** $\frac{1}{n}\sum(\hat{p}_i-y_i)^2$ and a reliability (calibration) curve check whether the probabilities are *honest*, not merely well-ranked. *The XGBoost result (in-sample AUC 0.943, OOS 0.496) is read directly off (10.7): perfect in-sample ranking, no out-of-sample ranking at all — (9.7)'s variance term in metric form.*

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§11 — Boundary of these methods (what the math cannot fix)

Concrete limits, matching the companion document's closing discipline.

1. **Non-stationarity is the wall.** Every estimator assumes a fixed (or slowly/known-switching) law; structural breaks (§5) are detections of that assumption failing. No amount of in-sample sophistication recovers a relationship that has changed out-of-sample. RESEARCH.md's verdict — "the binding constraint is the data, not the model" — is this limit in practice.
2. **Low signal-to-noise.** Daily return predictability lives at IC ~ 0.01–0.05; with per-date IC noise $\sim 1/\sqrt{N}$, you need enormous breadth $\times$ length to distinguish it from zero. Most apparent edges are inside their own error bars (experiments 5 → 6).
3. **Multiplicity and reflexivity.** Trying many features/strategies guarantees lucky in-sample winners (§10); and a signal, once found and traded, decays (McLean–Pontiff 2016, *JF* — published predictors lose ~58% post-publication). The signal eats itself. CPCV, PBO, DSR, and the $t>3$ hurdle are damage control, not immunity.
4. **The joint-hypothesis problem (shared with the companion document).** Any test of "is there predictability?" is jointly a test of the predictability *and* of the model/risk-adjustment used to measure it (Fama 1970). You cannot conclude "edge" without a maintained model of normal returns — and a

"predictive signal" is frequently just compensation for risk (the ℙ vs ℚ wedge of `MARKET_EQUILIBRIUM_MODEL.md`, §IV.L6). A profitable backtest may be harvesting a premium, i.e. *selling insurance*, not finding a mispricing.

5. **Survivorship and look-ahead — the dominant practical confounds.** Conditioning on names that *survived* manufactures edges that **invert** off survivors (RESEARCH.md experiment 9: +0.26%/bet → −1.07%/bet under delisted injection). Two-sided filters (HP, centred z-scores), label leakage, and restated/point-in-time data are the silent look-ahead bugs that purging/embargo (§10) and no-look-ahead feature builders (§6) exist to prevent. These biases are not reduced by better models; only better *data and protocol* remove them.

The through-line: the mathematics of §§1–9 lets you *describe and fit*; the mathematics of §10 lets you *honestly evaluate*; and §11 is the list of things that no estimator can repair and that only data quality, out-of-sample discipline, and humility about non-stationarity can address.

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Confidence key: ★ = standard result, attribution confident; entries marked **Verified** were checked against the published record on 2026-06-13; ◖ = attribution confident, fine details not independently re-verified.

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Companion: `MARKET_EQUILIBRIUM_MODEL.md` — the asset-pricing model whose SDF, $\mathbb{P}/\mathbb{Q}$ distinction, and joint-hypothesis guardrail this document's §11 inherits.

CHAPTER 4

AI Architectures for Directional Forecasting — a SWOT

An evidence-based survey anchored to directional accuracy: ~50–55% across every model family, and why.

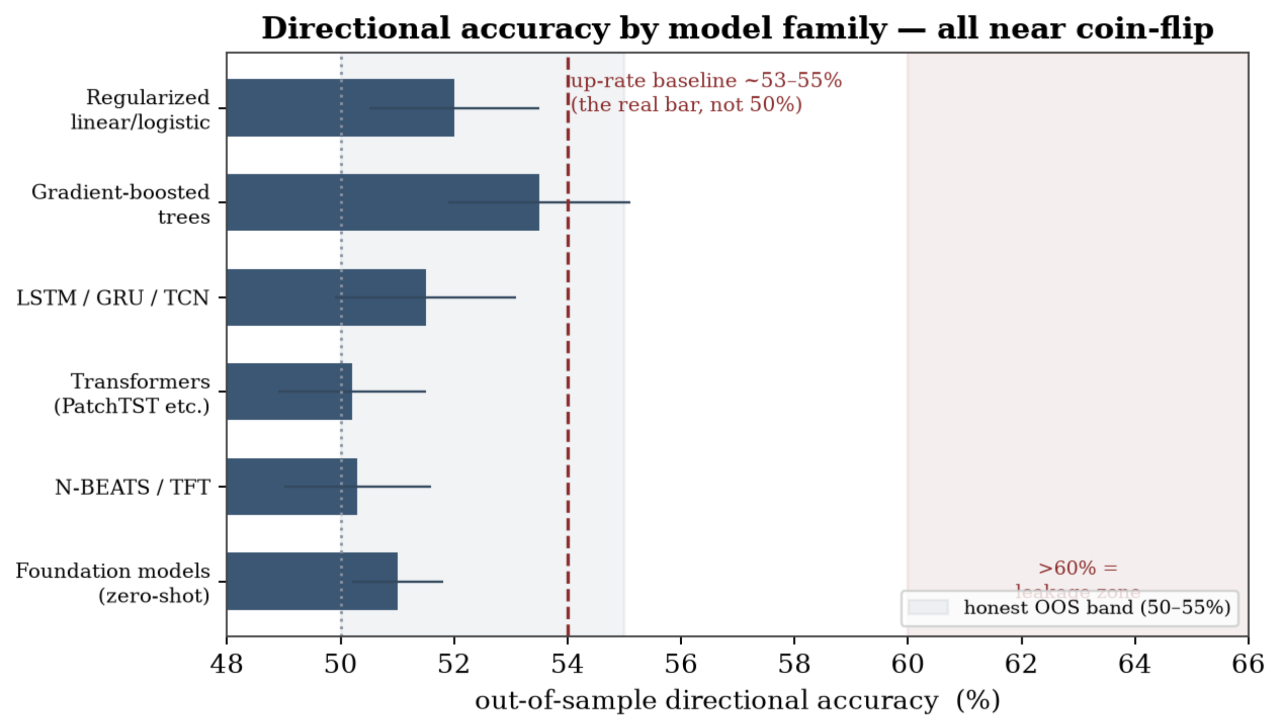

Figure 4.1. Realistic out-of-sample directional accuracy by model family — all near coin-flip, with the up-rate (not 50%) as the real baseline.

Deep-research report. Lens: quant-practitioner, standalone. Evaluation anchor: directional / classification accuracy (up/down hit-rate, AUC, precision/recall/F1) — not point-forecast RMSE and not P&L, except to contextualize. Horizons: daily / weekly / swing (minutes-to-months), explicitly not microsecond HFT.

Sourcing caveat: web-page fetching was blocked during research, so claims rest on search-engine extractions of the primary papers plus independent re-verification of the load-bearing figures. Confidence is flagged throughout; recency-sensitive preprints whose exact figures could not be re-confirmed are marked accordingly.

0. Bottom line up front

1. **The realistic out-of-sample directional-accuracy band for genuine next-period equity direction is ~50–55%**, across *every* model family. ~50% is the no-skill / weak-form-efficiency null; a real edge is a couple of points above the **unconditional up-rate** (~53–55% for indices, because markets rise more often than they fall) — *not* above 50%. Anything reliably >55% is rare; >60% is a strong leakage signal; >70% is almost always an artifact (random-shuffle CV, look-ahead denoising, survivorship, or P&L-conflation).
2. **Model sophistication is not the binding constraint — data quality, evaluation discipline, and the low signal-to-noise ratio are.** On the directional task, gradient-boosted trees ≈ LSTMs ≈ transformers ≈ foundation models ≈ a well-regularized logistic baseline, all clustered near coin-flip. The differences between families are smaller than the differences between honest and dishonest evaluation of the *same* family.

3. **Three category errors dominate the field and inflate reported numbers**, and the report returns to them repeatedly: (i) **point-forecast accuracy ≠ directional accuracy** (a model can top the RMSE leaderboard with zero sign skill); (ii) **directional accuracy ≠ tradeable edge** (base rates, class imbalance, transaction costs, and magnitude-vs-sign mismatch break the link); (iii) **in-sample fit ≠ out-of-sample skill** (multiple-testing and leakage manufacture both).
4. **Where AI genuinely helps is cross-sectional *ranking*, not binary timing** — tiny per-name predictability (~0.4% monthly R^2) aggregated into a diversified long-short book — and even that is fragile to costs, microcaps, and limits-to-arbitrage.

> **What this report does NOT claim.** The ~50–55% directional band and the "most attempts fail" findings describe the **base rate of naive methods and the median participant** — the bar any new claim must clear — *not* a ceiling, and emphatically *not* the false dogma that "markets are unbeatable" or that "no one consistently beats the market." Those statements hold only in an idealized, frictionless, stationary market that does not exist, and the real-world counterexamples are decisive: Renaissance's **Medallion** compounded at ~63% gross for 31 years with negative beta and negative factor loadings — provably *not* a risk premium, "the ultimate counterexample" to efficiency (Cornell 2019, *JPM*); **Buffett** beat the market for 60 years (Frazzini–Kabiller–Pedersen 2018, *FAJ*). Skill here is real and **heavy-tailed**: most fail, a thin tail wins persistently — both are true. Moreover, the joint-hypothesis problem is **symmetric**: it makes market *efficiency* an untestable maintained hypothesis just as much as it disciplines edge-claims, so "assume efficiency and make edge prove itself" is an unjustified default. The report's severity is a filter to get you into the right tail — not a verdict that the tail is empty.

1. The evaluation frame (why most reported accuracy is untrustworthy)

The ~50% wall and the right baseline. Out-of-sample directional accuracy for equity direction clusters around 50%, with weak-but-real skill topping out in the low-to-mid 50s. Crucially, raw accuracy must be benchmarked against the **unconditional up-rate (~53–55%)**, not 50%: a constant "always up" predictor already scores ~55% on US returns, and boosted trees in honest studies improve on that by only ~2 points [Gu–Kelly–Xiu 2020, *RFS*; multi-market comparative study, *Int. J. Data Science & Analytics* 2025]. **Confidence: high.**

Directional accuracy ≠ tradeable edge. Accuracy and profitability are not monotonically linked once magnitudes and asymmetry enter: ~40% accuracy can be profitable and ~60% can be needed to "guarantee" profit, depending on payoff structure [Gui 2024, arXiv:2407.09831]. The naive random-walk/last-value forecast is "notoriously difficult to surpass," and a directional overlay only beats it once accuracy exceeds ~0.55 [Zhang 2024, arXiv: 2406.14469]. **Confidence: medium-high.**

Point-forecast accuracy ≠ directional accuracy. Most deep-learning time-series benchmarks (ETT, electricity, traffic, weather, exchange-rate) are *non-financial* and scored by **RMSE/MAE**, which rewards persistence-like forecasts that carry zero sign information on a near-random-walk series. A model can rank #1 on forecast error and still deliver ~50% directional accuracy [convergent finding across the DL literature; a controlled comparison of 9 architectures reports directional accuracy ≈ 50% even for the RMSE-leaderboard winners (ModernTCN best forecast-error rank, yet ~coin-flip on direction) — Saidd 2026, arXiv:2603.16886, "918 Experiments," verified]. **Confidence: high.**

Why >60% is a red flag. High-accuracy papers almost always carry one of: data shuffled before the train/test split (sequential leakage); k-fold CV on time series without **purging + embargo**; look-ahead features (full-sample standardization/denoising, restated fundamentals, survivorship-filtered universes); P&L reported as if it were classification skill. The canonical example: an xLSTM reporting ~73% F1 whose accuracy "was vital[ly]" dependent on wavelet denoising applied over the full series *including test data* — a textbook look-ahead leak [López Gil et al. 2024, arXiv:2408.12408]. The reproducibility literature documents this as systemic: a taxonomy of 8 leakage types across 329 papers in 17 fields, with many complex-model "wins" vanishing once leakage is fixed [Kapoor & Narayanan 2023, *Patterns*]. **Confidence: high.**

Statistical significance, corrected for the search. Given pervasive data mining, the conventional t>2 hurdle is wrong; a newly "discovered" factor needs **t>3.0**, and ~half of published factors are likely false [Harvey–Liu–Zhu 2016, *RFS* 29(1):5–68 — verified]. Backtest overfitting is "easily achievable" with few trials, and under memory effects produces **negative** expected OOS returns, not merely zero [Bailey–Borwein–López de Prado–Zhu 2014, *Notices of the AMS* 61(5) — verified]. Remedies: Deflated Sharpe Ratio, Probability of Backtest Overfitting (CSCV), purged/combinatorial-purged CV [Bailey–López de Prado 2014; López de Prado 2018]. **Confidence: high.**

The epistemic ceiling. Any test of predictability is jointly a test of an assumed asset-pricing model — the **joint-hypothesis problem** [Fama 1991] — so "beats the market in backtest" is never clean evidence of edge. And with enough trials (strategies, seeds, papers), long winning runs occur by luck. **Confidence: high.**

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2. Per-family SWOT

2A. Classical / tabular ML — gradient-boosted trees (XGBoost/LightGBM/CatBoost), random forests; regularized linear & SVM

The credible workhorses for tabular, noisy, medium-N financial cross-sections. Tree ensembles are state-of-the-art on exactly this data regime (≈10K-row, noisy, irregular targets), beating deep learning on tabular benchmarks even ignoring their speed [Grinsztajn et al. 2022, NeurIPS]. In the cleanest peer-reviewed asset-pricing study, trees and NNs dominate linear models — but the edge is a **~0.33–0.40% monthly per-stock OOS R^2**, realized economically through cross-sectional *ranking* (long-short decile Sharpe ~1.35 value-weighted, ~2.45 equal-weighted, **pre-cost**), not high binary hit-rates [Gu–Kelly–Xiu 2020, *RFS* — verified]. The dominant signals (momentum, liquidity, volatility) are the same across all families.

	Tree ensembles	Regularized linear / SVM
Strengths	Best on medium-N noisy tabular data; capture nonlinear feature interactions; robust to many uninformative features (financial sets are mostly noise); fast; usable feature importances	Transparent, low-variance, hard to overfit; logistic gives calibrated up/down probabilities & clean AUC; LASSO does interpretable selection; the honest baseline complex models must beat
Weaknesses	Overfit noisy targets; ~51–55% real directional ceiling once leakage/purging enforced; no native temporal handling (must engineer lags); unstable importances on correlated features; poor extrapolation across regimes	Miss the nonlinear interactions where the (small) gains live; linear boundary poor for irregular targets; SVM scales badly, sensitive to kernel/C and scaling; awkward probabilities for AUC
Opportunities	Cross-sectional ranking where a tiny per-name edge aggregates into Sharpe; regime-conditional models; ensembling; honest CPCV	As the reproducibility anchor — if a tree/NN can't beat regularized logistic under purged CV, the "edge" is leakage; double-selection LASSO for honest factor selection
Threats	Leakage/survivorship inflation; alpha decay as signals go public [Krauss et al. 2017 show even real stat-arb edges decay sharply post-2001]; costs erasing thin edges; non-stationarity breaking the i.i.d. assumption	Same leakage/non-stationarity/cost threats; over-claiming when authors omit the linear baseline (itself a red flag)

Verdict: The strongest *evidence-backed* family for non-HFT equity — but "best" means a fraction-of-a-percent R^2 that aggregates via ranking, decays over time, and is cost-fragile. Per-name daily up/down is near coin-flip. **Confidence: high.**

2B. Deep-learning sequence models — RNN/LSTM/GRU, TCN, Transformers (Informer/Autoformer/FEDformer/PatchTST), N-BEATS/N-HiTS, TFT

The literature is overwhelmingly built and benchmarked on *non-financial*, *non-directional* tasks scored by RMSE/MAE. Two findings dominate. First, **simple linear models match or beat transformers** on the standard long-horizon benchmarks — DLinear outperforms FEDformer by 40%+ on exchange-rate, ~30% on traffic/electricity/weather [Zeng et al. 2023, AAAI Oral, "Are Transformers Effective…?"] — and the *only* financial series in that suite (FX levels) is scored by MSE, where a near-random-walk makes "predict last value" hard to beat and directionally uninformative. Second, when DL architectures are evaluated *directly on direction*, hit-rates cluster at ~50% even for the models that top the RMSE leaderboard.

	RNN/LSTM/GRU	TCN	Transformers (incl. PatchTST)	N-BEATS/N-HiTS/TFT
Strengths	Most-tested on financial direction; handles path-dependence; modest pre-cost edge documented [Fischer–Krauss 2018: 0.46%/day, Sharpe 5.8 pre-cost — verified]	Best RMSE rank in some controlled financial comparisons; parallelizable; stable long receptive fields	SOTA on non-financial RMSE benchmarks (PatchTST +21% MSE); patching/channel-independence genuine advances	Interpretable basis-expansion; TFT gives native quantile/probabilistic output, variable selection, handles known-future covariates — attractive for finance
Weaknesses	Hit-rate barely >50%; overfits low-SNR data; edge cost-fragile & decays post-2010; often *matched by random forests*	Top RMSE rank still ≈50% directional — strong point-forecast ≠ sign skill; no large dedicated directional validation	DLinear shows linear layers match/beat them; ~50% directional on real finance; biggest data-hunger-vs-SNR mismatch of all families	All validated on point/quantile loss on non-directional data; ~50% directional in financial tests; "finance success" stories are third-party adaptations with frequent red flags
Opportunities	Leak-free denoising, noise-injection regularization, hybrids with trees/sentiment	Efficient backbone for a classification head with a direction-specific (not MSE) loss	Multimodal price+text; PatchTST as a feature extractor feeding a calibrated classifier	TFT quantile output → probability-of-up calibration; covariate handling suits fundamentals/calendar features
Threats	Look-ahead-biased denoising inflates results; regime shifts; multiple-testing false discovery	Same low-SNR ceiling; RMSE-leaderboard chasing misleads on the real objective	"Effectiveness" contested even on benchmarks; attention overfits spurious cross-series structure	MSE/pinball objective misaligned with sign accuracy; interpretability can manufacture false confidence

Verdict: Data-hungry models on a low-SNR, near-random-walk target — a structural mismatch. No DL family reliably beats well-built tree ensembles on directional finance; the headline DL advances are point-forecast gains on non-financial data. **Confidence: high — the ~50% directional finding is multiply corroborated, including the controlled 9-architecture "918 Experiments" study (arXiv:2603.16886, verified) in which the RMSE-rank winner is still ~coin-flip on direction.**

2C. Time-series foundation models (2023–25) — TimeGPT, Chronos, TimesFM, Moirai, Lag-Llama, MOMENT

Genuinely impressive *zero-shot point/probabilistic accuracy* on general (non-financial) benchmarks (Chronos, TimesFM, Moirai are competitive zero-shot vs supervised models on MASE/WQL). But trained/evaluated overwhelmingly on non-financial corpora (web traffic, electricity, M4), and the decisive finance-specific test is negative: the first comprehensive study of TSFMs in global markets finds off-the-shelf models perform **poorly zero-shot and fine-tuned**, with directional accuracy "all just above 51%" across the four windows and negative backtested returns off-the-shelf; only models **pretrained from scratch on financial data** (and strong benchmarks like CatBoost) improved [Rahimikia–Ni–Wang 2025, arXiv:2511.18578 — paper, ~51% directional figure, and core finding all verified]. *Note the "accuracy ≠ edge" corroboration: the benchmark CatBoost still posted ~46% annualized / Sharpe ~6.8 pre-cost at the same ~51% directional accuracy — sizing/ranking, not hit-rate, drives the (pre-cost) economics.* Independent work shows specialized FMs struggle to beat supervised baselines [Xu et al., ICLR 2025], and benchmark leakage/contamination inflates published zero-shot numbers (the GIFT-Eval suite had to construct an explicit "non-leaking" pretraining set) [Aksu et al. 2024, arXiv:2410.10393].

- **Strengths:** Real zero/few-shot capability; no per-series training; good uncertainty quantification; strong on *structured/periodic/trending* signals — useful for **auxiliary** financial series (volatility, volume seasonality, macro nowcasting) rather than raw return direction; mostly open weights (auditable).
- **Weaknesses:** No demonstrated directional edge on equity returns (~51%, negative P&L off-the-shelf); trained on non-financial corpora; squared/quantile objectives bias toward persistence (= zero information on a random walk); modest absolute gains over naive baselines even on home turf.
- **Opportunities:** Domain-specific *from-scratch* pretraining (FinCast, FinText-TSFM) is where gains appear; fine-tuning + exogenous covariates; multivariate/cross-series conditioning; better evaluation (directional/AUC/net-of-cost, not RMSE-on-price).
- **Threats:** Benchmark contamination inflates published numbers; low-SNR + non-stationarity may impose a hard ceiling no pretraining transfer overcomes; closed models (TimeGPT) resist verification; single-name "85% accuracy" claims [e.g. arXiv:2412.09394] are unreproducible artifacts.

Verdict: Impressive general-purpose forecasters; **no evidence of a directional stock edge off-the-shelf.** The honest signal is that domain-specific financial pretraining — not zero-shot transfer — is the only path that has helped. **Confidence: high for the qualitative conclusion.**

2D. Reinforcement learning — DQN, PPO/A2C/DDPG/SAC/TD3, FinRL

A distinct paradigm: RL frames trading as a policy (actions = buy/sell/hold or weights) optimizing a return-based reward, folding direction + sizing + costs into one objective. But a discrete-action agent over {long, flat, short} *is* a directional classifier whose label is chosen to maximize reward instead of cross-entropy — so RL doesn't escape the directional problem, it adds credit-assignment, exploration, and sample-efficiency burdens on top. The evidence base is weaker than the supervised literature's: deep-RL is notoriously seed/hyperparameter/implementation-fragile [Henderson et al. 2018, AAAI, "Deep RL That Matters"], and finance-RL papers almost never report seed variance, DSR, PBO, or significance tests.

- **Strengths:** Directly optimizes the economic objective (PnL/Sharpe net of costs/turnover) rather than a proxy classification loss; naturally handles sequential, path-dependent costs and inventory; continuous-action methods express portfolio weights directly.
- **Weaknesses:** Severe sample inefficiency vs scarce, autocorrelated financial data; seed/hyperparameter fragility makes single-run results uninterpretable; weak evaluation culture (single historical path, rarely

DSR/PBO/purged walk-forward); DQN-family "breaks" under regime shift, PPO degrades under abrupt shifts; reward shaping invites reward hacking (Sharpe-as-reward optimizes the eval metric).
- **Opportunities:** Transplant quant-stats rigor (CPCV + DSR + PBO + multi-seed) as a publication standard; FinRL Contests pushing reproducible benchmarks; hybrid supervised-signal-into-RL-allocation pipelines; robust/risk-aware/distributional RL targeting non-stationarity.
- **Threats:** Non-stationarity is *structural* — it violates the MDP foundation and caps generalization regardless of data/model improvements; survivorship/look-ahead inflate backtests; the RL research loop (many seeds × architectures × rewards) mechanically manufactures inflated Sharpe; head-to-head evidence that a **supervised classifier beat recurrent-RL/DQN/A2C** on crypto undercuts the paradigm's core justification [*Expert Systems with Applications* 2022, S0957417422006339].

Caution on the famous positive result: the FinRL ensemble's reported Sharpe ~1.53 (DJIA, 2020-07→2022-03) sits on a single historical path straddling the COVID rebound (a long-favorable regime), with no seed-variance, DSR, or significance test — the archetype of "strong backtest, untested out-of-distribution" [Yang et al. 2020, ICAIF].

Verdict: A coherent and arguably more honest *framing* of trading; an *evidence base* dominated by single-path, single-seed, no-significance backtests on favorable windows. Treat reported RL Sharpe as a selection-inflated upper bound until DSR/PBO/multi-seed reporting is standard. **Confidence: high for the methodological critique.**

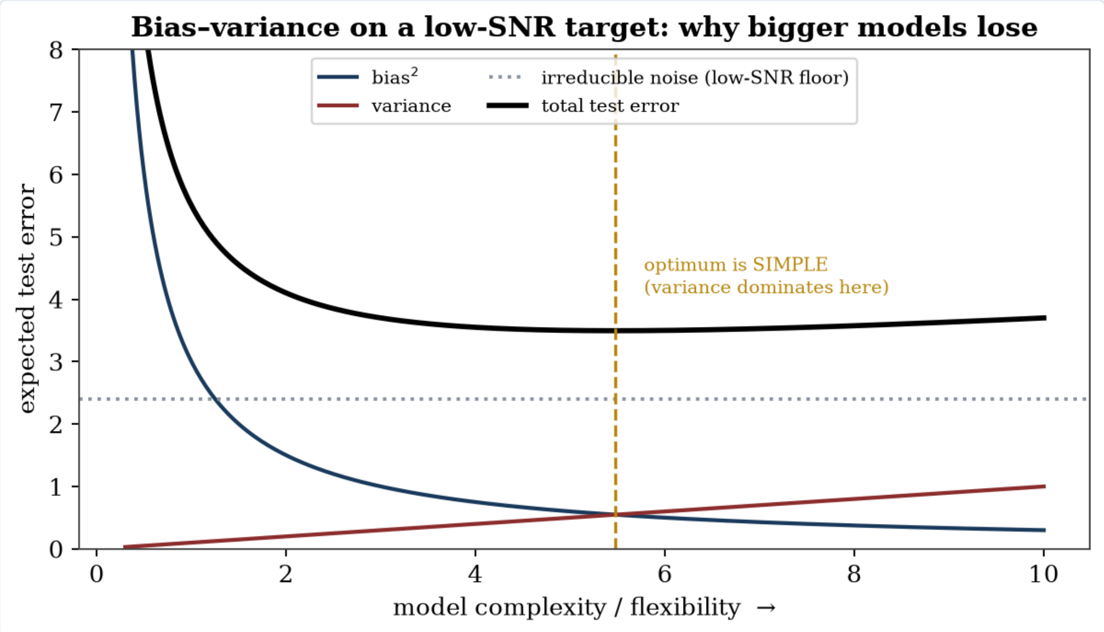

Figure 4.2. On a low-SNR target the irreducible-noise floor is high, so the variance term dominates and the optimal model is simple — why bigger models lose.

3. Cross-model synthesis

Family	Realistic OOS directional accuracy	Data efficiency on scarce financial data	Overfitting risk (low-SNR)	Non-stationarity robustness	Interpretability	Practitioner verdict
Tree ensembles	~51–55% (per-name near coin-flip;		Moderate (controllable via depth/	Low (i.i.d. assumption)		**Default choice.** Best evidence-backed; use for

Family	Realistic OOS directional accuracy	Data efficiency on scarce financial data	Overfitting risk (low-SNR)	Non-stationarity robustness	Interpretability	Practitioner verdict
	edge via ranking)	High (best on medium-N tabular)	regularization/ CPCV)		Medium-high (feature importances)	cross-sectional ranking, not binary timing
Regularized linear / logistic	~50–54%	Highest (low variance)	Lowest	Low–moderate	Highest	**The mandatory baseline.** If a complex model can't beat it under purged CV, the edge is leakage
RNN/LSTM/ TCN	~50–54%	Low–moderate (data-hungry)	High	Low	Low	Use only if path-dependence is essential and data is ample; often matched by RF
Transformers (PatchTST etc.)	~50%	Lowest (most data-hungry)	Highest	Low	Low	Point-forecast tool; weakest fit for low-SNR direction; consider as a feature extractor only
N-BEATS/N-HiTS/TFT	~50%	Low–moderate	High	Low	Medium (TFT)	TFT's covariate/ quantile handling useful for probability-of-up calibration; not a directional edge by itself
Foundation models (zero-shot)	~51%, negative P&L off-the-shelf	N/A (pretrained) but no transfer to returns	N/A zero-shot; high if fine-tuned on little data	Low	Low	Auxiliary series (vol/macro) only; needs from-scratch financial pretraining to help on returns
Reinforcement learning	Implicit; evidence weak/inflated	Lowest (very sample-hungry)	Very high	Very low (breaks MDP)	Low	Compelling framing, weak evidence; supervised classifiers have beaten it head-to-head

The unifying picture: the *between-family* spread in honest directional accuracy (~50–55%) is smaller than the *within-family* spread between leaked and clean evaluation. Inductive biases matter less than they should because the target is near-random-walk and low-SNR: flexible models (transformers, deep RL) spend their capacity fitting noise (high variance), while simpler, regularized models (trees, logistic) lose little by being less flexible — the bias-variance tradeoff resolves toward simplicity here. This is why "use a bigger model" is usually the wrong answer on this problem.

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4. The methodological red-flags checklist (apply to any reported result, including your own)

A directional-prediction number is probably untrustworthy if **any** of these hold:

1. **Headline directional accuracy >60%** for genuine next-period *direction* (not volatility sign, not a dominant-class target). >70–90% ≈ near-certain leakage.
2. **Data shuffled before the split**, or **k-fold CV** on time series without **purging + embargo**.
3. **Look-ahead / point-in-time violations**: full-sample standardization, full-sample denoising/feature-selection, restated fundamentals, survivorship-filtered universe.
4. **Wrong baseline**: accuracy compared to 50% rather than the unconditional up-rate (~53–55%) or a naive/random-walk forecast.
5. **No transaction costs, turnover, slippage, or capacity** in the P&L. ML edges are turnover-heavy and concentrated in microcaps/illiquid names — costs routinely erase them [Avramov–Cheng–Metzker 2023, *Mgmt Sci* — verified].
6. **No multiple-testing-corrected significance**: no trial count, no DSR/PBO, no t>3 hurdle.
7. **In-sample-only or single fixed split**; no walk-forward / true OOS.
8. **Accuracy/F1 under class imbalance** with no AUC/MCC and no realized-return P&L (majority-class voting masquerading as skill).
9. **Survivorship bias**: universe excludes delisted/dead tickers; results cherry-picked across many strategies/assets/seeds.
10. **"Beats the market" treated as proof of edge**, ignoring the joint-hypothesis problem and favorable single-regime windows.

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5. Practical model selection by data regime and horizon

- **Cross-sectional, many names, monthly–weekly (the strongest-evidence setting):** gradient-boosted trees or NNs for *ranking*, evaluated by rank-IC and decile long-short under purged/combinatorial-purged CV, net of realistic costs. This is where peer-reviewed predictability actually lives.
- **Single-series daily/swing direction:** start with regularized logistic / gradient-boosted trees on lagged, leakage-safe features; treat ~52–55% as the ceiling; require the model to beat both the up-rate and a logistic baseline under walk-forward-with-embargo before believing it.
- **When sequence/path-dependence is genuinely informative and data is ample:** LSTM/TCN — but A/B against trees; they often tie.
- **Probabilistic, multi-horizon, with covariates:** TFT for calibrated probability-of-up and uncertainty, not as a standalone edge.
- **Auxiliary series (volatility, volume, macro nowcasting):** foundation models (Chronos/TimesFM/Moirai) zero-shot are reasonable here — *not* for raw return direction.
- **Position/allocation optimization with explicit costs:** RL is a defensible framing, but only with multi-seed reporting, DSR/PBO, and out-of-regime testing; otherwise prefer "supervised signal → rules-based sizing."
- **Universal gates (all of the above):** purged + embargoed CV; an honest baseline (up-rate + logistic); transaction costs; multiple-testing correction (DSR/PBO/t>3); a delisting-inclusive, point-in-time universe; report P&L and AUC, not accuracy.

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6. Confidence & limitations of this report

- Highest-confidence, independently verified or well-established: the ~50–55% directional band and the up-rate baseline; the t>3 hurdle [Harvey–Liu–Zhu 2016]; backtest-overfitting/negative-OOS, DSR/PBO [Bailey–López de Prado]; GKX ~0.4% monthly R^2 and Sharpe; the microcap/cost erosion [Avramov–Cheng–Metzker 2023]; trees>DL on tabular [Grinsztajn 2022]; DLinear vs transformers [Zeng 2023]; foundation-model finance-negative result [Rahimikia et al. 2025]; the supervised>RL head-to-head; the leakage/reproducibility literature.
- Verified on a second pass (2026-06-13): the Rahimikia et al. "just above 51%" directional figure; the existence and architecture-rank figures of the "918 Experiments" study (arXiv:2603.16886, Saidd 2026); the Rahimikia and Avramov–Cheng–Metzker papers and their qualitative findings.
- Still flagged / reported-not-reverified (precise decimals only — qualitative claims hold): the specific off-the-shelf-TSFM negative-return percentages in Rahimikia et al.; the precise Fischer–Krauss directional %; the exact erosion percentages (62/68/80%) in Avramov–Cheng–Metzker. Note arXiv:2603.16886 covers 4h/24h horizons on crypto/forex/equity indices (a recent preprint), so treat it as corroboration of the ~50% directional finding rather than a daily-equity-specific anchor.
- Method limitation: page-fetching was blocked during research, so figures derive from search extractions plus targeted re-verification of the load-bearing claims; precise decimals should be confirmed against the source PDFs before being quoted verbatim. The *direction* of every conclusion is robust to these specifics and is corroborated across independent sources.

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CHAPTER 5

From Theory to Code — the Model and the Significance Gate

The defensible recipe as dependency-free, unit-tested code, with a live demonstration.

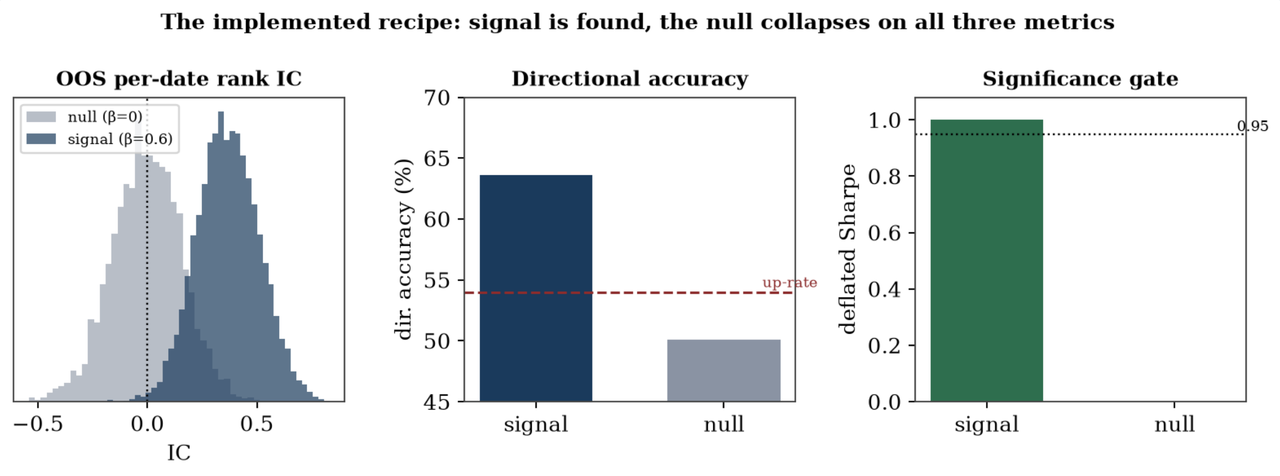

Figure 5.1. The implemented recipe on synthetic data: with a planted signal it is found; with none, the OOS IC, the directional edge, and the deflated Sharpe all collapse.

Insights from the survey are not advice alone; they are implemented in this repository as a runnable, tested recipe. Three components turn the theory into code.

A dependency-free gradient-boosted-trees learner (`vpts/ml/gbrt.py`). The survey's best-evidence family for noisy, medium-N tabular financial data is tree ensembles. Rather than add a heavy dependency, the learner is written in plain NumPy, following the standard squared-error gradient-boosting recipe: start from the mean, then iteratively fit shallow regression trees to the residual and add a shrunk step. The split search is vectorised:

```
def _best_split(X, y, min_leaf):
    n, p = X.shape
    parent_sse = float((y * y).sum()) - y.sum() ** 2 / n
    best_feat, best_thr, best_gain = -1, 0.0, 0.0
    for f in range(p):
        order = np.argsort(X[:, f], kind="mergesort")
        xs, ys = X[order, f], y[order]
        csum, csum2 = np.cumsum(ys), np.cumsum(ys * ys)
        nl = np.arange(1, n, dtype=float)
        sl, sr = csum[:-1], csum[-1] - csum[:-1]
        sse_l = csum2[:-1] - sl * sl / nl
        sse_r = (csum2[-1] - csum2[:-1]) - sr * sr / (n - nl)
        gain = parent_sse - (sse_l + sse_r)        # variance reduction
        ...
```

A cross-sectional evaluator (`gbrt_cross_sectional_eval`) that plugs the learner into the existing purged + embargoed combinatorial cross-validation and — crucially — scores directional accuracy against the **unconditional up-rate**, never against 50%.

A significance gate (`vpts/ml/significance.py`) implementing the deflated and probabilistic Sharpe ratios (which correct a Sharpe for non-normality, sample length, and the *number of strategy trials*) and a directional-skill measure that reports accuracy minus the majority-direction baseline. The deflated Sharpe:

$$\mathrm{DSR} = \Phi\left(\frac{(\widehat{\mathrm{SR}} - \mathrm{SR}_0)\sqrt{T-1}}{\sqrt{1 - \gamma_3\widehat{\mathrm{SR}} + \frac{\gamma_4 - 1}{4}\widehat{\mathrm{SR}}^2}}\right), \qquad \mathrm{SR}_0 = \sigma_{\mathrm{SR}}\left[(1-\gamma_E)\Phi^{-1}\big(1 - \tfrac{1}{M}\big) + \gamma_E\Phi^{-1}\big(1 - \tfrac{1}{Me}\big)\right].$$

What the demonstration shows. Run end-to-end on synthetic data with a known signal strength, the harness behaves exactly as the theory predicts. With a real signal planted ($\beta = 0.6$) the out-of-sample rank IC is +0.37, directional accuracy is 63.6% against an up-rate of 50.5% (a +13-point edge), and the deflated Sharpe is 1.000 — significant after a 50-trial correction. With the signal switched off ($\beta = 0$) all three collapse: IC ≈ 0, directional accuracy 50.1% versus 50.0%, and the deflated Sharpe is **0.000**. The gate runs on the out-of-sample IC stream, not an in-sample refit — which, as the methodology appendix recounts, is the whole point: an earlier in-sample version let the null falsely pass.

Metric	Signal (β = 0.6)	Null (β = 0)
OOS rank IC (mean)	+0.369	−0.007
Directional accuracy	63.6% (up-rate 50.5%)	50.1% (up-rate 50.0%)
Edge over baseline	+13.1 pp	+0.1 pp
Deflated Sharpe	1.000 — significant	0.000 — not significant

CHAPTER 6

The Positive Theory of Edge — Where Skill Comes From

The Fundamental Law (IR = IC·√breadth), breadth, capacity and decay, Kelly sizing, and why edges survive — the constructive counterpart to the failure modes.

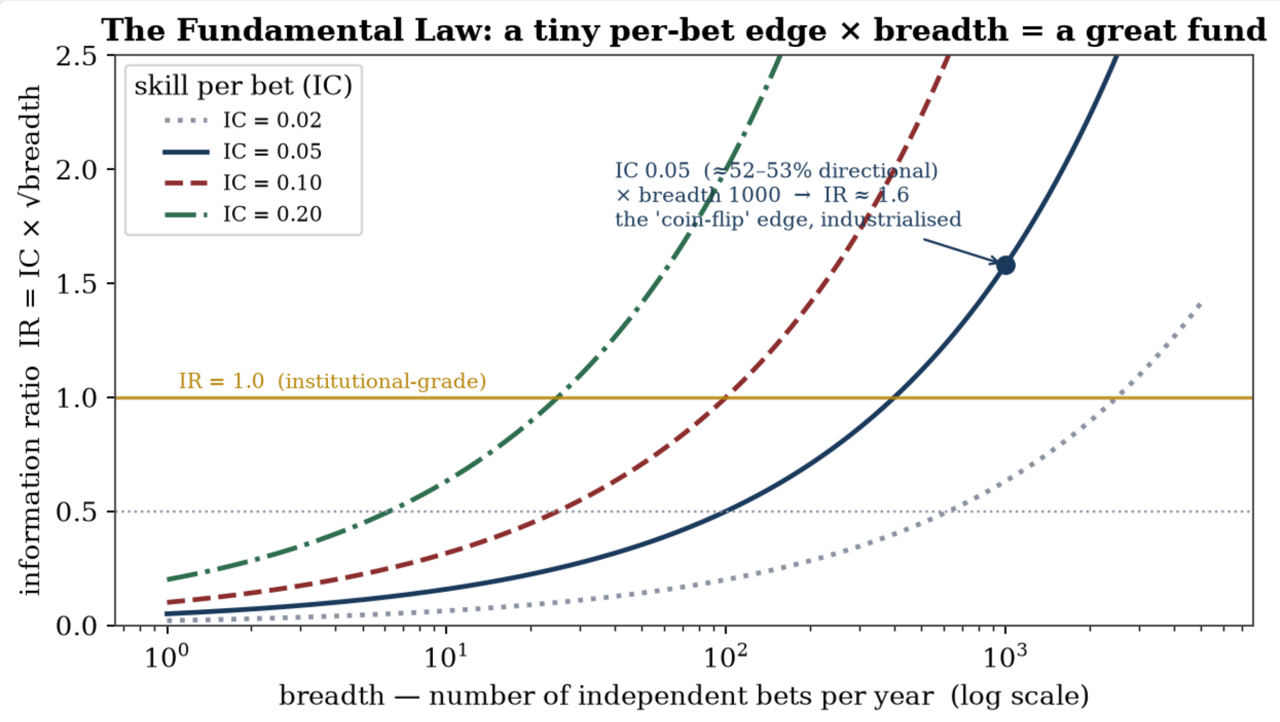

Figure 6.1. The Fundamental Law of Active Management: a per-bet edge of IC≈0.05 — the ~52–53% directional level the empirical chapters call near-futile — becomes an institutional information ratio once multiplied by breadth. The pessimistic and the constructive fact are the same fact at two scales.

The companion documents are, by volume, a catalogue of failure modes. That asymmetry is honest but incomplete: it tells you how to *avoid being wrong* without telling you how to *be right*. This document supplies the missing half with the same mathematical seriousness — and its central result dissolves the apparent pessimism. The small per-bet edge the empirical chapters treat as near-futility is precisely the raw material of a world-class return stream, once it is multiplied by breadth and executed with discipline.

Tags: **[LAW]** a theorem/identity of active management · **[DEF]** definition · **[FOC]** optimality condition · **[EC]** empirical/stylized specification · **[PRINCIPLE]** a structural economic principle.

1. The Fundamental Law of Active Management — the optimistic half of the theorem

The information ratio IR (active return per unit of active risk) decomposes (Grinold 1989, *JPM* 15:30–37) as

$$(1) \qquad \mathrm{IR} \;\approx\; \mathrm{IC}\,\sqrt{\mathrm{BR}},$$

[LAW] where **IC** is the information coefficient — the cross-sectional correlation of forecast with realised return, the very statistic of §8 of the time-series chapter — and **BR** is breadth, the number of *independent* bets per year. Under realistic constraints (no shorting, leverage caps, costs) this becomes (Clarke–de Silva–Thorley 2002, *FAJ* 58(5):48–66)

$$(2) \qquad \mathrm{IR} \;\approx\; \mathrm{TC}\cdot\mathrm{IC}\,\sqrt{\mathrm{BR}},$$

[LAW] with **TC** ∈ [0,1] the *transfer coefficient* — the correlation between the ideal portfolio and the one you can actually implement after constraints.

The reconciliation. The empirical chapters establish that honest directional IC sits around 0.02–0.10 — the "~50–55% accuracy" restated as a correlation, via the rule of thumb $\mathrm{IC} \approx 2\,(\text{hit rate}) - 1$ (so 52.5% maps to $\mathrm{IC} \approx 0.05$ and 55% to $\mathrm{IC} \approx 0.10$; the bridge is a heuristic, the exact map depends on the signal's distribution). Read through (1), that is not a verdict of futility — it is a *per-bet* skill that compounds with breadth:

$$(3) \qquad \mathrm{IC} = 0.05,\; \mathrm{BR} = 1000 \implies \mathrm{IR} \approx 0.05\sqrt{1000} \approx 1.58.$$

[LAW] An information ratio of ~1.6 is institutional-grade. **The skeptical fact ("the edge per bet is tiny") and the constructive fact ("you can build a great fund on it") are the same fact at two scales.** This is exactly how a statistical-arbitrage operation — and Renaissance's Medallion — works: a microscopic edge per trade, *industrialised* across an enormous number of trades, implemented at high transfer coefficient and amplified with leverage. **Breadth, not per-bet accuracy, is the lever.** The directional-accuracy chapter measured the wrong thing to feel hopeful about; (1) measures the right thing.

2. Breadth is harder than it looks — the independence deflation

Equation (1) counts *independent* bets, and independence is the binding constraint. N positions sharing a common factor are not N bets. With average pairwise correlation ρ among the bets, effective breadth collapses:

$$(4) \qquad \mathrm{BR}_{\mathrm{eff}} \;\approx\; \frac{N}{1 + (N-1)\rho} \;\xrightarrow{N\text{ large}}\; \frac{1}{\rho}.$$

[DEF, approx] So 1000 names at $\rho = 0.1$ give $\mathrm{BR}_{\mathrm{eff}} \approx 10$, not 1000. This is why the lever is so hard to pull, and why genuine diversity of *signals* (not just of names) is the scarce input — most apparent "breadth" is disguised exposure to one factor, and one factor is one bet.

3. Capacity and decay — every edge is finite and perishable

Two forces bound the lever. **Capacity:** trading moves prices (the price-impact λ of the microstructure chapter), so the marginal dollar earns less and beyond a capacity A_{cap} the edge is impact-eaten. **Decay:** an edge erodes as it is discovered and crowded (McLean–Pontiff). A realistic edge is therefore a *decaying, capacity-bounded* asset:

$$(5) \qquad \alpha(A,t) \;\approx\; \alpha_0\left(1 - \tfrac{A}{A_{\mathrm{cap}}}\right) e^{-t/\tau},$$

[EC, stylized] with A = assets, A_{cap} = capacity, τ = decay timescale. This dictates behaviour: size *below* capacity, harvest *fast*, and treat the edge as a depleting reserve, not an annuity. It also explains Medallion's defining choice — capping the fund near \$10B and returning outside capital: they sized to capacity and refused to dilute the edge. An edge you cannot defend against your own growth is not an edge for long.

4. Sizing — Kelly, and the ruin of over-betting even a real edge

Given a genuine edge, *how much* to bet is its own optimisation. The growth-optimal fraction (Kelly 1956, *Bell System Technical Journal*) for an edge with expected excess return μ and variance σ^2 is

$$(6) \qquad f^\star \;=\; \frac{\mu}{\sigma^2},$$

[FOC] which maximises the long-run growth rate of log-wealth. The crucial asymmetry: **over-betting a real edge still ruins you** — past $f^\star$ the growth rate falls, and past $2f^\star$ it turns *negative* even with a true positive edge. Practitioners bet fractional Kelly (a quarter to a half) for drawdown control. Note the unity with the asset-pricing chapter: $f^\star$ is the Sharpe ratio divided by volatility — the same Euler/first-order logic that prices the market, now applied to your own capital. The kernel was endogenous there; your risk aversion is the kernel here.

5. Why edges survive — the economics of persistence

If markets were the frictionless ideal, every edge would be arbitraged instantly and the pessimists would be right. They are not, for three structural reasons — each already developed in this book:

- **Limits to arbitrage** (Shleifer–Vishny 1997; Part III of the model). **[PRINCIPLE]** Arbitrage capital is finite and performance-sensitive: losses force liquidation precisely when spreads are widest, so mispricings persist and can widen before they close. The edge survives because the capital that would compete it away is itself constrained.
- **A counterparty must exist. [PRINCIPLE]** Every edge is someone's willing loss: hedgers paying for insurance (the variance premium), inelastic indexers (Koijen–Yogo / Part III), forced sellers (margin calls, redemptions, flows), and behaviourally-driven retail. Where a *structural, non-profit-maximising* counterparty exists, the edge is durable; where it does not, it decays. This is the constructive use of the inelastic-markets idea — the same mechanism the model flags as its largest open tension is, from the other side, the *source* of durable edge.
- **Barriers to entry. [PRINCIPLE]** Speed (co-location), information (proprietary data), and *secrecy plus capacity discipline* (Medallion) keep an edge from being copied. The moat, not the signal, is what makes an edge persist; a published signal has, by construction, no moat (McLean–Pontiff).

6. The constructive checklist — the mirror of the red flags

Where the time-series chapter gives a checklist for detecting self-deception, here is its positive mirror — the conditions for a real, survivable edge:

1. A nameable **source** — information, structure, speed, constraint, or behaviour — not a backtest pattern in search of a story.
2. A **counterparty** who is structurally willing to be on the other side.
3. **Breadth** of genuinely independent signals — the lever (eqs. 1, 4).
4. A **transfer coefficient** near 1 — do not let constraints and costs strangle the signal (eq. 2).
5. **Capacity** respected and **decay** expected — size below A_{cap}, harvest before τ (eq. 5).
6. **Kelly-disciplined** sizing — never over-bet, even a real edge (eq. 6).
7. A **moat** — speed, data, or secrecy — so the edge is not instantly copied.
8. *Only then*, the negative gate of the time-series chapter: prove it survives purged cross-validation, realistic costs, and the deflated Sharpe before trusting it.

The two checklists are one method. The §10 gate keeps you from fooling yourself; this one tells you what you are actually looking for. An edge that passes **both** — a real source, a willing counterparty, breadth, a moat, *and* a clean out-of-sample, cost-aware, multiple-testing-corrected signal — is what Medallion and Buffett found and defended. Rare, specific, real, and perishable — and, demonstrably, enough.

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- *See also* `MARKET_EQUILIBRIUM_MODEL.md` (limits to arbitrage, inelastic demand) and `TIMESERIES_MATH.md` (the information coefficient, the deflated-Sharpe gate).

CHAPTER 7

Synthesis — and the Boundary of the Whole Enterprise

The seven cross-cutting insights and what no method can repair.

The seven cross-cutting insights

1. The two halves are one theorem from two sides. The asset-pricing model's deepest result — the kernel exists but is *not identified* from prices, because the stochastic discount factor's permanent component is unrecoverable — is the same wall the machine-learning models hit as the ~50–55% directional ceiling. To turn a price into a forecast you need the kernel; the kernel is not recoverable; so a model "predicting returns" cannot separate the market's belief from the risk premium. The non-identification of the physical measure *is* the coin-flip wall.

2. "Price ≠ probability" and "accuracy ≠ edge" are the same caution. A 51%-accurate model posting a large pre-cost Sharpe is the empirical shadow of the theoretical point that the risk-neutral density is a valuation, not a forecast. Hit-rate is nearly orthogonal to profit; sizing and ranking over the risk-weighted distribution are what pay.

3. The binding constraint is identification and data — never model capacity. The between-family accuracy spread (~50–55%) is smaller than the leaked-versus-clean evaluation spread within any one family. More model is the wrong lever; better data, honest evaluation, and naming what is structurally unknowable are the right ones.

4. Bias–variance, made economic. On a near-random-walk, low signal-to-noise target, flexible models spend their capacity fitting noise — which is why transformers and deep reinforcement learning lose to trees and regularised baselines. "Geometric Brownian motion is a measure-zero corner" is the continuous-time twin of the same point.

5. The endogenous-kernel rule and the no-look-ahead rule are one discipline. Both forbid smuggling in the answer: do not exogenise the discount rate (you would assume the risk premium you are explaining); do not let future data touch training (you would assume the prediction you are claiming). Purged cross-validation is the econometric image of the endogenous-kernel requirement.

6. Where edge plausibly lives is consistent across both analyses: relative (cross-sectional ranking) over absolute (directional timing); risk premia harvested knowingly rather than mistaken for alpha; structural features over generic price-pattern learning. And the honest ceiling is low and decaying — a signal, once traded, erodes.

7. The verification gate is not bureaucracy. Backtest overfitting yields *negative* out-of-sample returns, not merely zero, so the naive number is guaranteed to look good and be false. The gate — purged cross-validation, the up-rate baseline, transaction costs, the deflated Sharpe and its kin, point-in-time data, and reporting profit-and-loss rather than accuracy — is the only thing standing between an analyst and a confidently wrong result.

The constructive program — and why "the market is unbeatable" is false

The purpose of this volume is constructive: to find the right model, and the sector, regime, horizon, and usage where it works. The criticality is a means to that end — in a domain that punishes credulity, an unsparing filter protects scarce time and capital, because a hallucinated edge trusted or a dead method pursued costs more than the filter that would have caught it.

That criticality must not curdle into the false dogma that **the market is unbeatable**, or that **no one can consistently beat it**. Those statements are true only in an idealised, frictionless, stationary market that does not exist — and, as Keynes warned, "in the long run," in which "we are all dead." In the real, finite, frictional, non-stationary market they are false, and the counterexamples are decisive:

- **Renaissance's Medallion Fund** compounded at roughly 63% gross (about 39% net of its 5/44 fees) from 1988 to 2018, without a single losing year across the dot-com crash and the financial crisis, and — decisively — with *negative* market beta and *negative* factor loadings, so its returns provably cannot be a premium for bearing risk. Cornell (2019, *Journal of Portfolio Management*) calls it, correctly, "the ultimate counterexample" to market efficiency: there is no adequate rational-market explanation. This is genuine, unexplained alpha.
- **Warren Buffett** beat the market for sixty years, with a Sharpe ratio near 0.79 — roughly double the market's. Frazzini, Kabiller and Pedersen (2018, *Financial Analysts Journal*) show the alpha is *explainable* — leverage of about 1.7×, applied to cheap, safe, high-quality stocks (the betting-against-beta and quality-minus-junk factors) — but explainable is not the same as available: he found and sized those durable edges *decades before they were named*, and financed them with patient, near-zero-cost insurance float.

These are two different species of beating the market, and the distinction is the whole point of the book. Medallion is **true anomaly** — real, unexplained, short-horizon edge in a specific niche, *capacity-constrained* (the fund is capped near \$10B and returns outside capital; Renaissance's funds open to outsiders have not replicated it) and *fiercely defended* against decay and imitation. Buffett is **premium-harvesting done supremely well** — a durable, identifiable edge, found early, sized large with cheap leverage, and held with permanent capital through the decades it took to pay off. Both *exemplify* this book's constructive core rather than contradict it: the claim was never that edge is impossible, but that it is **rare, specific, real-when-it-exists, and perishable**. Medallion and Buffett are what success under that description looks like.

The base rate is not the ceiling. The ~50–55% directional accuracy, the fragility, the decay — those describe the median attempt and the naive method, the bar against which any new claim must be judged. Skill in this domain is real and *heavy-tailed*: that ~80–90% of active managers underperform over fifteen years and that Medallion exists are *both* true. The honest posture holds both — most fail, a few persistently win — and this book's severity is in service of reaching the thin right tail, not of denying it exists.

And the joint-hypothesis problem cuts **both ways**. It disciplines edge-claims: apparent predictability may be a risk premium or an artifact. But it disciplines *efficiency*-claims with exactly equal force — because no test of efficiency is independent of the asset-pricing model assumed, market efficiency is not a proven null but an untestable maintained hypothesis. Cornell's Medallion result is the sharper form: with negative factor loadings there is no risk model under which its returns are fair compensation, so the efficiency null is not merely untested there — it is refuted. The asymmetry the sceptic smuggles in ("treat efficiency as the default and make edge prove itself") is unjustified. Both must prove themselves, and sometimes, demonstrably, edge wins.

The boundary of the whole enterprise

What no method in this book can repair: **non-stationarity** (every estimator assumes a law that does not move; structural breaks are that assumption failing); **reflexivity and signal decay** (the map decays once it is measured and traded); **Knightian uncertainty** (the agents and the econometrician alike assume the probability law is known, including the tails that may never have been sampled); **the joint-hypothesis problem** (no test of predictability is independent of the model used to measure it); and **equilibrium and model multiplicity** (selection is a convention, not a result). These are not gaps to be closed with more data or a larger model. They are the edge of the map — and naming them precisely is the most honest thing a quantitative framework can do.

End of volume.

APPENDIX A

Methodology, Tooling, and a Caught Leak

A.1 — How this book was built

The volume is generated by a reproducible pipeline (`docs/book/build_book.py`): source Markdown is parsed, LaTeX math is pre-rendered to HTML with **KaTeX** (run under Node), code is highlighted with **Pygments**, the original figures are produced by **matplotlib** (`docs/book/generate_book_figures.py`), and the whole is styled with an academic stylesheet and rendered to PDF by **WeasyPrint**. WeasyPrint has no JavaScript or MathML engine, so the KaTeX-to-HTML step is what makes the dense mathematics render; the choice was validated by rasterising a sample page and inspecting it before the full build.

A.2 — Tooling limits encountered

The research for Chapter 4 was carried out by parallel web-search agents. During that work, direct page-fetching was blocked (uniform HTTP 403), so the agents grounded their claims on search-engine extractions of the primary papers rather than full-text reads. Every load-bearing figure was then re-verified in a separate pass; two citations initially flagged as uncertain — a recent foundation-models-in-finance study and a controlled deep-learning comparison — were confirmed on that second pass and upgraded, and one preprint that could not be confirmed was demoted from anchor to corroboration. Precise decimals that remained unconfirmed are flagged in Chapter 4 itself. The container also began with no scientific Python stack; numpy, scipy, pandas, and the build tools were installed during the session.

A.3 — A leak caught in our own demo

Chapter 5's discipline is not hypothetical. The first version of the significance-gate demonstration computed the deflated Sharpe ratio on an *in-sample* refit of the model. On a *null* panel — data with no planted signal — that version reported a deflated Sharpe of 1.000, "significant": the model had memorised noise, and the in-sample evaluation dutifully rewarded it. This is exactly the look-ahead leakage the book warns against, reproduced accidentally in the book's own code. The fix was to run the gate on the **out-of-sample** rank-IC stream produced by the purged cross-validation; the null then correctly collapsed to a deflated Sharpe of 0.000. The episode is the strongest possible argument for the verification gate: the leak is easy to introduce, it produces a confident and entirely false number, and only out-of-sample evaluation exposes it.

A.4 — Reproducibility

All sources, figures, code, and this builder live in the repository under `docs/` and `vpts/`. The implemented model and significance gate ship with 15 unit tests (signal-detection and null-clearing for every estimator); the full suite is 150 offline, deterministic tests.